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**Ying et al.**

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(45) **Date of Patent: Sep. 9, 2025**

(54) **A1 POLICY FUNCTIONS FOR OPEN RADIO ACCESS NETWORK (O-RAN) SYSTEMS**

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(73) Assignee: **Intel Corporation**, Santa Clara, CA (US)

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(86) PCT No.: **PCT/US2022/036130**

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(2) Date: **Sep. 12, 2023**

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PCT Pub. Date: **Jan. 12, 2023**

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(60) Provisional application No. 63/218,861, filed on Jul. 6, 2021.

(51) **Int. Cl.**
***H04L 41/0894*** (2022.01)

(52) **U.S. Cl.**
CPC .................................. ***H04L 41/0894*** (2022.05)

(58) **Field of Classification Search**
CPC .................................................... H04L 41/0894
See application file for complete search history.

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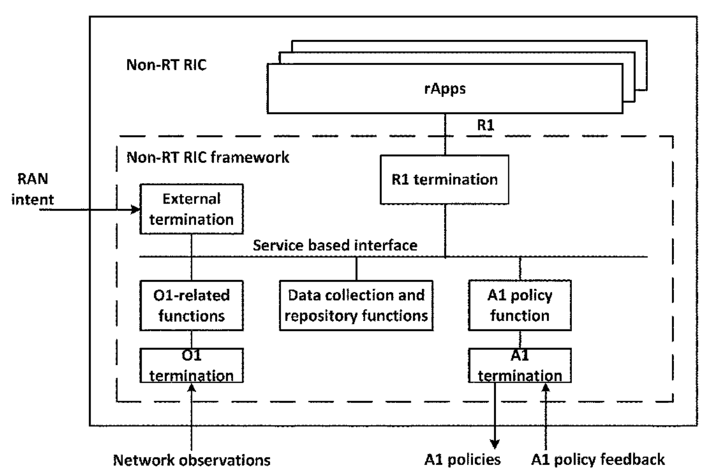
(57) **ABSTRACT**

The present invention relates to an apparatus comprising: memory to store policy statement information for a plurality of radio access network (RAN) automation applications (rApps); and processing circuitry, coupled with the memory, to: retrieve the policy statement information from the memory, wherein the policy statement information includes respective policy scope identifiers for respective rApps in the plurality of rApps; identify a conflict associated with common or overlapping policy scope identifiers between two or more rApps from the plurality of rApps; modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict; and notify the two or more rApps of the modification of the one or more A1 policies.

**20 Claims, 30 Drawing Sheets**



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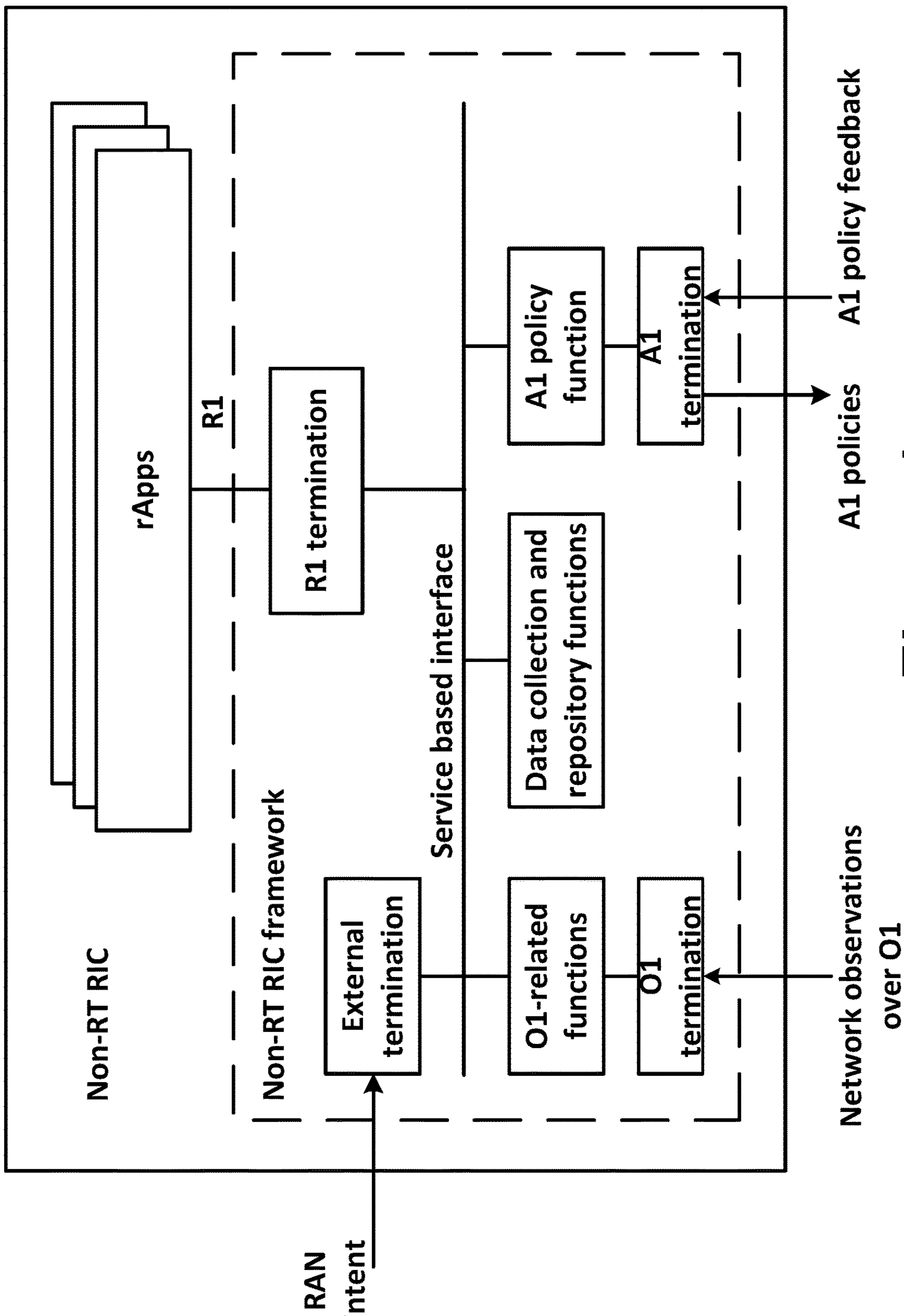

Non-RT RIC
rApps
R1
R1 termination
Non-RT RIC framework
External termination
Service based interface
Data collection and repository functions
A1 policy function
A1 termination
O1-related functions
O1 termination
RAN intent
Network observations over O1
A1 policies
A1 policy feedback

Figure 1

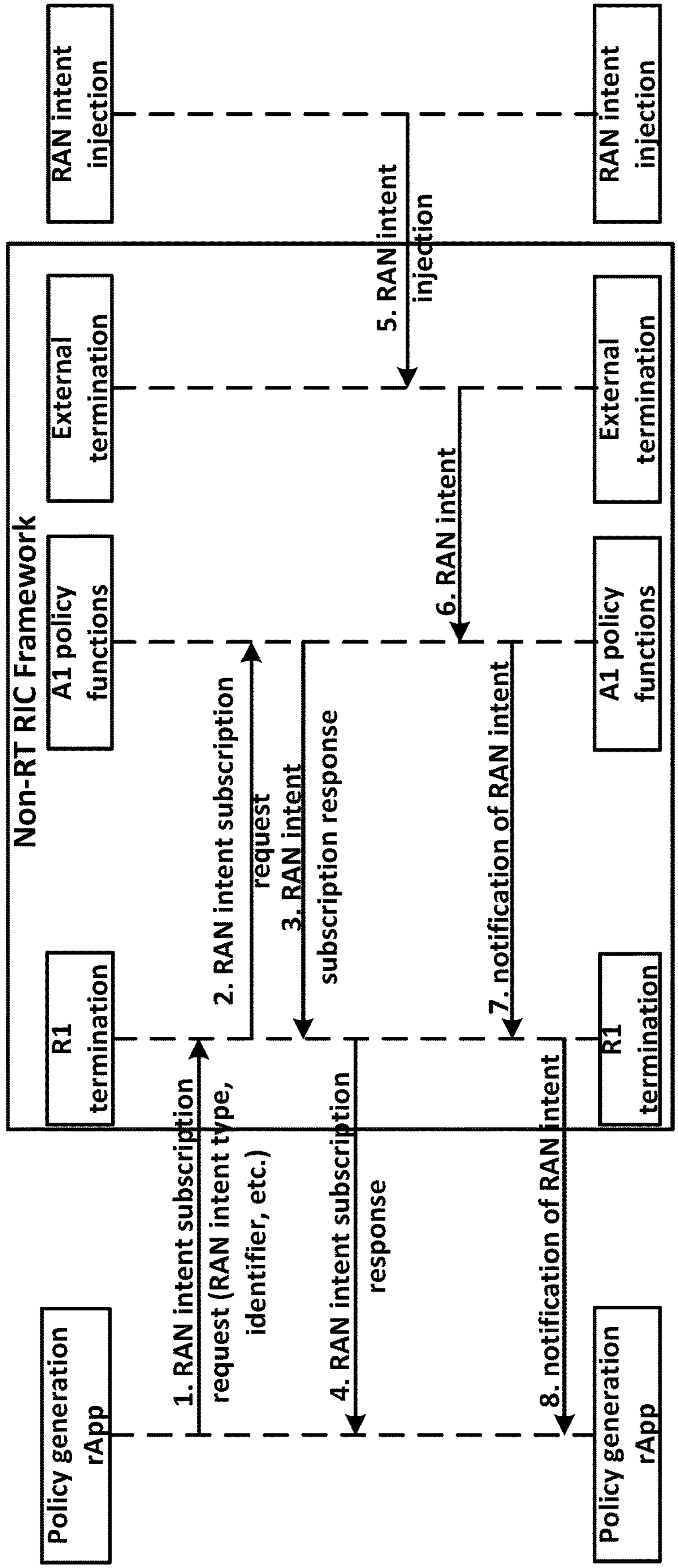

Non-RT RIC Framework
Policy generation rApp
R1 termination
A1 policy functions
External termination
RAN intent injection
1. RAN intent subscription request (RAN intent type, identifier, etc.)
2. RAN intent subscription request
3. RAN intent subscription response
4. RAN intent subscription response
5. RAN intent injection
6. RAN intent
7. notification of RAN intent
8. notification of RAN intent

Figure 2

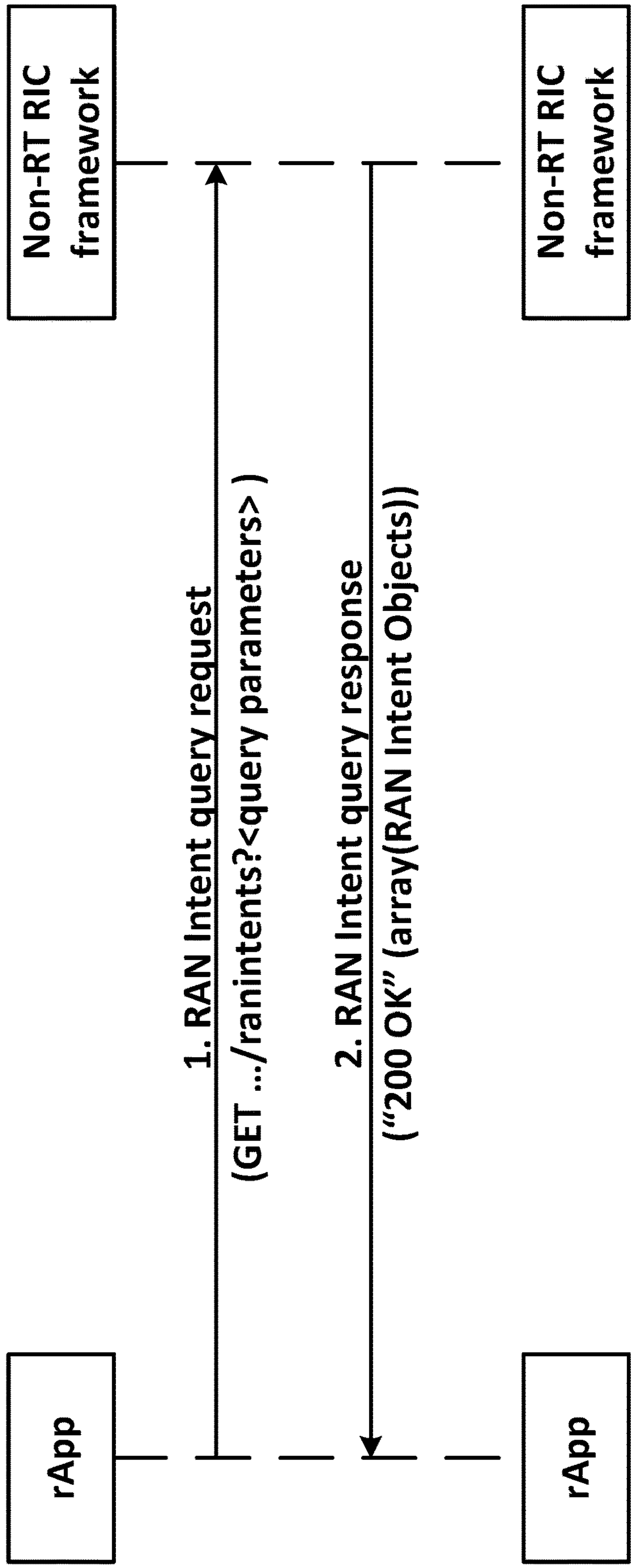

Non-RT RIC framework
1. RAN Intent query request
(GET .../ranintents?<query parameters> )
2. RAN Intent query response
("200 OK" (array(RAN Intent Objects))
Non-RT RIC framework
rApp
rApp

Figure 3

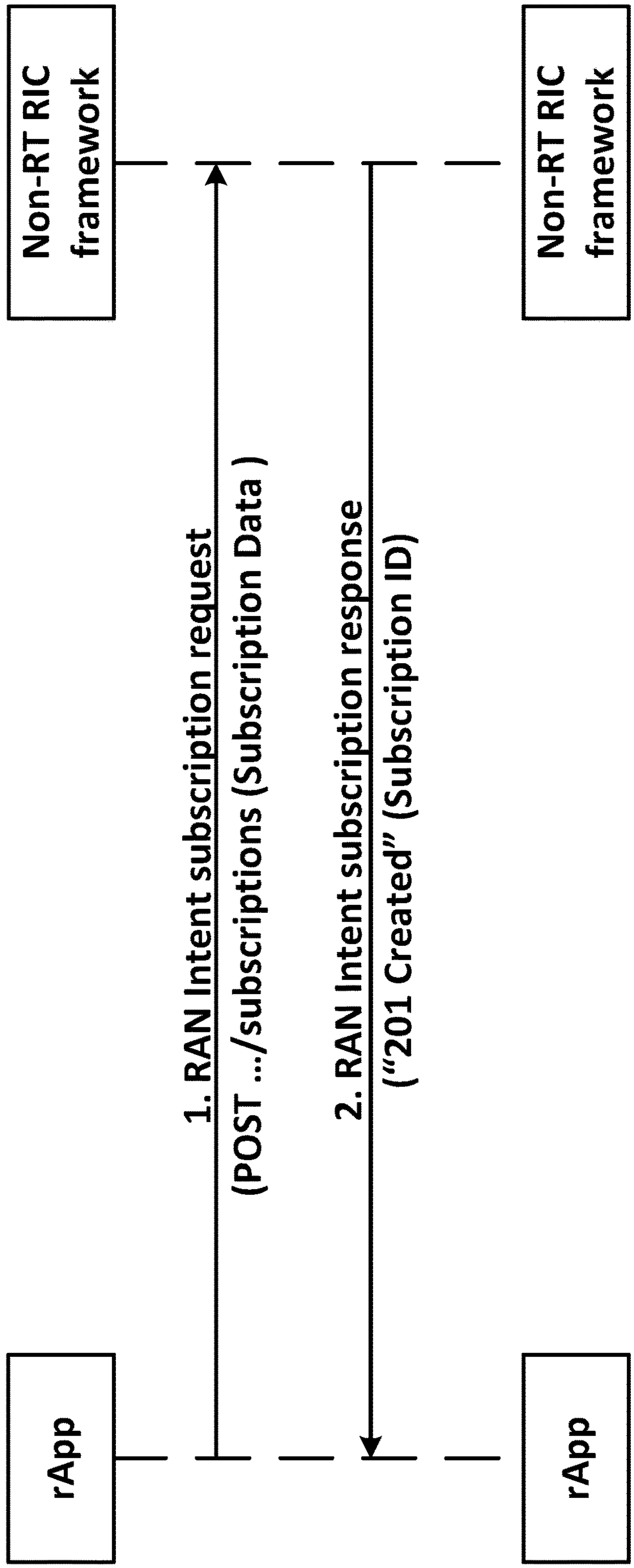

rApp
Non-RT RIC framework
1. RAN Intent subscription request
(POST .../subscriptions (Subscription Data )
2. RAN Intent subscription response
("201 Created" (Subscription ID)
rApp
Non-RT RIC framework

Figure 4

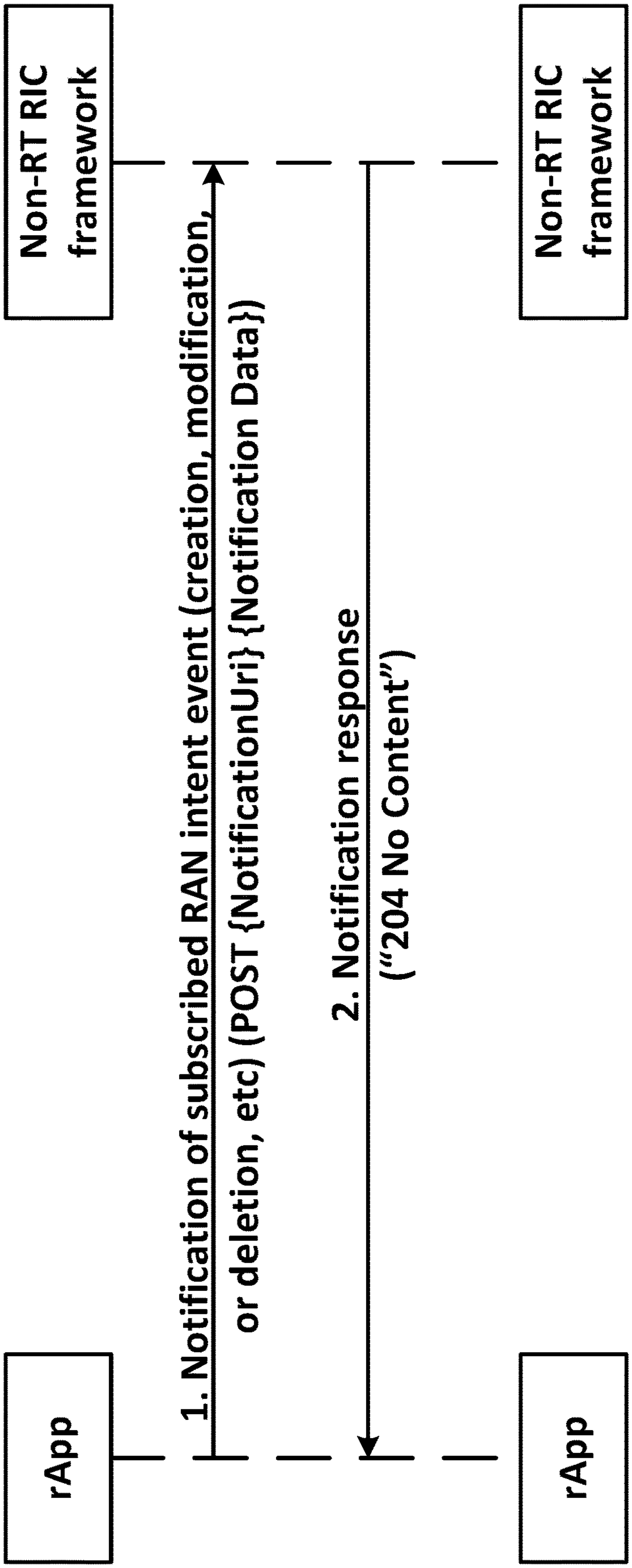

rApp
Non-RT RIC framework
1. Notification of subscribed RAN intent event (creation, modification, or deletion, etc) (POST {NotificationUri} {Notification Data})
2. Notification response ("204 No Content")
rApp
Non-RT RIC framework

Figure 5

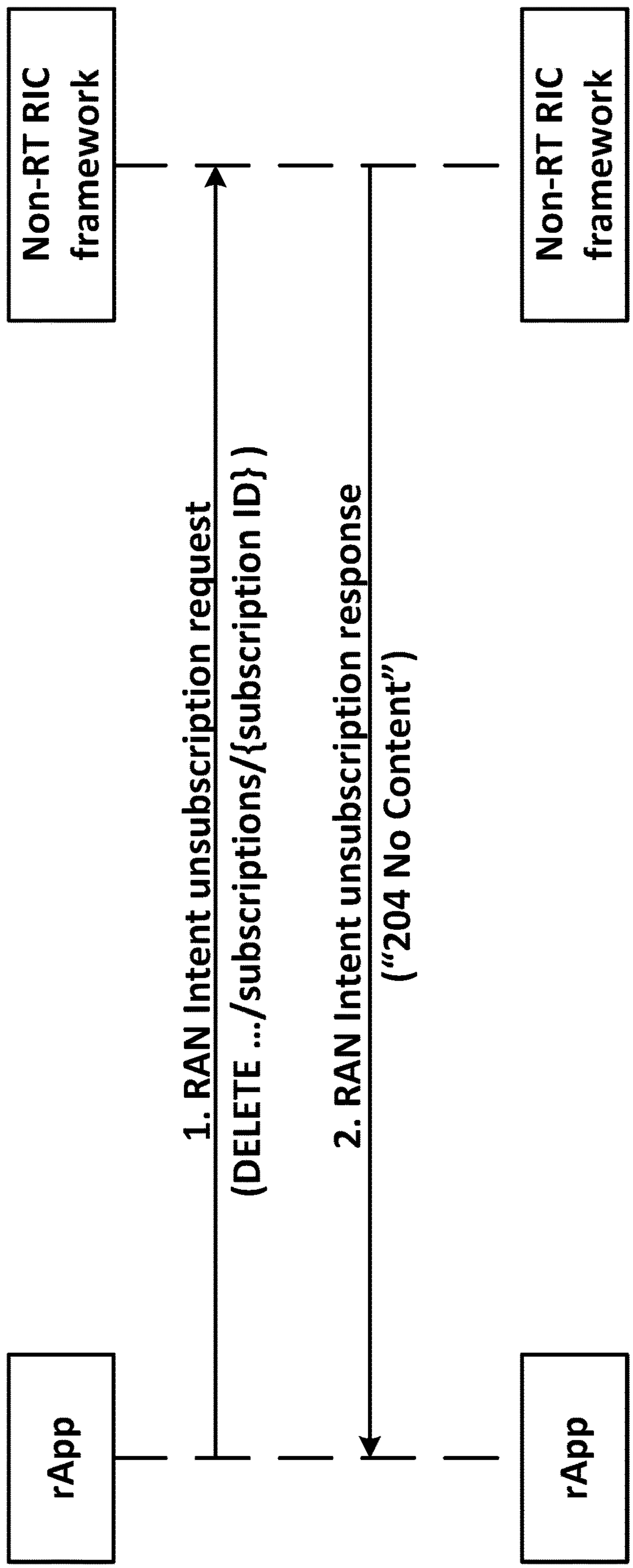

rApp
Non-RT RIC framework
1. RAN Intent unsubscription request
(DELETE …/subscriptions/{subscription ID} )
2. RAN Intent unsubscription response
("204 No Content")
rApp
Non-RT RIC framework

Figure 6

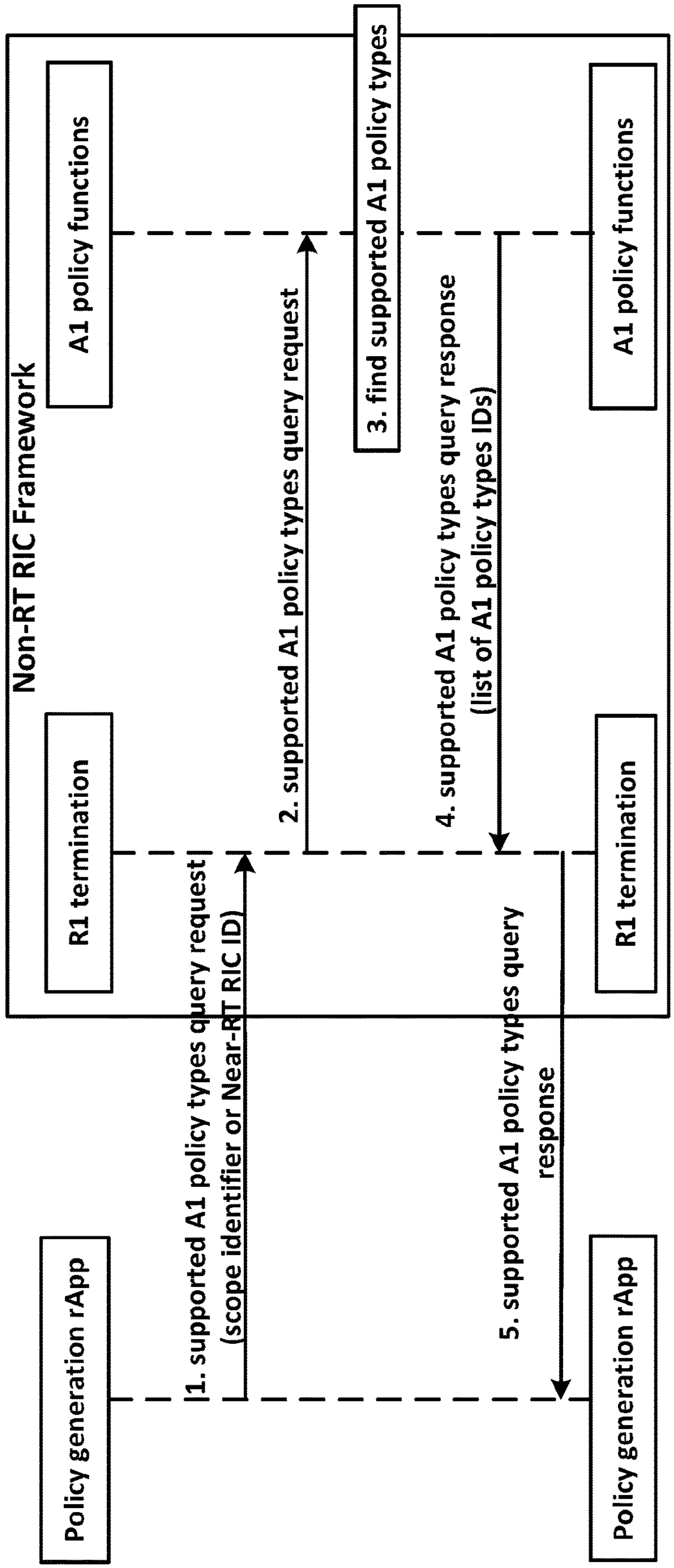

Non-RT RIC Framework
Policy generation rApp
R1 termination
A1 policy functions
1. supported A1 policy types query request
(scope identifier or Near-RT RIC ID)
2. supported A1 policy types query request
3. find supported A1 policy types
4. supported A1 policy types query response
(list of A1 policy types IDs)
5. supported A1 policy types query
response
Policy generation rApp
R1 termination
A1 policy functions

Figure 7

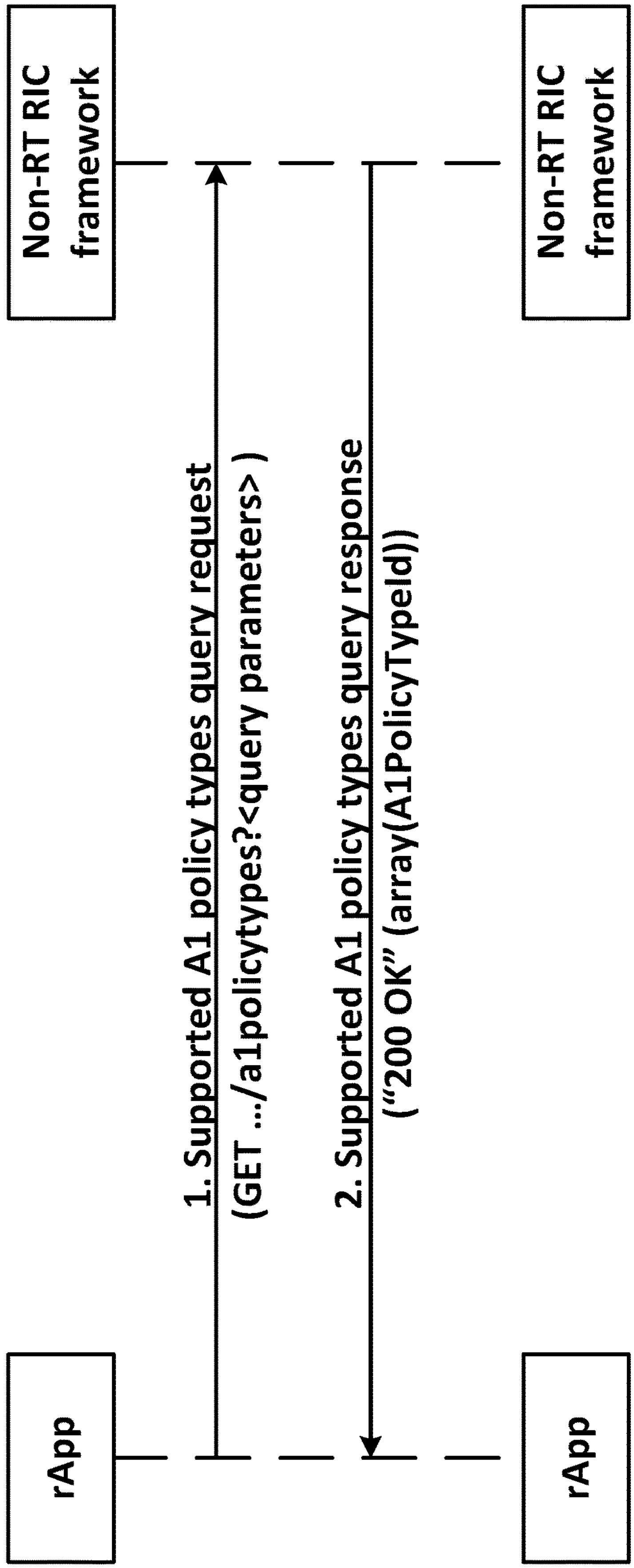

rApp
Non-RT RIC framework
1. Supported A1 policy types query request
(GET …/a1policytypes?<query parameters> )
2. Supported A1 policy types query response
("200 OK" (array(A1PolicyTypeId))
rApp
Non-RT RIC framework

Figure 8

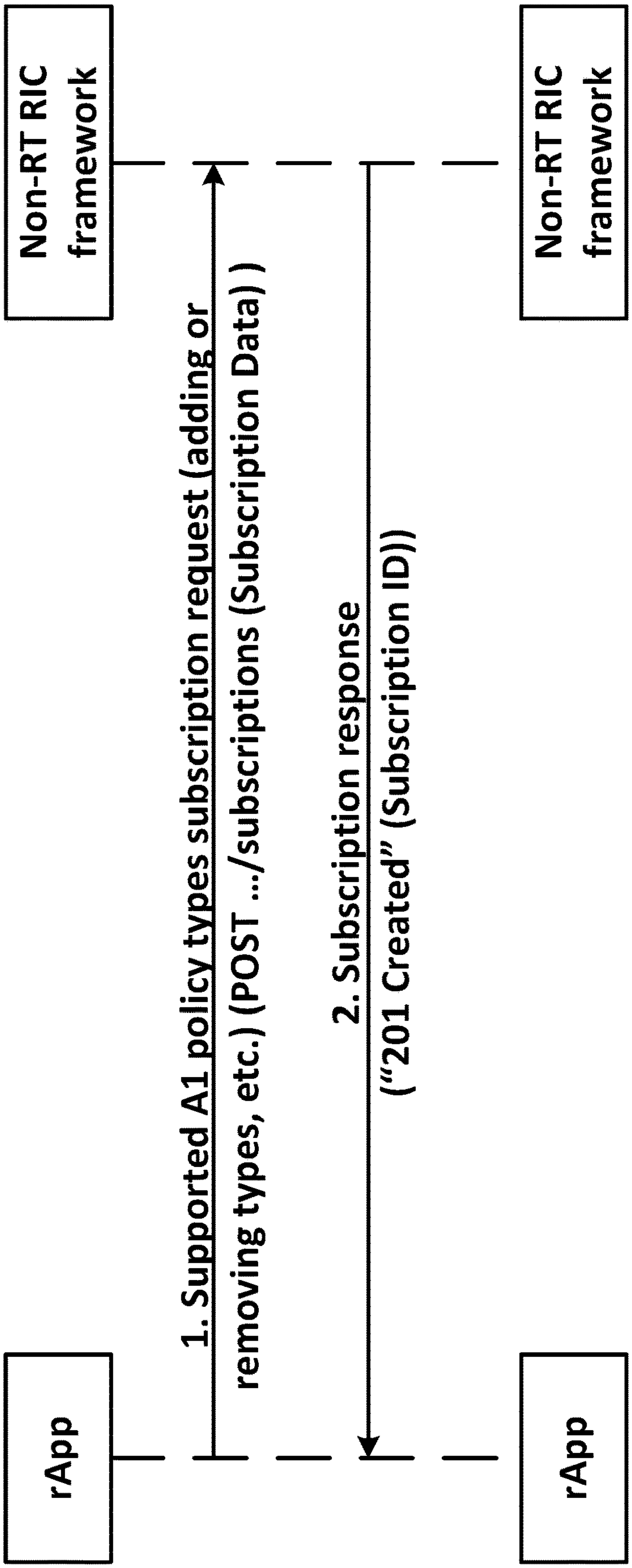

rApp
Non-RT RIC framework
1. Supported A1 policy types subscription request (adding or removing types, etc.) (POST .../subscriptions (Subscription Data) )
2. Subscription response ("201 Created" (Subscription ID))
rApp
Non-RT RIC framework

Figure 9

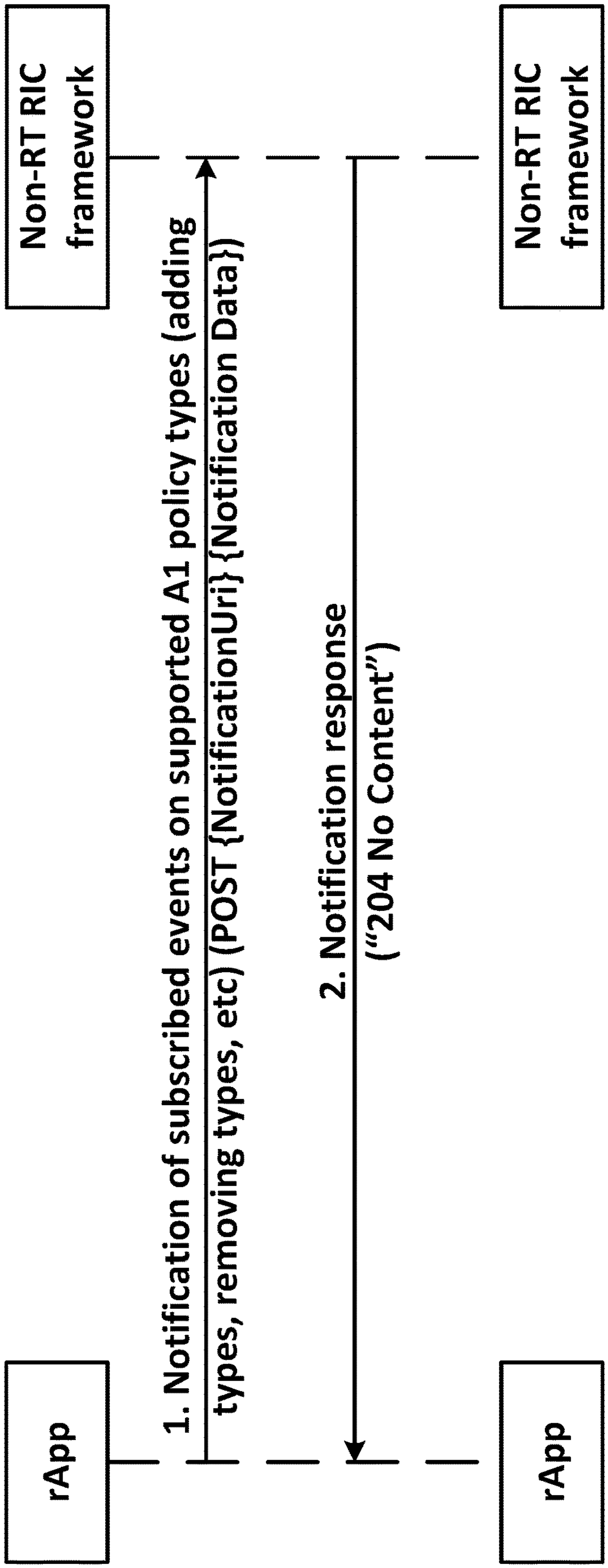

Non-RT RIC framework
rApp
1. Notification of subscribed events on supported A1 policy types (adding types, removing types, etc) (POST {NotificationUri} {Notification Data})
2. Notification response ("204 No Content")
Non-RT RIC framework
rApp

Figure 10


rApp
Non-RT RIC framework
1. Supported A1 policy types unsubscription request
(DELETE …/subscriptions/{subscriptionID} )
2. Unsubscription response
("204 No Content")
rApp
Non-RT RIC framework

Figure 11

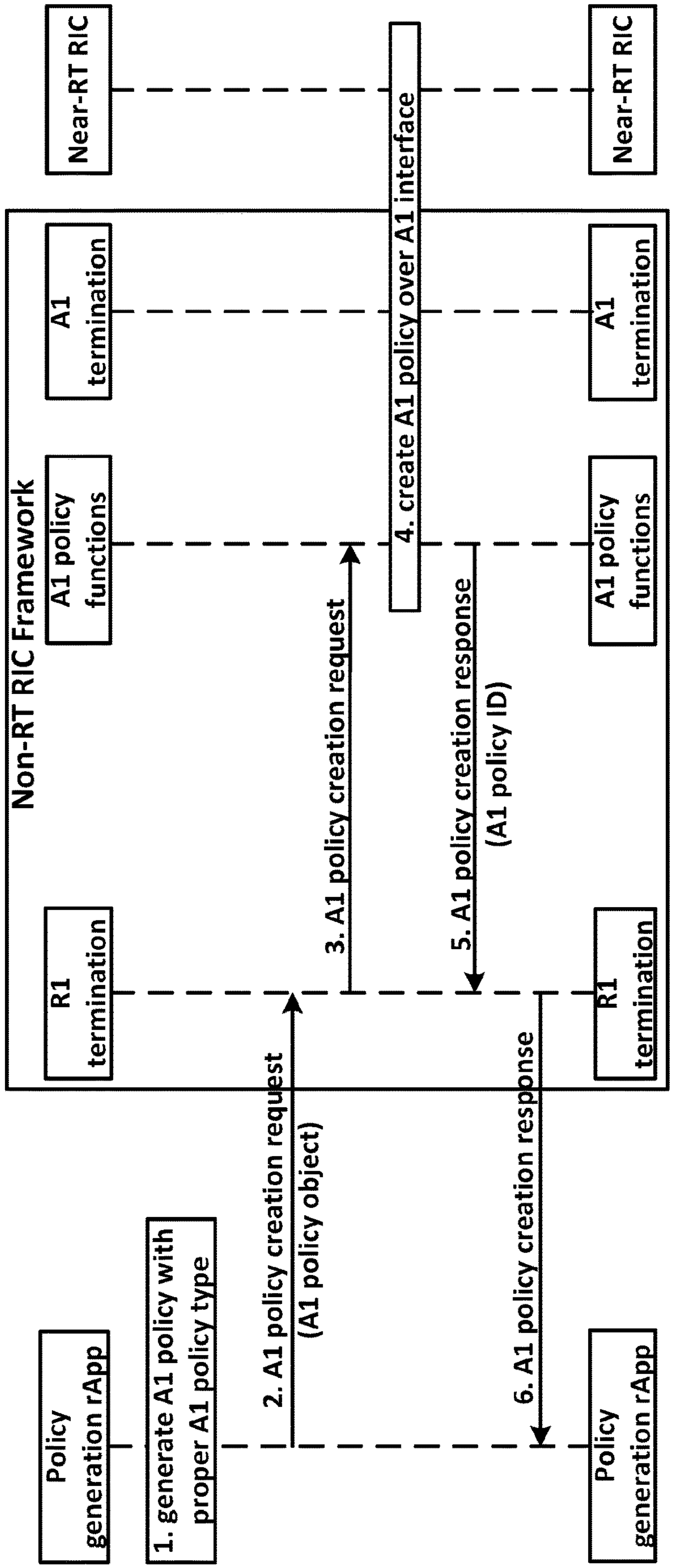

Policy generation rApp
1. generate A1 policy with proper A1 policy type
2. A1 policy creation request (A1 policy object)
6. A1 policy creation response
Policy generation rApp
Non-RT RIC Framework
R1 termination
3. A1 policy creation request
5. A1 policy creation response (A1 policy ID)
R1 termination
A1 policy functions
4. create A1 policy over A1 interface
A1 policy functions
A1 termination
A1 termination
Near-RT RIC
Near-RT RIC

Figure 12

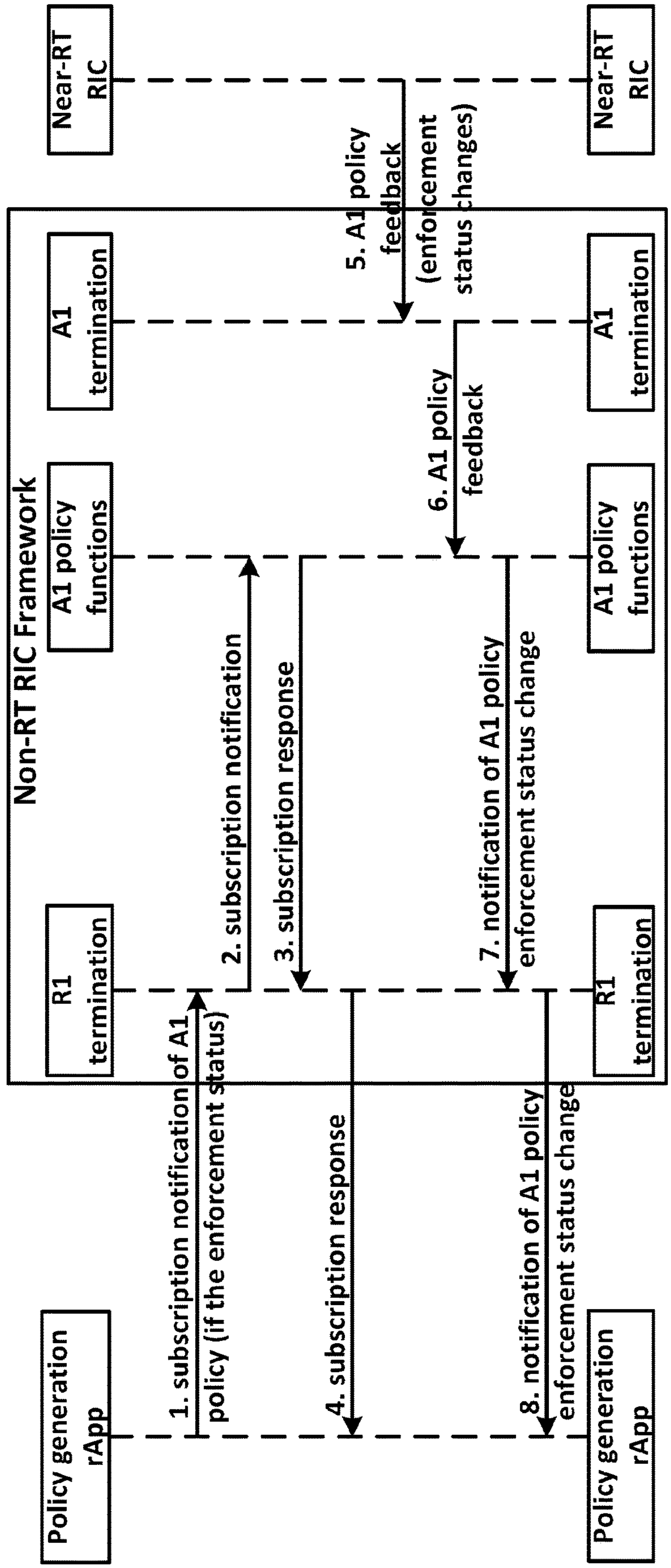

Policy generation rApp
Non-RT RIC Framework
R1 termination
A1 policy functions
A1 termination
Near-RT RIC
1. subscription notification of A1 policy (if the enforcement status)
2. subscription notification
3. subscription response
4. subscription response
5. A1 policy feedback (enforcement status changes)
6. A1 policy feedback
7. notification of A1 policy enforcement status change
8. notification of A1 policy enforcement status change
Policy generation rApp
R1 termination
A1 policy functions
A1 termination
Near-RT RIC

Figure 13

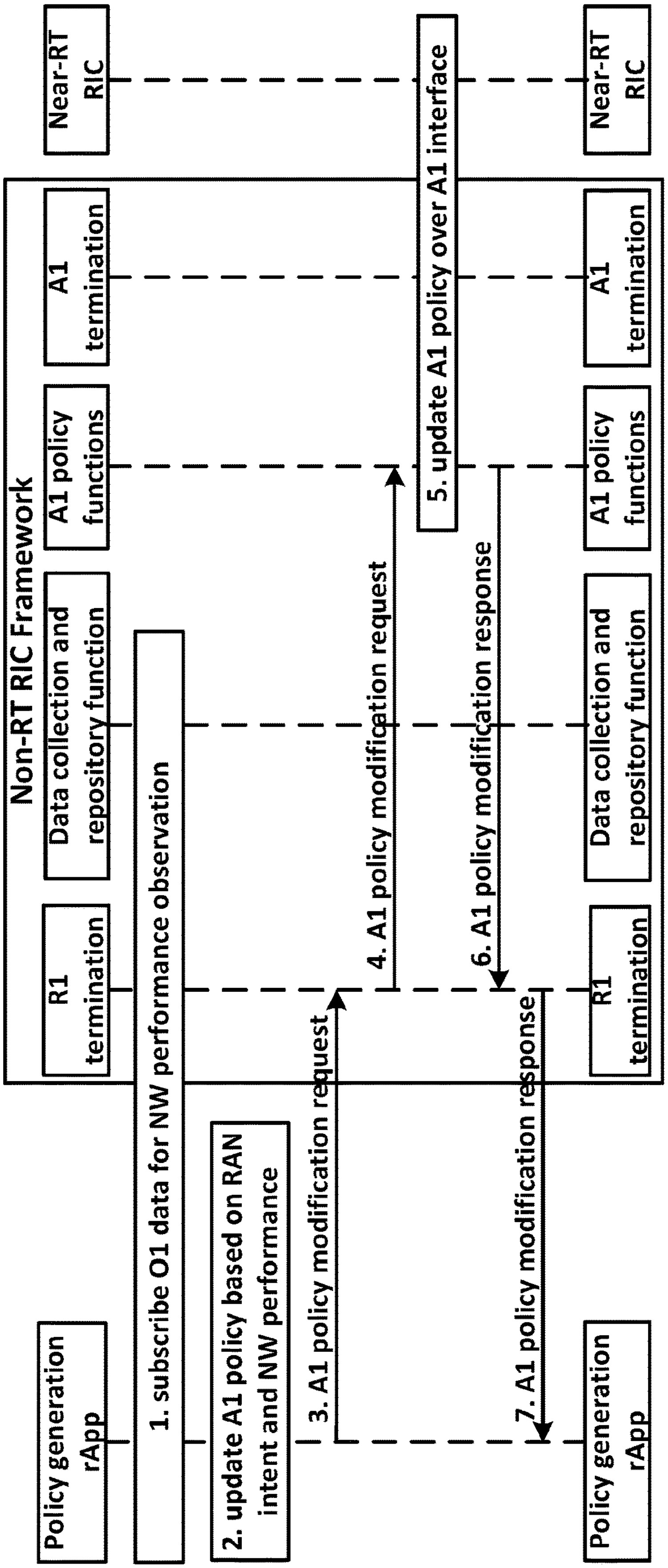

Policy generation rApp
Non-RT RIC Framework
R1 termination
Data collection and repository function
A1 policy functions
A1 termination
Near-RT RIC
1. subscribe O1 data for NW performance observation
2. update A1 policy based on RAN intent and NW performance
3. A1 policy modification request
4. A1 policy modification request
5. update A1 policy over A1 interface
6. A1 policy modification response
7. A1 policy modification response
Policy generation rApp
R1 termination
Data collection and repository function
A1 policy functions
A1 termination
Near-RT RIC

Figure 14

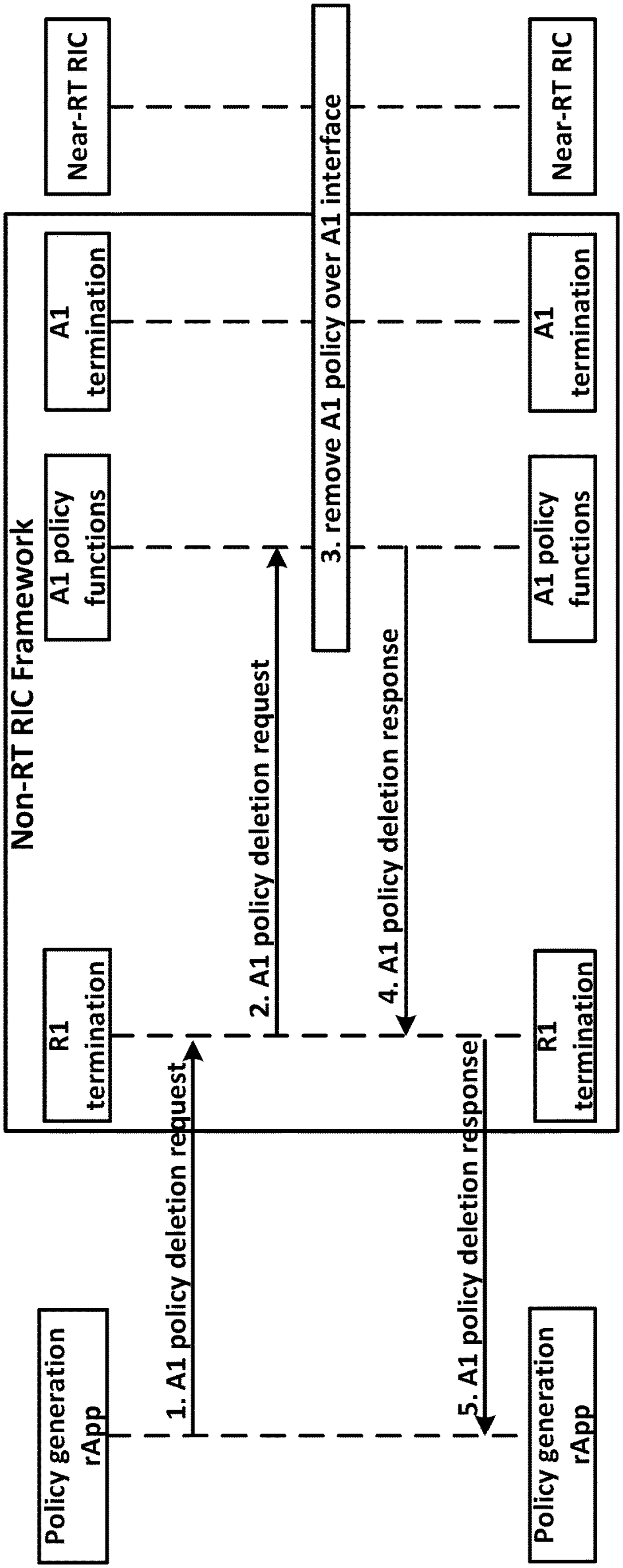

Non-RT RIC Framework
Policy generation rApp
R1 termination
A1 policy functions
A1 termination
Near-RT RIC
1. A1 policy deletion request
2. A1 policy deletion request
3. remove A1 policy over A1 interface
4. A1 policy deletion response
5. A1 policy deletion response
Policy generation rApp
R1 termination
A1 policy functions
A1 termination
Near-RT RIC

Figure 15

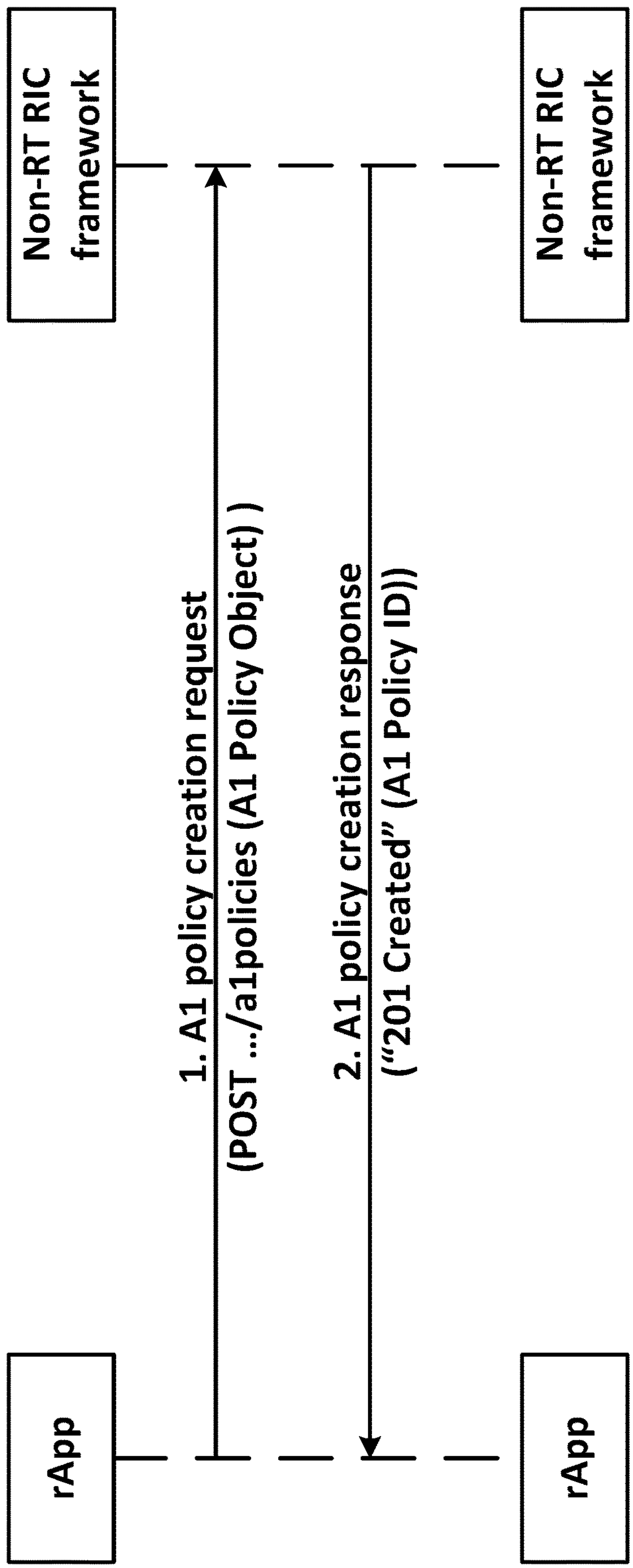

Non-RT RIC framework
rApp
1. A1 policy creation request
(POST .../a1policies (A1 Policy Object) )
2. A1 policy creation response
("201 Created" (A1 Policy ID))
Non-RT RIC framework
rApp

Figure 16

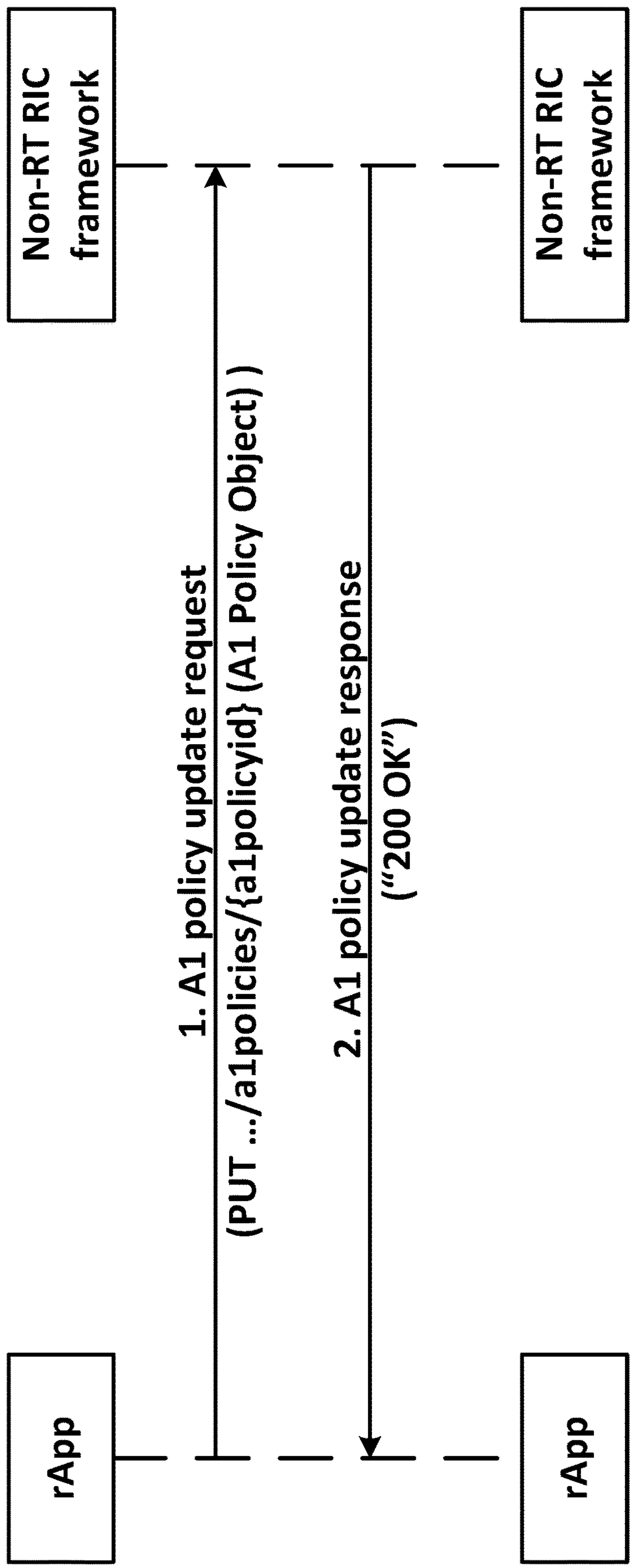

Non-RT RIC framework
rApp
1. A1 policy update request
(PUT …/a1policies/{a1policyid} (A1 Policy Object) )
2. A1 policy update response
("200 OK")
Non-RT RIC framework
rApp

Figure 17

Non-RT RIC framework
Non-RT RIC framework
rApp
rApp
1. A1 policy deletion request
(DELETE .../a1policies/{a1policyid} )
2. A1 policy deletion response
("204 No Content")


Figure 18

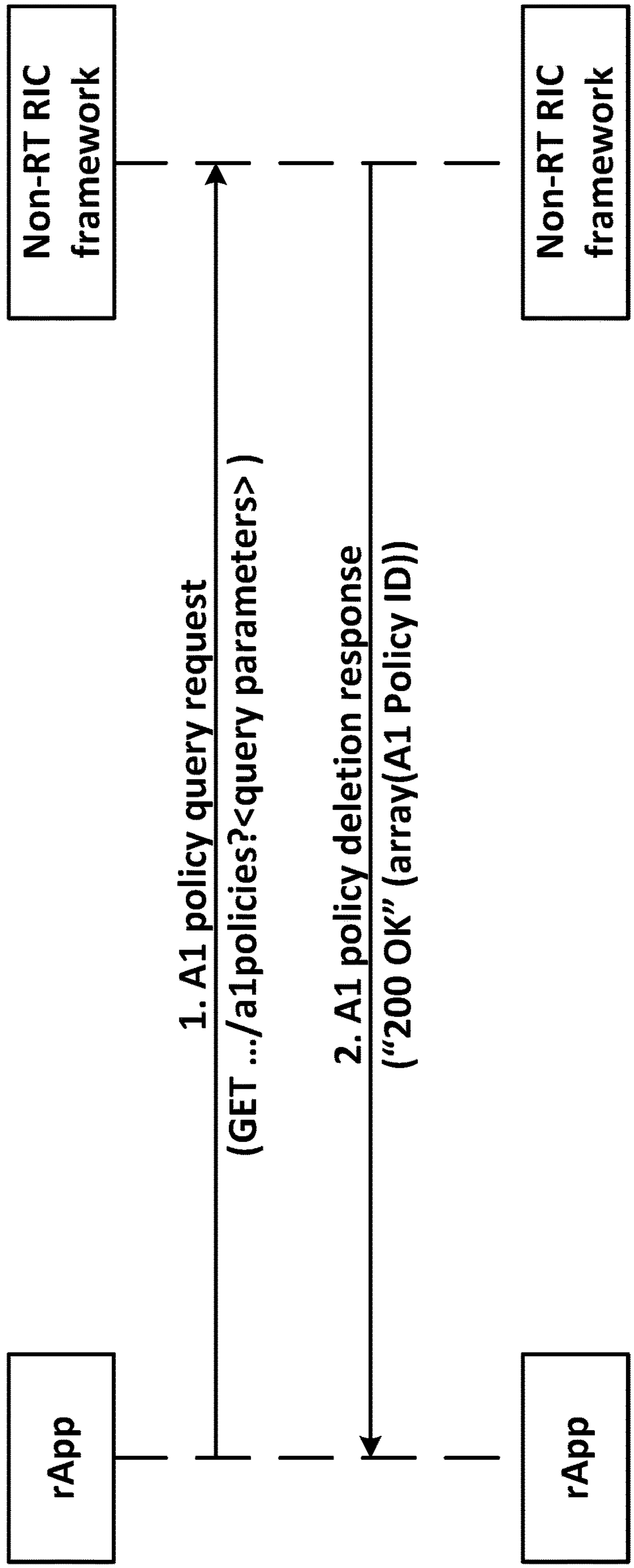

Non-RT RIC framework
Non-RT RIC framework
1. A1 policy query request
(GET .../a1policies?<query parameters> )
2. A1 policy deletion response
("200 OK" (array(A1 Policy ID))
rApp
rApp

Figure 19

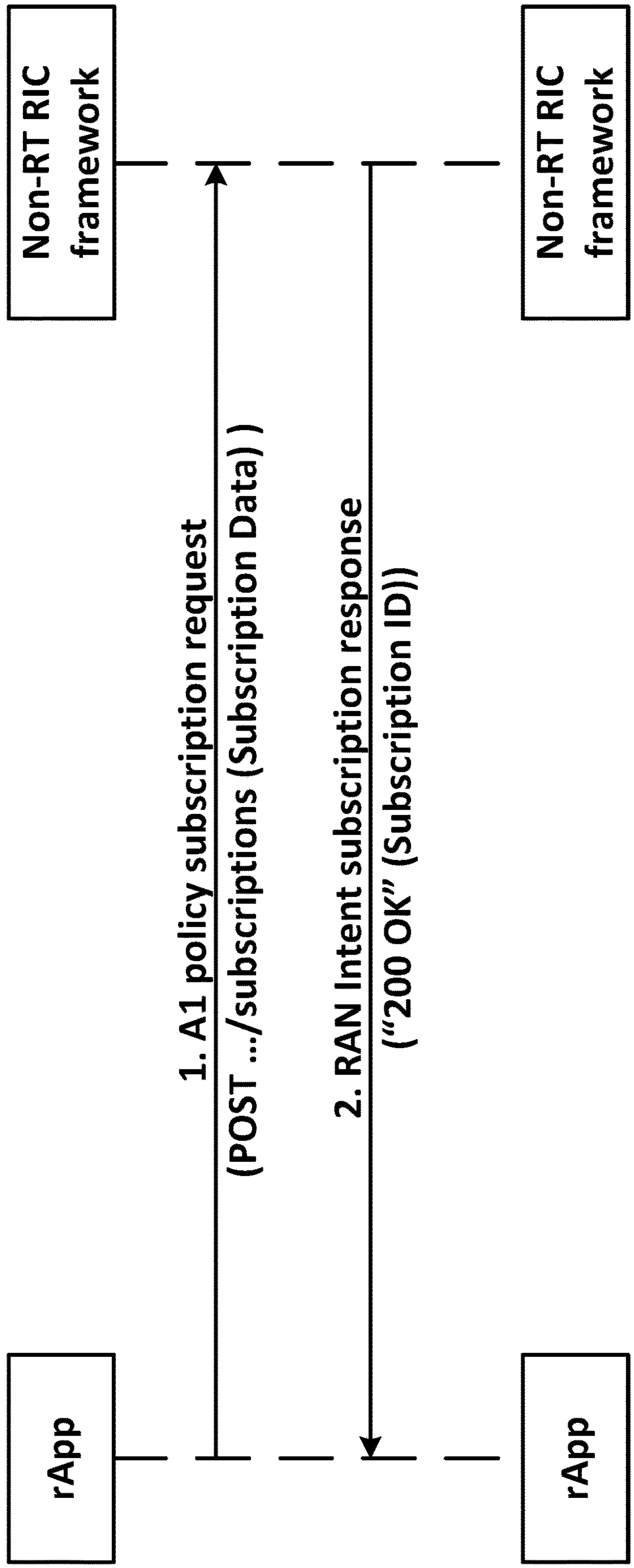

rApp
Non-RT RIC framework
1. A1 policy subscription request
(POST .../subscriptions (Subscription Data) )
2. RAN Intent subscription response
("200 OK" (Subscription ID))
rApp
Non-RT RIC framework

Figure 20

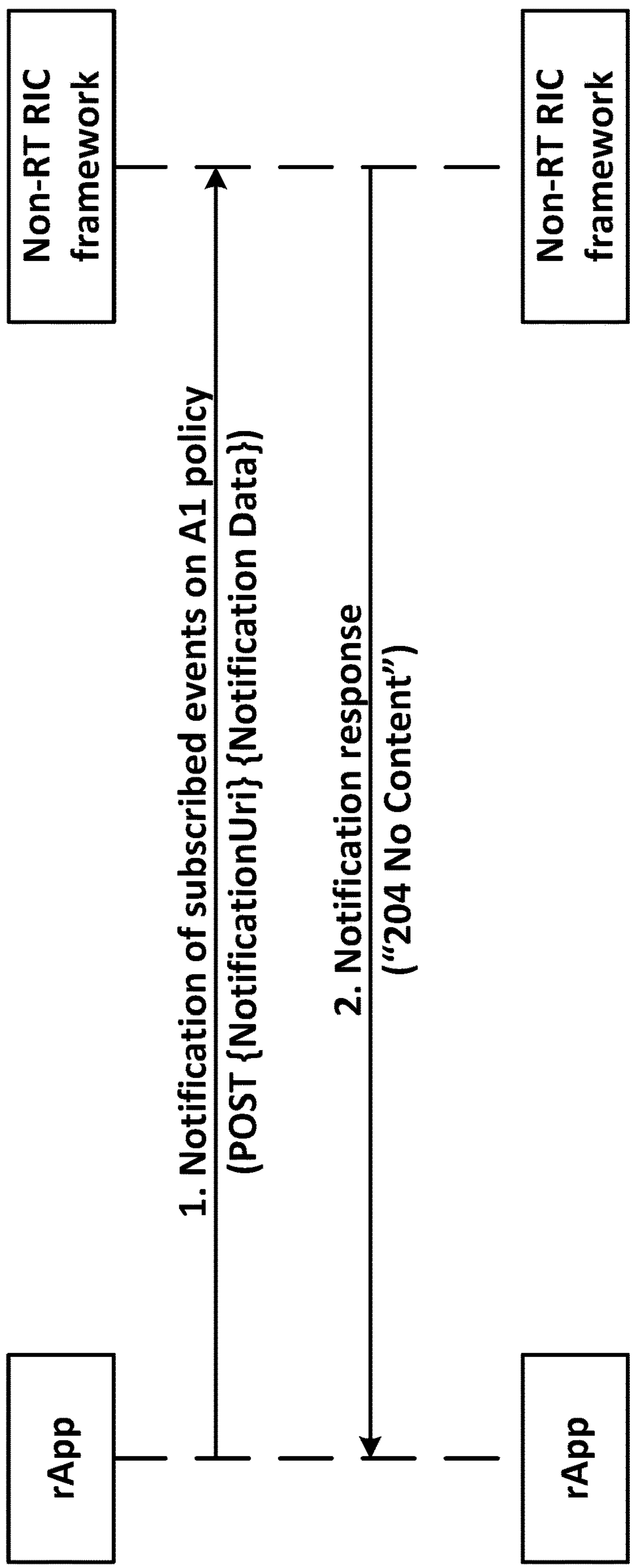

Non-RT RIC framework
Non-RT RIC framework
rApp
rApp
1. Notification of subscribed events on A1 policy
(POST {NotificationUri} {Notification Data})
2. Notification response
("204 No Content")

Figure 21

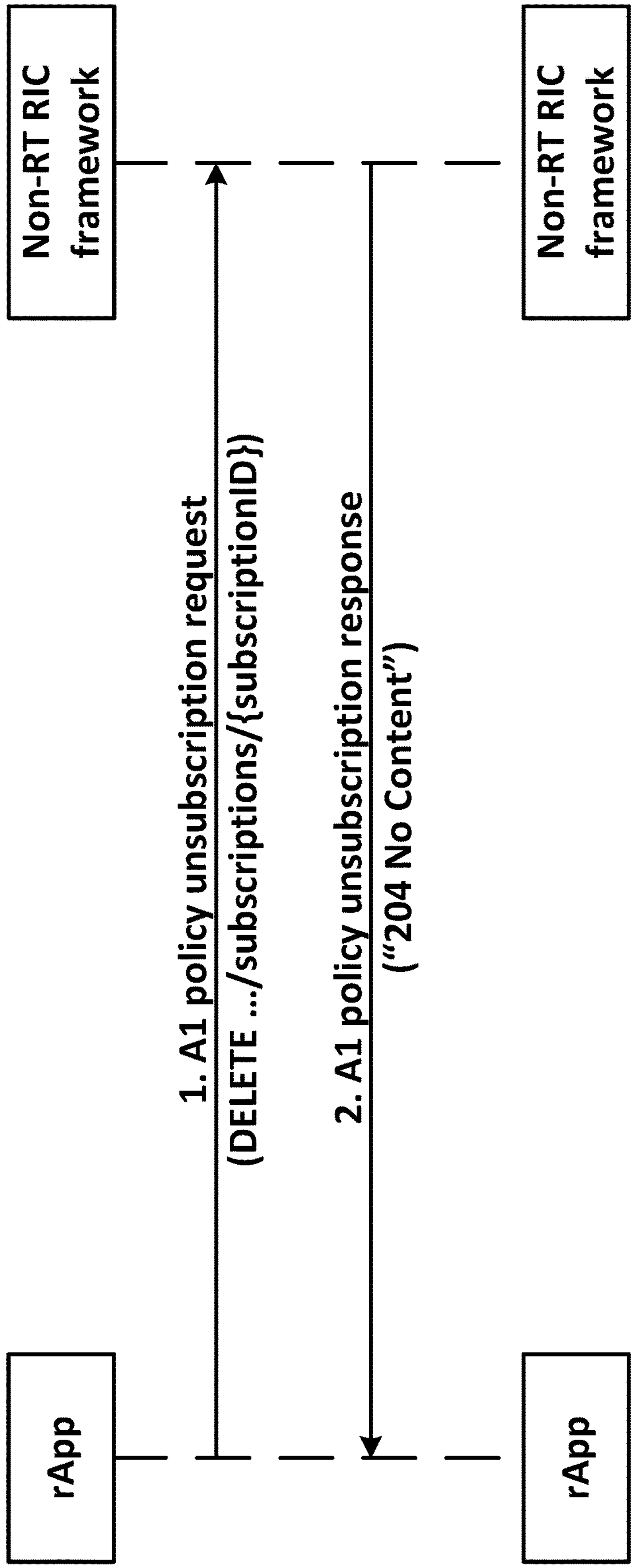
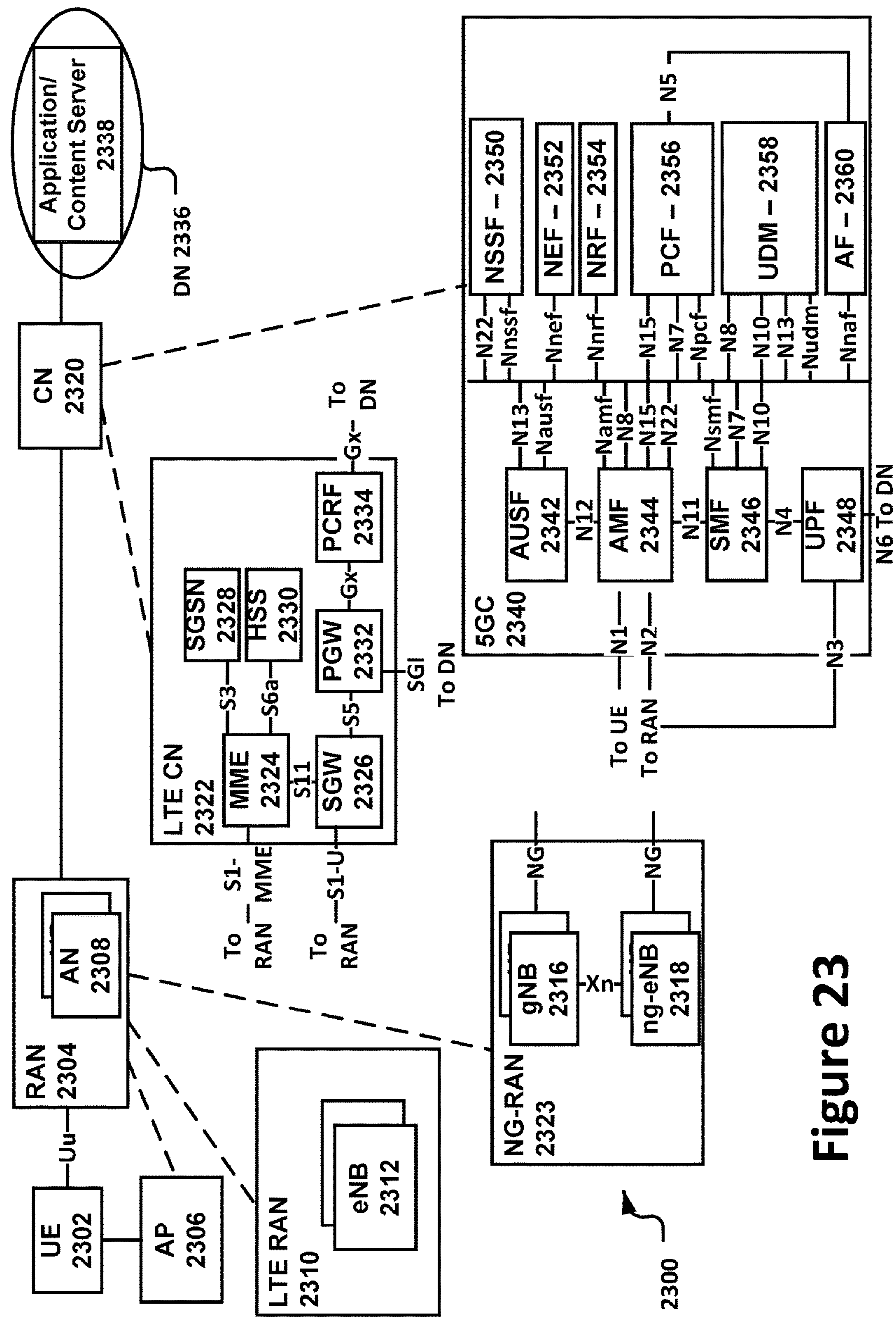

Non-RT RIC framework
Non-RT RIC framework
rApp
rApp
1. A1 policy unsubscription request
(DELETE .../subscriptions/{subscriptionID})
2. A1 policy unsubscription response
("204 No Content")

Figure 22



Application/ Content Server 2338
DN 2336
CN 2320
RAN 2304
AN 2308
UE 2302
AP 2306
Uu
LTE RAN 2310
eNB 2312
LTE CN 2322
MME 2324
SGW 2326
SGSN 2328
HSS 2330
PGW 2332
PCRF 2334
S3
S6a
S11
S5
Gx
SGi To DN
To DN
To RAN S1-MME
To RAN S1-U
NG-RAN 2323
gNB 2316
ng-eNB 2318
Xn
NG
5GC 2340
AUSF 2342
AMF 2344
SMF 2346
UPF 2348
N12
N11
N4
N6 To DN
N1
N2
N3
To UE
To RAN
NSSF – 2350
NEF – 2352
NRF – 2354
PCF – 2356
UDM – 2358
AF – 2360
N5
N22
Nnssf
Nnef
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N15
N7
Npcf
N8
N10
N13
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N8
N15
N22
Nsmf
N7
N10
N13
2300

Figure 23

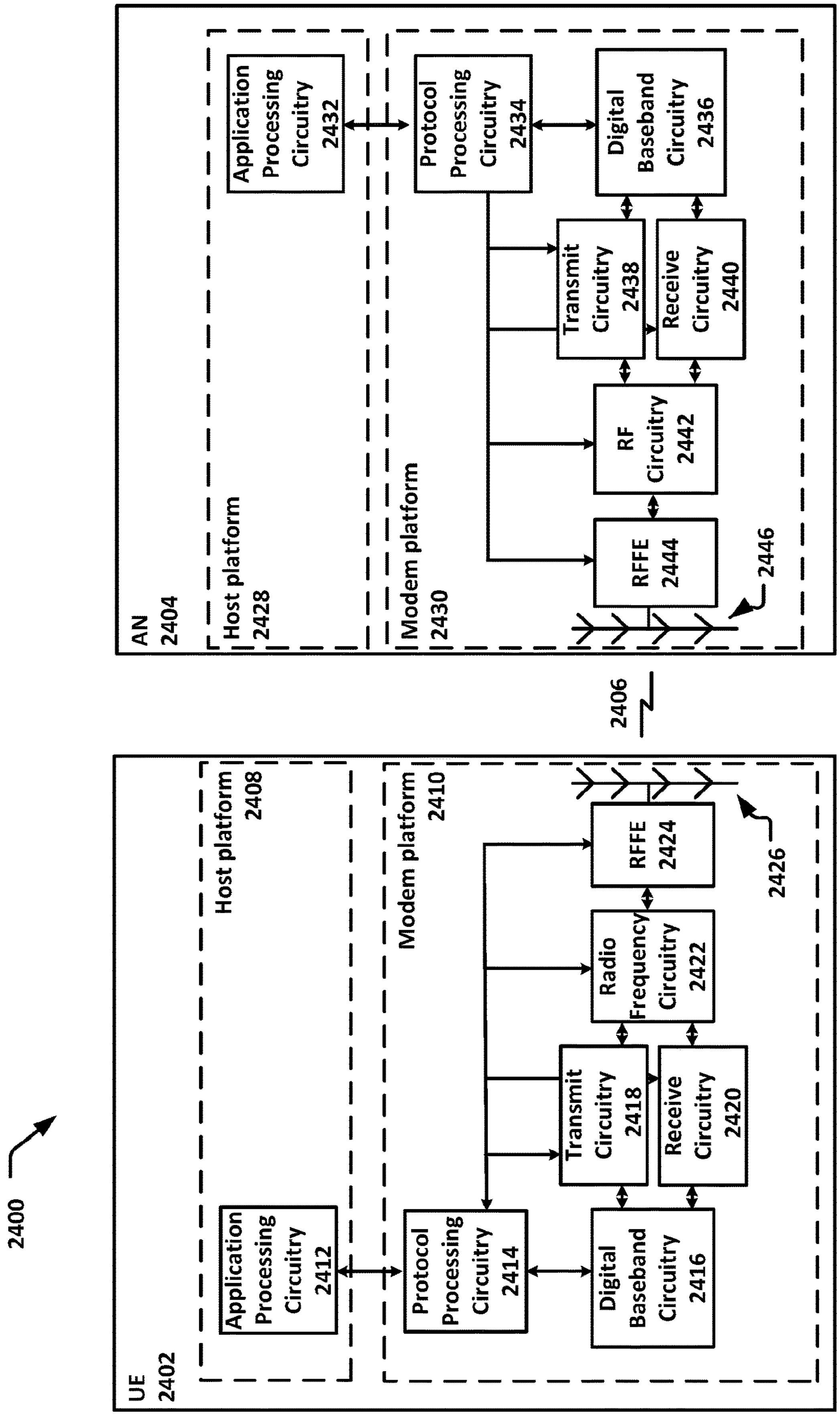
2400
UE
2402
Host platform
2408
Application Processing Circuitry 2412
Modem platform
2410
Protocol Processing Circuitry 2414
Digital Baseband Circuitry 2416
Transmit Circuitry 2418
Receive Circuitry 2420
Radio Frequency Circuitry 2422
RFFE 2424
2426
2406
AN
2404
Host platform
2428
Application Processing Circuitry 2432
Modem platform
2430
Protocol Processing Circuitry 2434
Digital Baseband Circuitry 2436
Transmit Circuitry 2438
Receive Circuitry 2440
RF Circuitry 2442
RFFE 2444
2446

Figure 24

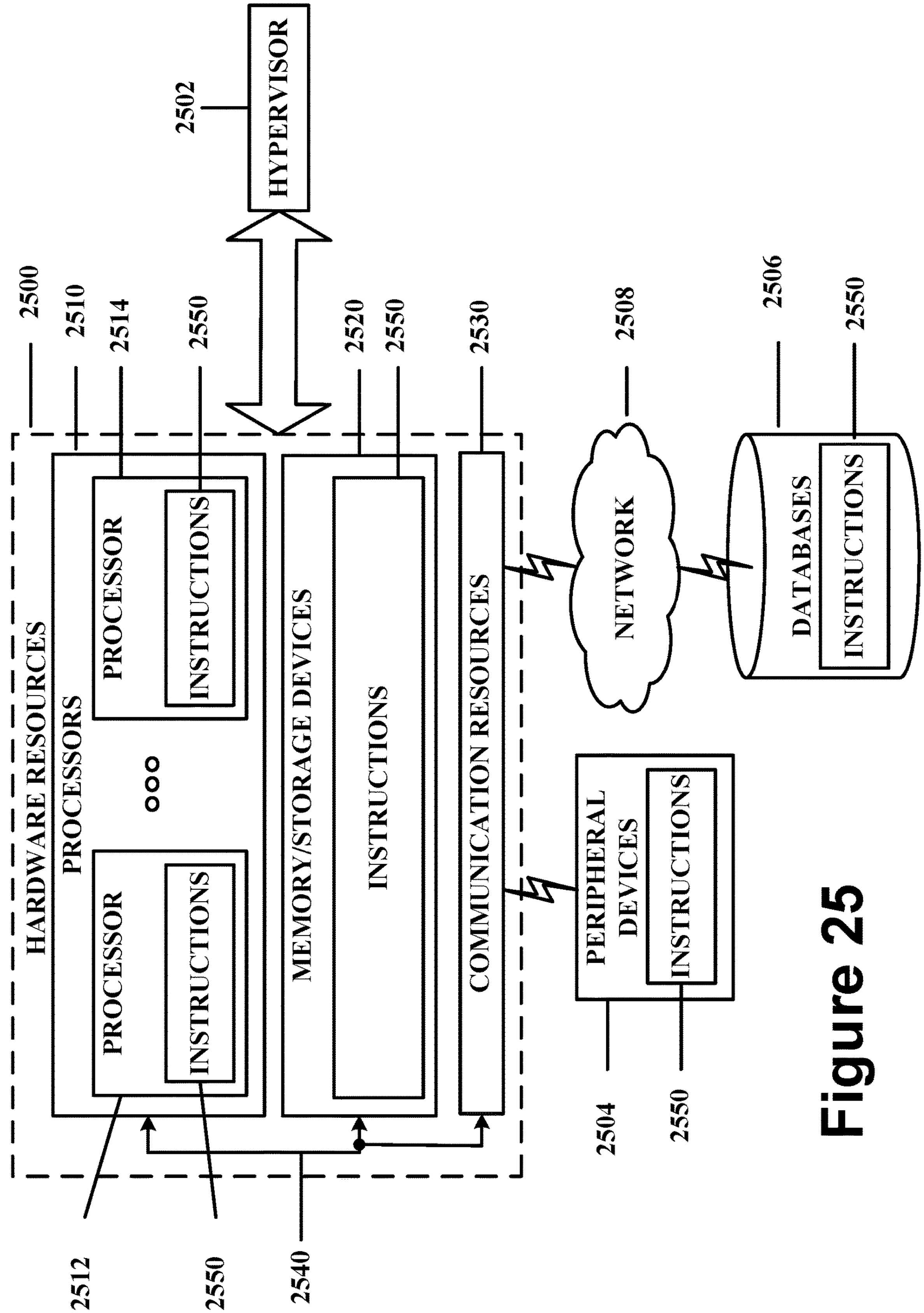
2500
HARDWARE RESOURCES
2510
PROCESSORS
2514
PROCESSOR
2550
INSTRUCTIONS
PROCESSOR
INSTRUCTIONS
2512
2550
2540
2520
2550
MEMORY/STORAGE DEVICES
INSTRUCTIONS
2530
COMMUNICATION RESOURCES
2502
HYPERVISOR
2508
NETWORK
2506
2550
DATABASES
INSTRUCTIONS
PERIPHERAL DEVICES
INSTRUCTIONS
2504
2550

Figure 25

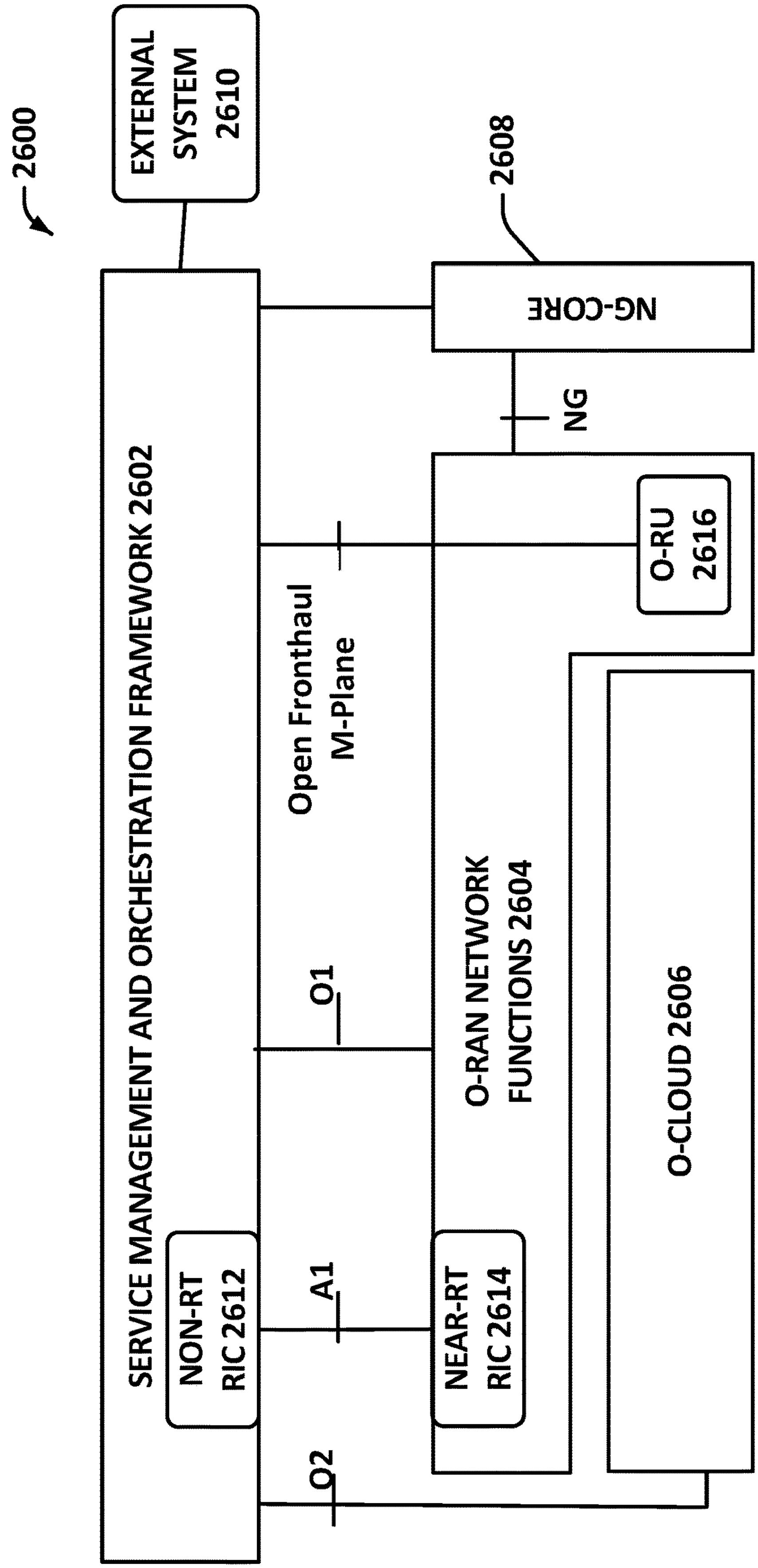
2600
EXTERNAL SYSTEM 2610
2608
NG-CORE
NG
SERVICE MANAGEMENT AND ORCHESTRATION FRAMEWORK 2602
NON-RT RIC 2612
Open Fronthaul M-Plane
O-RU 2616
O1
A1
O2
O-RAN NETWORK FUNCTIONS 2604
NEAR-RT RIC 2614
O-CLOUD 2606

Figure 26

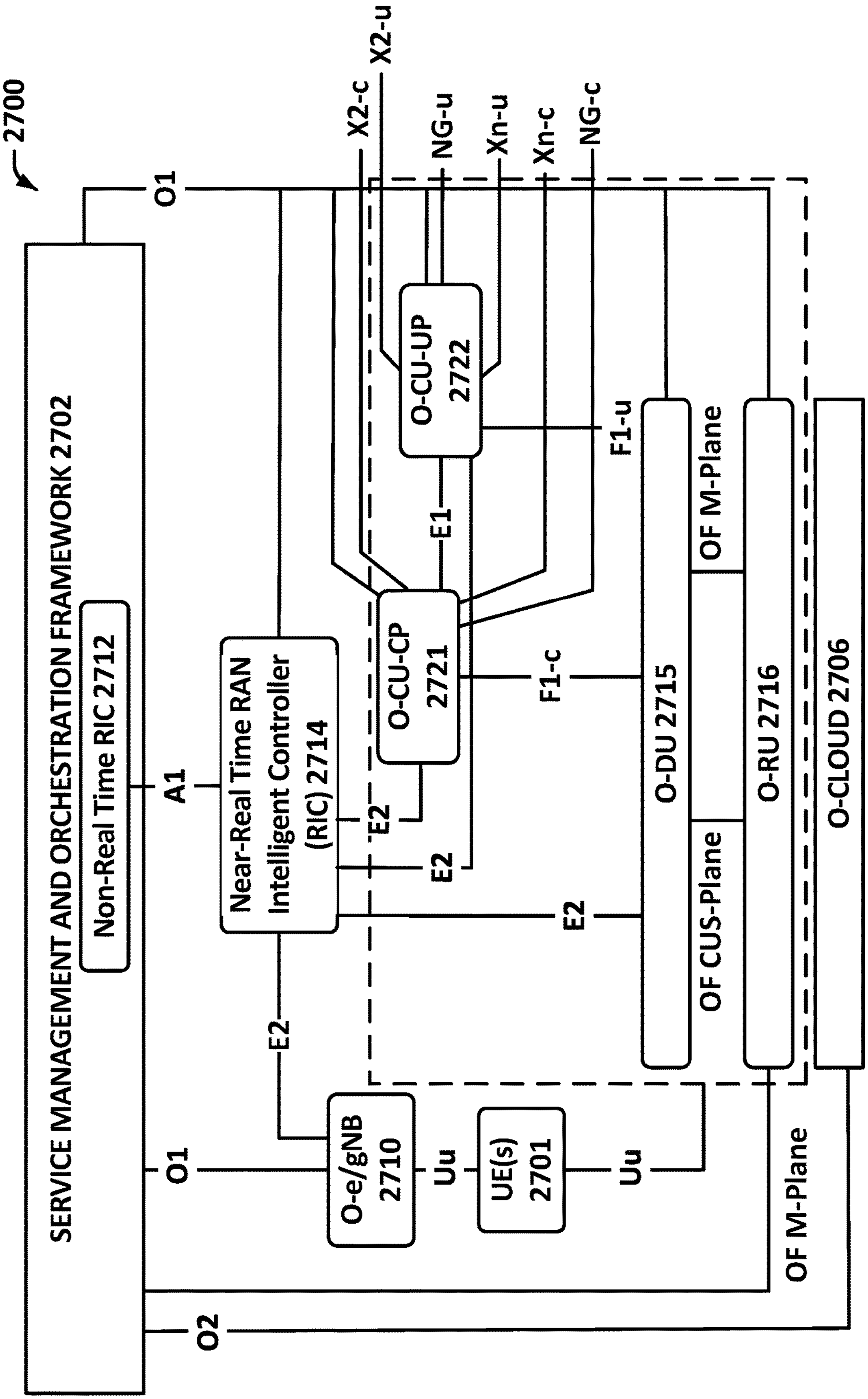
2700
SERVICE MANAGEMENT AND ORCHESTRATION FRAMEWORK 2702
Non-Real Time RIC 2712
A1
Near-Real Time RAN Intelligent Controller (RIC) 2714
O1
O1
O2
E2
E2
E2
E2
O-CU-CP 2721
O-CU-UP 2722
E1
F1-c
F1-u
X2-c
X2-u
NG-u
Xn-u
Xn-c
NG-c
O-DU 2715
O-RU 2716
O-CLOUD 2706
OF M-Plane
OF CUS-Plane
OF M-Plane
O-e/gNB 2710
UE(s) 2701
Uu
Uu

Figure 27

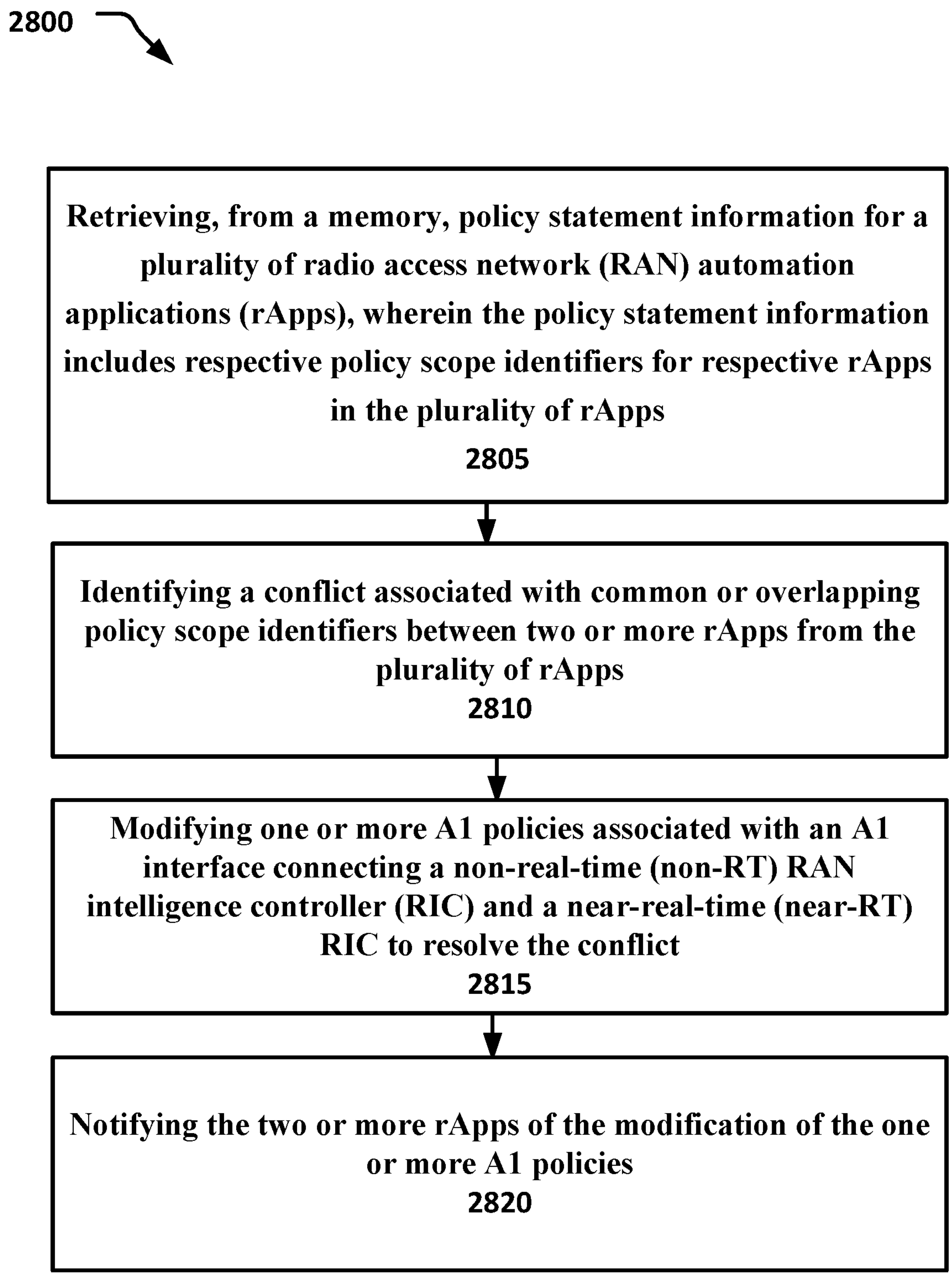
2800
Retrieving, from a memory, policy statement information for a plurality of radio access network (RAN) automation applications (rApps), wherein the policy statement information includes respective policy scope identifiers for respective rApps in the plurality of rApps
2805
Identifying a conflict associated with common or overlapping policy scope identifiers between two or more rApps from the plurality of rApps
2810
Modifying one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict
2815
Notifying the two or more rApps of the modification of the one or more A1 policies
2820

Figure 28

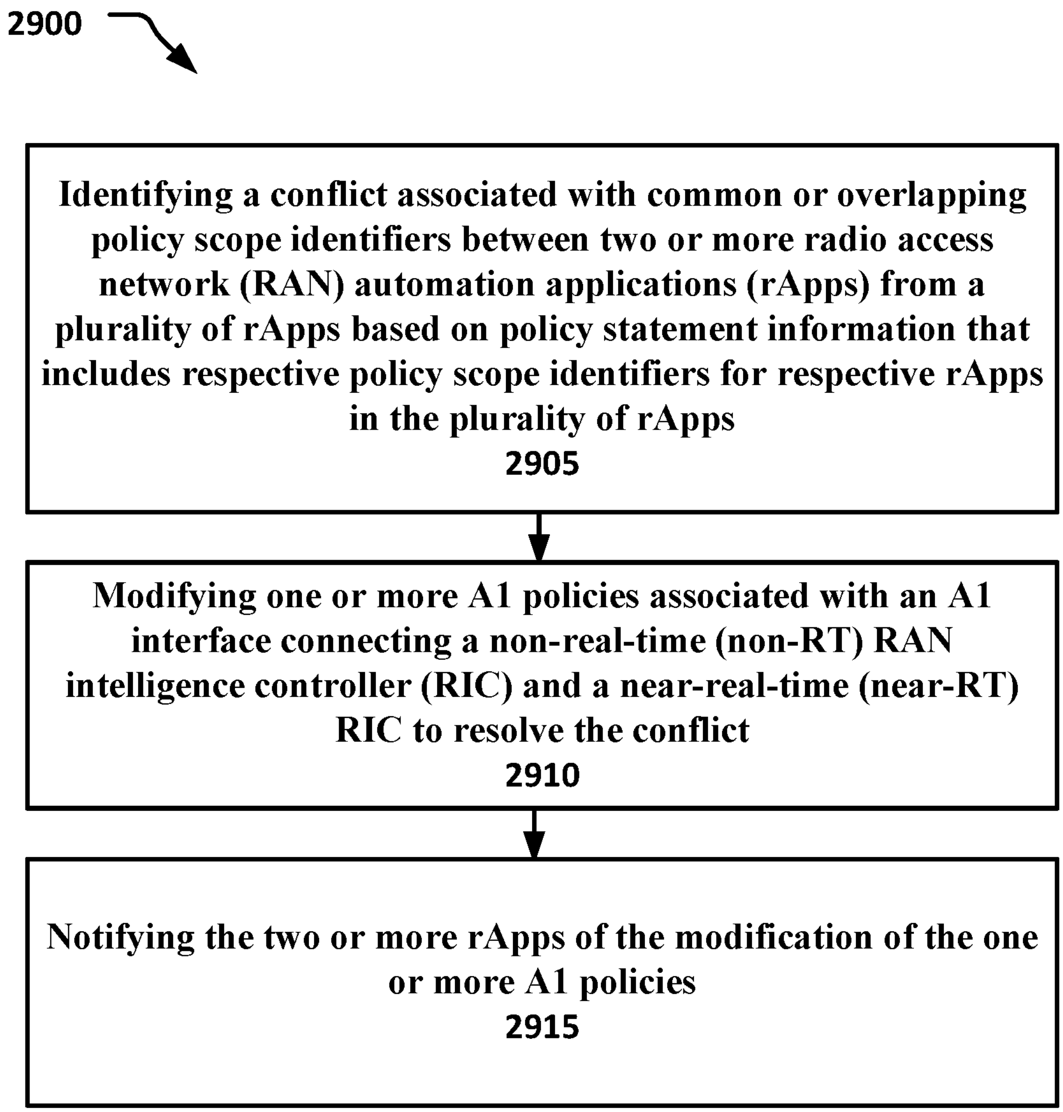
2900
Identifying a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps
2905
Modifying one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict
2910
Notifying the two or more rApps of the modification of the one or more A1 policies
2915

Figure 29

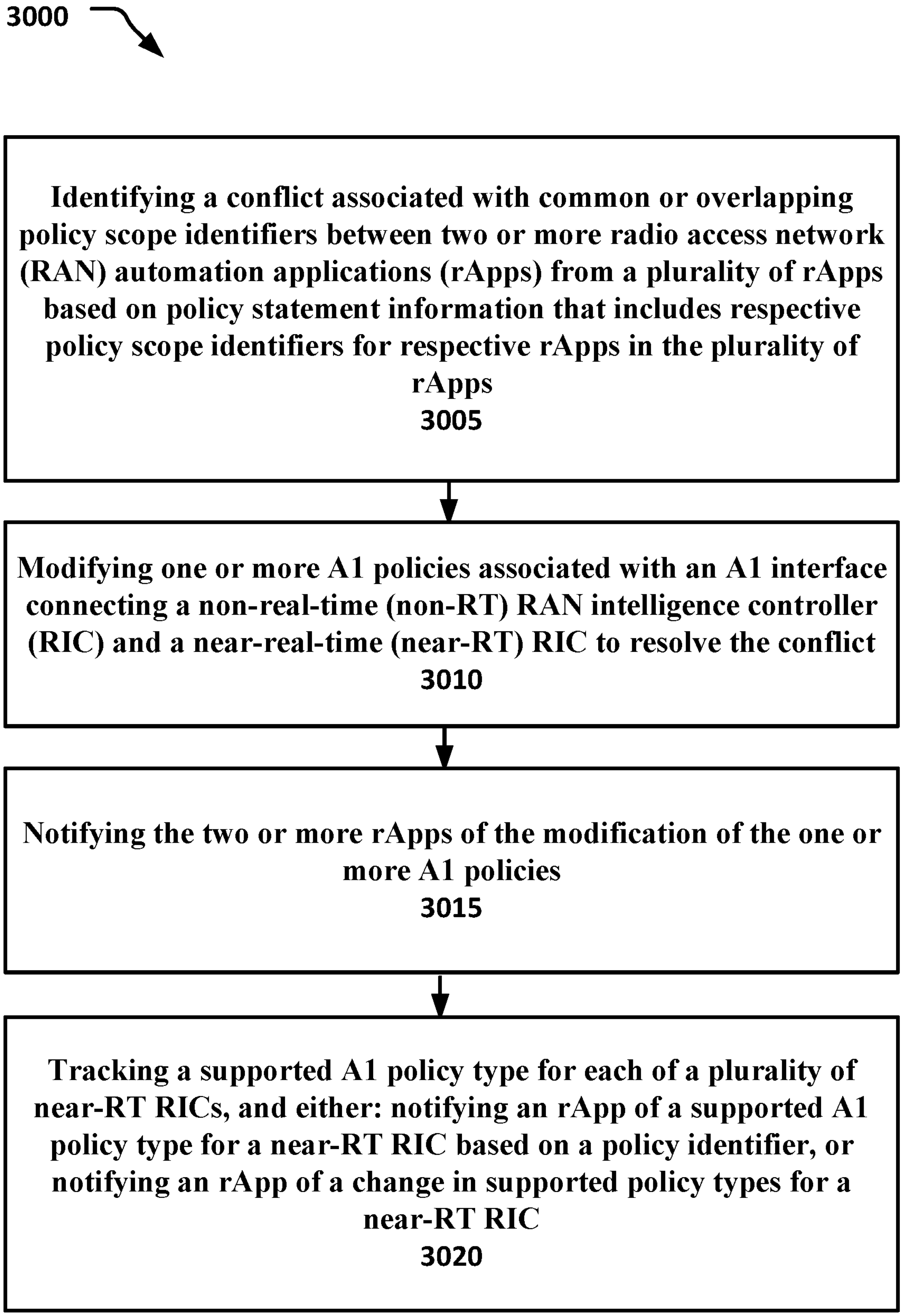
3000
Identifying a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps
3005
Modifying one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict
3010
Notifying the two or more rApps of the modification of the one or more A1 policies
3015
Tracking a supported A1 policy type for each of a plurality of near-RT RICs, and either: notifying an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier, or notifying an rApp of a change in supported policy types for a near-RT RIC
3020

Figure 30

# A1 POLICY FUNCTIONS FOR OPEN RADIO ACCESS NETWORK (O-RAN) SYSTEMS

## CROSS REFERENCE TO RELATED APPLICATION

The present application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/US2022/036130, filed Jul. 5, 2022, entitled "A1 POLICY FUNCTIONS FOR OPEN RADIO ACCESS NETWORK (O-RAN) SYSTEMS," which claims priority to U.S. Provisional Patent Application No. 63/218,861, which was filed Jul. 6, 2021, the entire disclosures of which are hereby incorporated by reference.

## FIELD

Various embodiments generally may relate to the field of wireless communications. For example, some embodiments may relate to features associated with A1 policy functions for open radio access network (O-RAN) systems.

## BACKGROUND

Among other things, the Open Radio Access Network (O-RAN) architecture helps enable intelligent radio access network (RAN) operation and optimization using artificial intelligence (AI) and machine learning (ML) in wireless communication networks. RAN intelligence controllers (RIC) are developed to manage AI/ML-assisted solutions for RAN functions. In the overall O-RAN architecture, A1 policies enable the Non-RT (Non-Real-Time) RIC to guide its connected Near-RT (Near-Real-Time) RICs to fulfil the RAN intent via the A1 interface.

## BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate like structural elements. Embodiments are illustrated by way of example and not by way of limitation in the figures of the accompanying drawings.

FIG. **1** illustrates an example of a non-RT RIC architecture with A1 policy function in accordance with various embodiments.

FIG. **2** illustrates an example of a procedure for RAN intent subscription in accordance with various embodiments.

FIG. **3** illustrates an example of rApp querying RAN intent in a Non-RT RIC framework in accordance with various embodiments.

FIG. **4** illustrates an example of an rApp subscription notification of the creation, modification, and deletion of RAN intents in accordance with various embodiments.

FIG. **5** illustrates an example of a Non-RT RIC notifying an rApp of a subscribed event for RAN intent in accordance with various embodiments.

FIG. **6** illustrates an example of an rApp unsubscribing its subscribed events for RAN intent in accordance with various embodiments.

FIG. **7** illustrates an example of a procedure for an rApp querying supported A1 policy types in accordance with various embodiments.

FIG. **8** illustrates an example of an rApp querying supported A1 policy types in accordance with various embodiments.



FIG. **9** illustrates an example of in accordance with various embodiments.

FIG. **10** illustrates an example of an rApp subscription notification on supported A1 policy types (e.g., adding types, remove types) in accordance with various embodiments.

FIG. **11** illustrates an example of an rApp unsubscribing from its subscribed notification in accordance with various embodiments.

FIG. **12** illustrates an example of a procedure for an rApp to create an A1 policy in accordance with various embodiments.

FIG. **13** illustrates an example of a procedure for an rApp to subscribe for notification of A1 policy enforcement status changes in accordance with various embodiments.

FIG. **14** illustrates an example of a procedure for an rApp to update an A1 policy in accordance with various embodiments.

FIG. **15** illustrates an example of a procedure for an rApp to remove an A1 policy in accordance with various embodiments.

FIG. **16** illustrates an example of an rApp creating an A1 policy in accordance with various embodiments.

FIG. **17** illustrates an example of an rApp updating an A1 policy in accordance with various embodiments.

FIG. **18** illustrates an example of an rApp deleting an A1 policy in accordance with various embodiments.

FIG. **19** illustrates an example of an rApp querying A1 policies in accordance with various embodiments.

FIG. **20** illustrates an example of an rApp subscribing for notifications (e.g., new policy created, policy update or removal, policy enforcement status change) in accordance with various embodiments.

FIG. **21** illustrates an example of a Non-RT RIC framework notifying subscribed events to an rApp in accordance with various embodiments.

FIG. **22** illustrates an example of an rApp unsubscribing from subscribed events for A1 policies in accordance with various embodiments.

FIG. **23** schematically illustrates a wireless network in accordance with various embodiments.

FIG. **24** schematically illustrates components of a wireless network in accordance with various embodiments.

FIG. **25** is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein.

FIG. **26** provides an example of a high-level view of an Open RAN (O-RAN) architecture in accordance with various embodiments.

FIG. **27** illustrates an example of a Uu interface in accordance with various embodiments.

FIGS. **28**, **29**, and **30** depict examples of procedures for practicing the various embodiments discussed herein.

## DETAILED DESCRIPTION

The following detailed description refers to the accompanying drawings. The same reference numbers may be used in different drawings to identify the same or similar elements. In the following description, for purposes of explanation and not limitation, specific details are set forth such as particular structures, architectures, interfaces, techniques, etc. in order to provide a thorough understanding of the various aspects of various embodiments. However, it will be apparent to those skilled in the art having the benefit of the present disclosure that the various aspects of the various embodiments may be practiced in other examples that depart from these specific details. In certain instances, descriptions of well-known devices, circuits, and methods are omitted so as not to obscure the description of the various embodiments with unnecessary detail. For the purposes of the present document, the phrases "A or B" and "A/B" mean (A), (B), or (A and B).

Various embodiments herein provide techniques for A1 policy functions in the Non-RT RIC framework to produce A1-P related services in the R1 interface. In some embodiments, an A1 policy function hosts the A1 policy repository, which stores A1 policies generated by an rApp, also referred to as a RAN automation application. Among other things, rApps may help optimize network traffic. Additionally, rApps can subscribe to notifications of A1 policy updates and supported A1 policy types of a specific policy scope identifier.

Some embodiments provide an A1 policy function in the Non-RT RIC. In some embodiments the A1 policy function may host an A1 policy repository and track A1 policy enforcement status. The A1 policy function may also track supported A1 policy types in connected Near-RT RICs. Among other things, embodiments of the present disclosure provide solutions for the Non-RT RIC architecture (including A1 policy functions) and related procedures, which have not been specified yet in O-RAN WG2.

A1 Policy

An A1 policy is a declarative policy, e.g., it expresses the goals of the policy, but it does not restrict any implementations on how the goals are achieved. In O-RAN, an A1 policy is used to enable Non-RT RIC/SMO to guide Near-RT RIC towards the fulfilment of the RAN intent.

An A1 policy includes a scope identifier and one or more policy statements. The scope identifier specifies the subjects of the policy statements, e.g., UEs and cells. The policy statement represents the policy goals, and it can contain policy objectives and policy resources. Policy objectives are statements with objectives to be reached in the policy. Policy resources are statements specifying usage of RAN resources.

An A1 policy may be created, modified, and deleted by the Non-RT RIC. Until a policy's deletion, it should be enforced in the Near-RT RIC. If the enforcement status changes, the Near-RT RIC would inform the Non-RT RIC.

A1 Policy Function

FIG. 1 illustrates an example of a non-RT RIC architecture with A1 policy function.

In one embodiment, the A1 policy function creates, updates, and deletes A1 policies in the Non-RT RIC. A1 policy function subscribes data analytics provided by rApps via interfacing with data collection and repository functions.

In another embodiment, policy generation rApps create, update, and delete A1 policies in the Non-RT RIC:

- Based on the injected RAN intent, the rApp creates A1 policies.
- Based on the network observation over the O1 interface, the rApp decides whether the RAN intent is fulfilled or not.
  - If the RAN intent is not fulfilled, the rApp modifies/updates the A1 policies.
  - If the RAN intent is not fulfilled, the rApp can delete the A1 policies.

The A1 policy function provides the following functionalities:

- It hosts RAN intent repository.
  - Policy generation rApp can subscribe/unsubscribe notifications of the creation, modification, and deletion of RAN intents.
- It hosts A1 policy repository.
  - Policy generation rApp store generated policy in the policy repository, and A1 policy function assign an ID to each stored policy. Note that this ID is not necessary as same as the Policy Id used in the A1 spec.
  - Service consumer rApp can query A1 policies with specific policy scope identifiers (e.g., A1 policies applied to a specific UE or specific cell, etc.) or Near-RT RIC (e.g., A1 policies in a specific Near-RT RIC)
  - Service consumer rApp can subscribe/unsubscribe notifications of the creation, modification, and deletion of A1 policies with specific policy scope identifiers (e.g., A1 policies applied to a specific UE or specific cell, etc.) or Near-RT RIC (e.g., A1 policies in a specific Near-RT RIC).
- It performs conflict mitigation.
  - If A1 policy generation rApps set contradicting policy statements to the same or overlapping policy scope identifiers, the A1 policy function resolves the conflicts by modifying one or more A1 policies. A1 policy function notifies the change of A1 policies to their generation rApp. Examples of conflicting policy statements are
    - Examples for policy objective conflicts: same UE with different priority Level in QoS optimization, same slice with different qoeScore in QoE optimization, etc.
    - Example for policy resource conflicts: same UE with different preference on the same cell in traffic steering policy, etc.
- It tracks the enforcement status of all A1 policies created in the Non-RT RIC. Service consumer rApp can subscribe the enforcement status of A1 policies. A1 policy function notifies rApp when the enforcement status of subscribed A1 policy changes.
- It tracks the supported A1 policy types of each connected Near-RT RIC.
  - In one embodiment, policy generation rApp can request and subscribe/unsubscribe supported policy types given specific policy identifier. A1 policy function identify the right Near-RT RIC(s) based on the policy identifier. A1 policy function notifies supported policy types of subscribed policy identifier, based on the supported policy types of identified Near-RT RIC(s).
  - In another embodiment, policy generation rApp request and subscribe/unsubscribe supported policy types of Near-RT RICs. A1 policy function notify policy generation rApp when the supported policy types of subscribed Near-RT RIC changes.
- It creates, updates, and deletes A1 policies in the Near-RT RIC via A1 termination and A1 interface. It queries A1 policy status in the Near-RT RIC, and it receives A1 policy feedback from Near-RT RIC
  - This functionality may also include related south-bound services toward A1 termination.

A1-P Related Services in the R1 Interface

In one embodiment, Table 1 shows the A1-P related service in the R1 interface, produced by the A1-policy function.

TABLE 1

| A1-P related services in the R1 interface produced by the A1-policy function | | | |
|---|---|---|---|
| Service Name | Service operation | Operation semantics | Description |
| R1_RanIntentDiscovery | RanIntentQuery | Request/response | Policy generation rApp requests RAN intent in the A1 policy function |
| | RanIntentSubscribe | Subscribe/notify | Policy generation rApp subscribes to be notified, if interested RAN intents is created, updated, and removed. |
| | RanIntentNotify | | A1 policy function notifies Policy generation rApp of its subscribed events on RAN intents |
| | RanIntentUnubscribe | | Policy generation rApp unsubscribes its subscribed events on RAN intents |
| R1_A1PolicyManagement | A1PolicyCreate | Request/response | Policy generation rApp creates and stores a generated A1 policy in the A1 policy function, based on the RAN intent. The A1 policy function responds with an ID for created policy |
| | A1PolicyUpdate | Request/response | Policy generation rApp modifies an A1 policy stored in the A1 policy function, based on the network observation over O1 interface. |
| | A1PolicyDelete | Request/response | Policy generation rApp removes an A1 policy stored in the A1 policy function |
| | A1PolicyQuery | Request/response | rApp requests A1 policies in the A1 policy function with specific scope identifier or Near-RT RIC |
| | A1PolicySubscribe | Subscribe/notify | rApp subscribes to be notified, if |

TABLE 1-continued

| A1-P related services in the R1 interface produced by the A1-policy function | | | |
|---|---|---|---|
| Service Name | Service operation | Operation semantics | Description |
| | | | A1 policies applied to specific scope identifier are created, updated, removed, and if their enforcement status changes. |
| | A1PolicyNotify | | A1 policy function notifies rApp of its subscribed events on A1 policies |
| | A1PolicyUnsubscribe | | rApp unsubscribes its subscribed events on A1 policies |
| R1_A1PolicyTypeDiscovery | A1PolicyTypeQuery | Request/response | Policy generation rApp requests supported A1 policy types with specific scope identifier or Near-RT RIC |
| | A1PolicyTypeSubscribe | Subscribe/notify | rApp subscribes to be notified, if supported A1 policy types of specific scope identifier or Near-RT RIC is changed. |
| | A1PolicyTypeNotify | | A1 policy function notifies rApp of its subscribed events on supported A1 policy types. |
| | A1PolicyTypeUnsubscribe | | rApp unsubscribes its subscribed events on supported A1 policy types. |

A1 Policy Related Procedures

RAN Intent Subscription

An example of a procedure for an rApp subscribing for RAN intent in A1 policy functions is illustrated in FIG. **2**.

Step 1: The A1 policy generation rApp send subscription request to the Non-RT RIC framework.

Step 2: The subscription request is forwarded to the A1 policy function through R1 termination.

Step 3: A1 policy function sends back subscription response.

Step 4: Subscription response is forwarded to the policy generation rApp through R1 termination.

Step 5: RAN intent is injected into the Non-RT RIC framework.

Step 6: RAN intent is forwarded to RAN intent repository in the A1 policy function through external termination.

Step 7: A1 policy function sends notification about the newly injected RAN intent.

Step 8: The notification is forwarded to policy generation rApp through R1 termination.

For an R1 interface, examples of the service operations of a R1_RanIntentDiscovery service are illustrated in FIGS. **3** through **6**.

Supported A1 Policy Type Query

An example of a procedure for an rApp querying supported A1 policy types is illustrated in FIG. **7**:

Step 1: rApp sends query request about supported A1 policy types to the Non-RT RIC. Policy generation rApp needs to know what types A1 policy is supported in the Near-RT RIC before generating the A1 policy.

The query can be filtered by scope identifiers (e.g., UE id, cell id, etc.) or the query can be made toward a specific Near-RT RIC.

Step 2: Query request is forwarded to A1 policy function through R1 termination

Step 3: A1 policy function figure out supported A1 policy types

Step 4: A1 policy function sends back query response with a list of supported A1 policy types.

Step 5: Query response is forwarded to rApp through R1 termination

For the R1 interface, examples of the service operations of R1_A1PolicyTypeDiscovery service are illustrated in FIGS. **8-11**.

A1 Policy Creation

An example of a procedure for creating an A1 policy is illustrated in FIG. **12**:

Step 1: Policy generation rApp generates A1 policy based on the RAN intent.

Step 2: Policy generation rApp sends A1 policy creation request to Non-RT RIC framework

Step 3: A1 policy creation request is forwarded to A1 policy function through R1 termination

Step 4: A1 policy function creates A1 policy in the Near-RT RIC via A1 interface

Step 5: A1 policy function sends back A1 policy creation response with an identification for created policy. Note that this identification is not same as A1 policy ID in A1 specification.

Step 6: A1 policy creation response is forwarded to rApp through R1 termination

A1 Policy Status Subscription

An example of a procedure for an rApp subscribing for notification of an A1 policy enforcement status change is illustrated in FIG. **13**:

Step 1: Policy generation rApp subscribes notification of A1 policy, if its enforcement status changes.

Step 2: Subscription requestion is forwarded to A1 policy function through R1 termination

Step 3: A1 policy function sends back subscription response.

Step 4: Subscription response is forwarded to the rApp through R1 termination.

Step 5, 6: A1 policy feedback is sent by the Near-RT RIC to indicate the change of A1 policy enforcement status to A1 policy function through A1 termination

Step 7: A1 policy function notifies rApp about the change of the subscribed A1 policy enforcement status

Step 8: Notification is forwarded to policy generation rApp through R1 termination.

A1 Policy Update

An example of a procedure for an rApp updating an A1 policy is illustrated in FIG. **14**:

Step 1: Policy generation rApp subscribes O1 data from data collection and repository function for network performance observation.

Step 2: Policy generation rApp updates A1 policy based on RAN intent and network performance.

Step 3: Policy generation rApp sends A1 policy modification request to the Non-RT RIC framework

Step 4: Modification requestion is forwarded to A1 policy function through R1 termination

Step 5: A1 policy function update A1 policy in the Near-RT RIC via A1 interface

Step 6: A1 policy function sends back policy modification response.

Step 7: Policy modification response is forwarded to the rApp through R1 termination.

A1 Policy Deletion

An example of a procedure for an rApp deleting an A1 policy is illustrated in FIG. **15**:

Step 1: Policy generation rApp sends A1 policy deletion request to the Non-RT RIC.

Step 2: Deletion request is forwarded to A1 policy function.

Step 3: A1 policy function remove A1 policy in the Near-RT RIC via A1 interface.

Step 4: A1 policy function send back policy deletion response

Step 5: Policy deletion response is forwarded to the policy generation rApp through R1 termination.

For an R1 interface, examples of the service operations of an R1_A1PolicyManagement service are illustrated in FIGS. **16-22**.

Systems and Implementations

FIGS. **23-27** illustrate various systems, devices, and components that may implement aspects of disclosed embodiments.

FIG. **23** illustrates a network **2300** in accordance with various embodiments. The network **2300** may operate in a manner consistent with 3GPP technical specifications for LTE or 5G/NR systems. However, the example embodiments are not limited in this regard and the described embodiments may apply to other networks that benefit from the principles described herein, such as future 3GPP systems, or the like.

The network **2300** may include a UE **2302**, which may include any mobile or non-mobile computing device designed to communicate with a RAN **2304** via an over-the-air connection. The UE **2302** may be communicatively coupled with the RAN **2304** by a Uu interface. The UE **2302** may be, but is not limited to, a smartphone, tablet computer, wearable computer device, desktop computer, laptop computer, in-vehicle infotainment, in-car entertainment device, instrument cluster, head-up display device, onboard diagnostic device, dashtop mobile equipment, mobile data terminal, electronic engine management system, electronic/engine control unit, electronic/engine control module, embedded system, sensor, microcontroller, control module, engine management system, networked appliance, machine-type communication device, M2M or D2D device, IoT device, etc.

In some embodiments, the network **2300** may include a plurality of UEs coupled directly with one another via a sidelink interface. The UEs may be M2M/D2D devices that communicate using physical sidelink channels such as, but not limited to, PSBCH, PSDCH, PSSCH, PSCCH, PSFCH, etc.

In some embodiments, the UE **2302** may additionally communicate with an AP **2306** via an over-the-air connection. The AP **2306** may manage a WLAN connection, which may serve to offload some/all network traffic from the RAN **2304**. The connection between the UE **2302** and the AP **2306** may be consistent with any IEEE 802.11 protocol, wherein the AP **2306** could be a wireless fidelity (Wi-Fi®) router. In some embodiments, the UE **2302**, RAN **2304**, and AP **2306** may utilize cellular-WLAN aggregation (for example, LWA/LWIP). Cellular-WLAN aggregation may involve the UE **2302** being configured by the RAN **2304** to utilize both cellular radio resources and WLAN resources.

The RAN **2304** may include one or more access nodes, for example, AN **2308**. AN **2308** may terminate air-interface protocols for the UE **2302** by providing access stratum protocols including RRC, PDCP, RLC, MAC, and L1 protocols. In this manner, the AN **2308** may enable data/voice connectivity between CN **2320** and the UE **2302**. In some embodiments, the AN **2308** may be implemented in a discrete device or as one or more software entities running on server computers as part of, for example, a virtual network, which may be referred to as a CRAN or virtual baseband unit pool. The AN **2308** be referred to as a BS, gNB, RAN node, eNB, ng-eNB, NodeB, RSU, TRxP, TRP, etc. The AN **2308** may be a macrocell base station or a low power base station for providing femtocells, picocells or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells.

In embodiments in which the RAN **2304** includes a plurality of ANs, they may be coupled with one another via an X2 interface (if the RAN **2304** is an LTE RAN) or an Xn interface (if the RAN **2304** is a 5G RAN). The X2/Xn interfaces, which may be separated into control/user plane interfaces in some embodiments, may allow the ANs to communicate information related to handovers, data/context transfers, mobility, load management, interference coordination, etc.

The ANs of the RAN **2304** may each manage one or more cells, cell groups, component carriers, etc. to provide the UE **2302** with an air interface for network access. The UE **2302** may be simultaneously connected with a plurality of cells provided by the same or different ANs of the RAN **2304**. For example, the UE **2302** and RAN **2304** may use carrier aggregation to allow the UE **2302** to connect with a plurality of component carriers, each corresponding to a Pcell or Scell. In dual connectivity scenarios, a first AN may be a master node that provides an MCG and a second AN may be secondary node that provides an SCG. The first/second ANs may be any combination of eNB, gNB, ng-eNB, etc.

The RAN **2304** may provide the air interface over a licensed spectrum or an unlicensed spectrum. To operate in the unlicensed spectrum, the nodes may use LAA, eLAA, and/or feLAA mechanisms based on CA technology with PCells/Scells. Prior to accessing the unlicensed spectrum, the nodes may perform medium/carrier-sensing operations based on, for example, a listen-before-talk (LBT) protocol.

In V2X scenarios the UE **2302** or AN **2308** may be or act as a RSU, which may refer to any transportation infrastructure entity used for V2X communications. An RSU may be implemented in or by a suitable AN or a stationary (or relatively stationary) UE. An RSU implemented in or by: a UE may be referred to as a "UE-type RSU": an eNB may be referred to as an "eNB-type RSU": a gNB may be referred to as a "gNB-type RSU": and the like. In one example, an RSU is a computing device coupled with radio frequency circuitry located on a roadside that provides connectivity support to passing vehicle UEs. The RSU may also include internal data storage circuitry to store intersection map geometry, traffic statistics, media, as well as applications/software to sense and control ongoing vehicular and pedestrian traffic. The RSU may provide very low latency communications required for high speed events, such as crash avoidance, traffic warnings, and the like. Additionally or alternatively, the RSU may provide other cellular/WLAN communications services. The components of the RSU may be packaged in a weatherproof enclosure suitable for outdoor installation, and may include a network interface controller to provide a wired connection (e.g., Ethernet) to a traffic signal controller or a backhaul network.

In some embodiments, the RAN **2304** may be an LTE RAN **2310** with eNBs, for example, eNB **2312**. The LTE RAN **2310** may provide an LTE air interface with the following characteristics: SCS of 15 KHz: CP-OFDM waveform for DL and SC-FDMA waveform for UL; turbo codes for data and TBCC for control: etc. The LTE air interface may rely on CSI-RS for CSI acquisition and beam management: PDSCH/PDCCH DMRS for PDSCH/PDCCH demodulation: and CRS for cell search and initial acquisition, channel quality measurements, and channel estimation for coherent demodulation/detection at the UE. The LTE air interface may operating on sub-6 GHZ bands.

In some embodiments, the RAN **2304** may be an NG-RAN **2314** with gNBs, for example, gNB **2316**, or ng-eNBs, for example, ng-eNB **2318**. The gNB **2316** may connect with 5G-enabled UEs using a 5G NR interface. The gNB **2316** may connect with a 5G core through an NG interface, which may include an N2 interface or an N3 interface. The ng-eNB **2318** may also connect with the 5G core through an NG interface, but may connect with a UE via an LTE air interface. The gNB **2316** and the ng-eNB **2318** may connect with each other over an Xn interface.

In some embodiments, the NG interface may be split into two parts, an NG user plane (NG-U) interface, which carries traffic data between the nodes of the NG-RAN **2314** and a UPF **2348** (e.g., N3 interface), and an NG control plane (NG-C) interface, which is a signaling interface between the nodes of the NG-RAN **2314** and an AMF **2344** (e.g., N2 interface).

The NG-RAN **2314** may provide a 5G-NR air interface with the following characteristics: variable SCS: CP-OFDM for DL. CP-OFDM and DFT-s-OFDM for UL: polar, repetition, simplex, and Reed-Muller codes for control and LDPC for data. The 5G-NR air interface may rely on CSI-RS. PDSCH/PDCCH DMRS similar to the LTE air interface. The 5G-NR air interface **20**) may not use a CRS, but may use PBCH DMRS for PBCH demodulation: PTRS for phase tracking for PDSCH: and tracking reference signal for time tracking. The 5G-NR air interface may operating on FR1 bands that include sub-6 GHz bands or FR2 bands that include bands from 24.25 GHz to 52.6 GHZ. The 5G-NR air interface may include an SSB that is an area of a downlink resource grid that includes PSS/SSS/PBCH.

In some embodiments, the 5G-NR air interface may utilize BWPs for various purposes. For example, BWP can be used for dynamic adaptation of the SCS. For example, the UE **2302** can be configured with multiple BWPs where each BWP configuration has a different SCS. When a BWP change is indicated to the UE **2302**, the SCS of the transmission is changed as well. Another use case example of BWP is related to power saving. In particular, multiple BWPs can be configured for the UE **2302** with different amount of frequency resources (for example, PRBs) to support data transmission under different traffic loading scenarios. A BWP containing a smaller number of PRBs can be used for data transmission with small traffic load while allowing power saving at the UE **2302** and in some cases at the gNB **2316**. A BWP containing a larger number of PRBs can be used for scenarios with higher traffic load.

The RAN **2304** is communicatively coupled to CN **2320** that includes network elements to provide various functions to support data and telecommunications services to customers/subscribers (for example, users of UE **2302**). The components of the CN **2320** may be implemented in one physical node or separate physical nodes. In some embodiments, NFV may be utilized to virtualize any or all of the functions provided by the network elements of the CN **2320** onto physical compute/storage resources in servers, switches, etc. A logical instantiation of the CN **2320** may be referred to as a network slice, and a logical instantiation of a portion of the CN **2320** may be referred to as a network sub-slice.

In some embodiments, the CN **2320** may be an LTE CN **2322**, which may also be referred to as an EPC. The LTE CN **2322** may include MME **2324**, SGW **2326**, SGSN **2328**, HSS **2330**, PGW **2332**, and PCRF **2334** coupled with one another over interfaces (or "reference points") as shown. Functions of the elements of the LTE CN **2322** may be briefly introduced as follows.

The MME **2324** may implement mobility management functions to track a current location of the UE **2302** to facilitate paging, bearer activation/deactivation, handovers, gateway selection, authentication, etc.

The SGW **2326** may terminate an SI interface toward the RAN and route data packets between the RAN and the LTE CN **2322**. The SGW **2326** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities may include lawful intercept, charging, and some policy enforcement.

The SGSN **2328** may track a location of the UE **2302** and perform security functions and access control. In addition, the SGSN **2328** may perform inter-EPC node signaling for mobility between different RAT networks: PDN and S-GW selection as specified by MME **2324**; MME selection for handovers: etc. The S3 reference point between the MME **2324** and the SGSN **2328** may enable user and bearer information exchange for inter-3GPP access network mobility in idle/active states.

The HSS **2330** may include a database for network users, including subscription-related information to support the network entities handling of communication sessions. The HSS **2330** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc. An S6a reference point between the HSS **2330**) and the MME **2324** may enable transfer of subscription and authentication data for authenticating/authorizing user access to the LTE CN **2320**.

The PGW **2332** may terminate an SGi interface toward a data network (DN) **2336** that may include an application/content server **2338**. The PGW **2332** may route data packets between the LTE CN **2322** and the data network **2336**. The PGW **2332** may be coupled with the SGW **2326** by an S5 reference point to facilitate user plane tunneling and tunnel management. The PGW **2332** may further include a node for policy enforcement and charging data collection (for example, PCEF). Additionally, the SGi reference point between the PGW **2332** and the data network **2336** may be an operator external public, a private PDN, or an intra-operator packet data network, for example, for provision of IMS services. The PGW **2332** may be coupled with a PCRF **2334** via a Gx reference point.

The PCRF **2334** is the policy and charging control element of the LTE CN **2322**. The PCRF **2334** may be communicatively coupled to the app/content server **2338** to determine appropriate QoS and charging parameters for service flows. The PCRF **2332** may provision associated rules into a PCEF (via Gx reference point) with appropriate TFT and QCI.

In some embodiments, the CN **2320** may be a 5GC **2340**. The 5GC **2340** may include an AUSF **2342**, AMF **2344**, SMF **2346**, UPF **2348**, NSSF **2350**, NEF **2352**, NRF **2354**, PCF **2356**, UDM **2358**, and AF **2360** coupled with one another over interfaces (or "reference points") as shown. Functions of the elements of the 5GC **2340** may be briefly introduced as follows.

The AUSF **2342** may store data for authentication of UE **2302** and handle authentication-related functionality. The AUSF **2342** may facilitate a common authentication framework for various access types. In addition to communicating with other elements of the 5GC **2340** over reference points as shown, the AUSF **2342** may exhibit an Nausf service-based interface.

The AMF **2344** may allow other functions of the 5GC **2340** to communicate with the UE **2302** and the RAN **2304** and to subscribe to notifications about mobility events with respect to the UE **2302**. The AMF **2344** may be responsible for registration management (for example, for registering UE **2302**), connection management, reachability management, mobility management, lawful interception of AMF-related events, and access authentication and authorization. The AMF **2344** may provide transport for SM messages between the UE **2302** and the SMF **2346**, and act as a transparent proxy for routing SM messages. AMF **2344** may also provide transport for SMS messages between UE **2302** and an SMSF. AMF **2344** may interact with the AUSF **2342** and the UE **2302** to perform various security anchor and context management functions. Furthermore. AMF **2344** may be a termination point of a RAN CP interface, which may include or be an N2 reference point between the RAN **2304** and the AMF **2344**; and the AMF **2344** may be a termination point of NAS (N1) signaling, and perform NAS ciphering and integrity protection. AMF **2344** may also support NAS signaling with the UE **2302** over an N3 IWF interface.

The SMF **2346** may be responsible for SM (for example, session establishment, tunnel management between UPF **2348** and AN **2308**); UE IP address allocation and management (including optional authorization); selection and control of UP function; configuring traffic steering at UPF **2348** to route traffic to proper destination; termination of interfaces toward policy control functions; controlling part of policy enforcement, charging, and QoS; lawful intercept (for SM events and interface to LI system); termination of SM parts of NAS messages; downlink data notification; initiating AN specific SM information, sent via AMF **2344** over N2 to AN **2308**; and determining SSC mode of a session. SM may refer to management of a PDU session, and a PDU session or "session" may refer to a PDU connectivity service that provides or enables the exchange of PDUs between the UE **2302** and the data network **2336**.

The UPF **2348** may act as an anchor point for intra-RAT and inter-RAT mobility, an external PDU session point of interconnect to data network **2336**, and a branching point to support multi-homed PDU session. The UPF **2348** may also perform packet routing and forwarding, perform packet inspection, enforce the user plane part of policy rules, lawfully intercept packets (UP collection), perform traffic usage reporting, perform QoS handling for a user plane (e.g., packet filtering, gating, UL/DL rate enforcement), perform uplink traffic verification (e.g., SDF-to-QoS flow mapping), transport level packet marking in the uplink and downlink, and perform downlink packet buffering and downlink data notification triggering. UPF **2348** may include an uplink classifier to support routing traffic flows to a data network.

The NSSF **2350** may select a set of network slice instances serving the UE **2302**. The NSSF **2350** may also determine allowed NSSAI and the mapping to the subscribed S-NSSAIs, if needed. The NSSF **2350** may also determine the AMF set to be used to serve the UE **2302**, or a list of candidate AMFs based on a suitable configuration and possibly by querying the NRF **2354**. The selection of a set of network slice instances for the UE **2302** may be triggered by the AMF **2344** with which the UE **2302** is registered by interacting with the NSSF **2350**, which may lead to a change of AMF. The NSSF **2350** may interact with the AMF **2344** via an N22 reference point; and may communicate with another NSSF in a visited network via an N31 reference point (not shown). Additionally, the NSSF **2350** may exhibit an Nnssf service-based interface.

The NEF **2352** may securely expose services and capabilities provided by 3GPP network functions for third party, internal exposure/re-exposure. AFs (e.g., AF **2360**), edge computing or fog computing systems, etc. In such embodiments, the NEF **2352** may authenticate, authorize, or throttle the AFs. NEF **2352** may also translate information exchanged with the AF **2360** and information exchanged with internal network functions. For example, the NEF **2352** may translate between an AF-Service-Identifier and an internal 5GC information. NEF **2352** may also receive information from other NFs based on exposed capabilities of other NFs. This information may be stored at the NEF **2352** as structured data, or at a data storage NF using standardized interfaces. The stored information can then be re-exposed by the NEF **2352** to other NFs and AFs, or used for other purposes such as analytics. Additionally, the NEF **2352** may exhibit an Nnef service-based interface.

The NRF **2354** may support service discovery functions, receive NF discovery requests from NF instances, and provide the information of the discovered NF instances to the NF instances. NRF **2354** also maintains information of available NF instances and their supported services. As used herein, the terms "instantiate." "instantiation." and the like may refer to the creation of an instance, and an "instance" may refer to a concrete occurrence of an object, which may occur, for example, during execution of program code. Additionally, the NRF **2354** may exhibit the Nnrf service-based interface.

The PCF **2356** may provide policy rules to control plane functions to enforce them, and may also support unified policy framework to govern network behavior. The PCF **2356** may also implement a front end to access subscription information relevant for policy decisions in a UDR of the UDM **2358**. In addition to communicating with functions over reference points as shown, the PCF **2356** exhibit an Npcf service-based interface.

The UDM **2358** may handle subscription-related information to support the network entities' handling of communication sessions, and may store subscription data of UE **2302**. For example, subscription data may be communicated via an N8 reference point between the UDM **2358** and the AMF **2344**. The UDM **2358** may include two parts, an application front end and a UDR. The UDR may store subscription data and policy data for the UDM **2358** and the PCF **2356**, and/or structured data for exposure and application data (including PFDs for application detection, application request information for multiple UEs **2302**) for the NEF **2352**. The Nudr service-based interface may be exhibited by the UDR **221** to allow the UDM **2358**. PCF **2356**, and NEF **2352** to access a particular set of the stored data, as well as to read, update (e.g., add, modify), delete, and subscribe to notification of relevant data changes in the UDR. The UDM may include a UDM-FE, which is in charge of processing credentials, location management, subscription management and so on. Several different front ends may serve the same user in different transactions. The UDM-FE accesses subscription information stored in the UDR and performs authentication credential processing, user identification handling, access authorization, registration/mobility management, and subscription management. In addition to communicating with other NFs over reference points as shown, the UDM **2358** may exhibit the Nudm service-based interface.

The AF **2360** may provide application influence on traffic routing, provide access to NEF, and interact with the policy framework for policy control.

In some embodiments, the 5GC **2340** may enable edge computing by selecting operator/3rd party services to be geographically close to a point that the UE **2302** is attached to the network. This may reduce latency and load on the network. To provide edge-computing implementations, the 5GC **2340** may select a UPF **2348** close to the UE **2302** and execute traffic steering from the UPF **2348** to data network **2336** via the N6 interface. This may be based on the UE subscription data. UE location, and information provided by the AF **2360**. In this way, the AF **2360** may influence UPF (re)selection and traffic routing. Based on operator deployment, when AF **2360** is considered to be a trusted entity, the network operator may permit AF **2360** to interact directly with relevant NFs. Additionally, the AF **2360** may exhibit an Naf service-based interface.

The data network **2336** may represent various network operator services, Internet access, or third party services that may be provided by one or more servers including, for example, application/content server **2338**.

FIG. **24** schematically illustrates a wireless network **2400** in accordance with various embodiments. The wireless network **2400** may include a UE **2402** in wireless communication with an AN **2404**. The UE **2402** and AN **2404** may be similar to, and substantially interchangeable with, like-named components described elsewhere herein.

The UE **2402** may be communicatively coupled with the AN **2404** via connection **2406**. The connection **2406** is illustrated as an air interface to enable communicative coupling, and can be consistent with cellular communications protocols such as an LTE protocol or a 5G NR protocol operating at mmWave or sub-6 GHZ frequencies.

The UE **2402** may include a host platform **2408** coupled with a modem platform **2410**. The host platform **2408** may include application processing circuitry **2412**, which may be coupled with protocol processing circuitry **2414** of the modem platform **2410**. The application processing circuitry **2412** may run various applications for the UE **2402** that source/sink application data. The application processing circuitry **2412** may further implement one or more layer operations to transmit/receive application data to/from a data network. These layer operations may include transport (for example UDP) and Internet (for example, IP) operations

The protocol processing circuitry **2414** may implement one or more of layer operations to facilitate transmission or reception of data over the connection **2406**. The layer operations implemented by the protocol processing circuitry **2414** may include, for example, MAC, RLC, PDCP. RRC and NAS operations.

The modem platform **2410** may further include digital baseband circuitry **2416** that may implement one or more layer operations that are "below" layer operations performed by the protocol processing circuitry **2414** in a network protocol stack. These operations may include, for example, PHY operations including one or more of HARQ-ACK functions, scrambling/descrambling, encoding/decoding, layer mapping/de-mapping, modulation symbol mapping, received symbol/bit metric determination, multi-antenna port precoding/decoding, which may include one or more of space-time, space-frequency or spatial coding, reference signal generation/detection, preamble sequence generation and/or decoding, synchronization sequence generation/detection, control channel signal blind decoding, and other related functions.

The modem platform **2410** may further include transmit circuitry **2418**, receive circuitry **2420**, RF circuitry **2422**, and RF front end (RFFE) **2424**, which may include or connect to one or more antenna panels **2426**. Briefly, the transmit circuitry **2418** may include a digital-to-analog converter, mixer, intermediate frequency (IF) components, etc.; the receive circuitry **2420** may include an analog-to-digital converter, mixer, IF components, etc.; the RF circuitry **2422** may include a low-noise amplifier, a power amplifier, power tracking components, etc.; RFFE **2424** may include filters (for example, surface/bulk acoustic wave filters), switches, antenna tuners, beamforming components (for example, phase-array antenna components), etc. The selection and arrangement of the components of the transmit circuitry **2418**, receive circuitry **2420**, RF circuitry **2422**, RFFE **2424**, and antenna panels **2426** (referred generically as "transmit/receive components") may be specific to details of a specific implementation such as, for example, whether communication is TDM or FDM, in mmWave or sub-6 gHz frequencies, etc. In some embodiments, the transmit/receive components may be arranged in multiple parallel transmit/receive chains, may be disposed in the same or different chips/modules, etc.

In some embodiments, the protocol processing circuitry **2414** may include one or more instances of control circuitry (not shown) to provide control functions for the transmit/receive components.

A UE reception may be established by and via the antenna panels **2426**, RFFE **2424**, RF circuitry **2422**, receive circuitry **2420**, digital baseband circuitry **2416**, and protocol processing circuitry **2414**. In some embodiments, the antenna panels **2426** may receive a transmission from the AN **2404** by receive-beamforming signals received by a plurality of antennas/antenna elements of the one or more antenna panels **2426**.

A UE transmission may be established by and via the protocol processing circuitry **2414**, digital baseband circuitry **2416**, transmit circuitry **2418**, RF circuitry **2422**, RFFE **2424**, and antenna panels **2426**. In some embodiments, the transmit components of the UE **2404** may apply a spatial filter to the data to be transmitted to form a transmit beam emitted by the antenna elements of the antenna panels **2426**.

Similar to the UE **2402**, the AN **2404** may include a host platform **2428** coupled with a modem platform **2430**. The host platform **2428** may include application processing circuitry **2432** coupled with protocol processing circuitry **2434** of the modem platform **2430**. The modem platform may further include digital baseband circuitry **2436**, transmit circuitry **2438**, receive circuitry **2440**, RF circuitry **2442**, RFFE circuitry **2444**, and antenna panels **2446**. The components of the AN **2404** may be similar to and substantially interchangeable with like-named components of the UE **2402**. In addition to performing data transmission/reception as described above, the components of the AN **2408** may perform various logical functions that include, for example, RNC functions such as radio bearer management, uplink and downlink dynamic radio resource management, and data packet scheduling.

FIG. **25** is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **25** shows a diagrammatic representation of hardware resources **2500** including one or more processors (or processor cores) **2510**, one or more memory/storage devices **2520**, and one or more communication resources **2530**, each of which may be communicatively coupled via a bus **2540** or other interface circuitry. For embodiments where node virtualization (e.g., NFV) is utilized, a hypervisor **2502** may be executed to provide an execution environment for one or more network slices/sub-slices to utilize the hardware resources **2500**.

The processors **2510** may include, for example, a processor **2512** and a processor **2514**. The processors **2510** may be, for example, a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a DSP such as a baseband processor, an ASIC, an FPGA, a radio-frequency integrated circuit (RFIC), another processor (including those discussed herein), or any suitable combination thereof.

The memory/storage devices **2520** may include main memory, disk storage, or any suitable combination thereof. The memory/storage devices **2520** may include, but are not limited to, any type of volatile, non-volatile, or semi-volatile memory such as dynamic random access memory (DRAM), static random access memory (SRAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Flash memory, solid-state storage, etc.

The communication resources **2530** may include interconnection or network interface controllers, components, or other suitable devices to communicate with one or more peripheral devices **2504** or one or more databases **2506** or other network elements via a network **2508**. For example, the communication resources **2530** may include wired communication components (e.g., for coupling via USB, Ethernet, etc.), cellular communication components, NFC components, Bluetooth® (or Bluetooth® Low Energy) components, Wi-Fi® components, and other communication components.

Instructions **2550** may comprise software, a program, an application, an applet, an app, or other executable code for causing at least any of the processors **2510** to perform any one or more of the methodologies discussed herein. The instructions **2550** may reside, completely or partially, within at least one of the processors **2510** (e.g., within the processor's cache memory), the memory/storage devices **2520**, or any suitable combination thereof. Furthermore, any portion of the instructions **2550** may be transferred to the hardware resources **2500** from any combination of the peripheral devices **2504** or the databases **2506**. Accordingly, the memory of processors **2510**, the memory/storage devices **2520**, the peripheral devices **2504**, and the databases **2506** are examples of computer-readable and machine-readable media.

FIG. **26** provides a high-level view of an Open RAN (O-RAN) architecture **2600**. The O-RAN architecture **2600** includes four O-RAN defined interfaces—namely, the A1 interface, the O1 interface, the O2 interface, and the Open Fronthaul Management (M)-plane interface—which connect the Service Management and Orchestration (SMO) framework **2602** to O-RAN network functions (NFs) **2604** and the O-Cloud **2606**. The SMO **2602** also connects with an external system **2610**, which provides data to the SMO **2602**. FIG. **26** also illustrates that the A1 interface terminates at an O-RAN Non-Real Time (RT) RAN Intelligent Controller (RIC) **2612** in or at the SMO **2602** and at the O-RAN Near-RT RIC **2614** in or at the O-RAN NFs **2604**. The

O-RAN NFS **2604** can be VNFs such as VMs or containers, sitting above the O-Cloud **2606** and/or Physical Network Functions (PNFs) utilizing customized hardware. All O-RAN NFs **2604** are expected to support the O1 interface when interfacing the SMO framework **2602**. The O-RAN NFs **2604** connect to the NG-Core **2608** via the NG interface (which is a 3GPP defined interface). The Open Fronthaul M-plane interface between the SMO **2602** and the O-RAN Radio Unit (O-RU) **2616** supports the O-RU **2616** management in the O-RAN hybrid model. The Open Fronthaul M-plane **20**) interface is an optional interface to the SMO **2602** that is included for backward compatibility purposes, and is intended for management of the O-RU **2616** in hybrid mode only. The management architecture of flat mode and its relation to the O1 interface for the O-RU **2616** is for future study. The O-RU **2616** termination of the O1 interface towards the SMO **2602**.

FIG. **27** shows an O-RAN logical architecture **2700** corresponding to the O-RAN architecture **2600** of FIG. **26**. In FIG. **27**, the SMO **2702** corresponds to the SMO **2602**. O-Cloud **2706** corresponds to the O-Cloud **2606**, the non-RT RIC **2712** corresponds to the non-RT RIC **2612**, the near-RT RIC **2714** corresponds to the near-RT RIC **2614**, and the O-RU **2716** corresponds to the O-RU **2616** of FIG. **27**, respectively. The O-RAN logical architecture **2700** includes a radio portion and a management portion.

The management portion/side of the architectures **2700** includes the SMO Framework **2702** containing the non-RT RIC **2712**, and may include the O-Cloud **2706**. The O-Cloud **2706** is a cloud computing platform including a collection of physical infrastructure nodes to host the relevant O-RAN functions (e.g., the near-RT RIC **2714**. O-CU-CP **2721**. O-CU-UP **2722**, and the O-DU **2715**), supporting software components (e.g., OSs. VMMs, container runtime engines, ML engines, etc.), and appropriate management and orchestration functions.

The radio portion/side of the logical architecture **2700** includes the near-RT RIC **2714**, the O-RAN Distributed Unit (O-DU) **2715**, the O-RU **2716**, the O-RAN Central Unit-Control Plane (O-CU-CP) **2721**, and the O-RAN Central Unit-User Plane (O-CU-UP) **2722** functions. The radio portion/side of the logical architecture **2700** may also include the O-e/gNB **2710**.

The O-DU **2715** is a logical node hosting RLC. MAC, and higher PHY layer entities/elements (High-PHY layers) based on a lower layer functional split. The O-RU **2716** is a logical node hosting lower PHY layer entities/elements (Low-PHY layer) (e.g., FFT/iFFT. PRACH extraction, etc.) and RF processing elements based on a lower layer functional split. Virtualization of O-RU **2716** is FFS. The O-CU-CP **2721** is a logical node hosting the RRC and the control plane (CP) part of the PDCP protocol. The O O-CU-UP **2722** is a logical node hosting the user plane part of the PDCP protocol and the SDAP protocol.

An E2 interface terminates at a plurality of E2 nodes. The E2 nodes are logical nodes/entities that terminate the E2 interface. For NR/5G access, the E2 nodes include the O-CU-CP **2721**. O-CU-UP **2722**, O-DU **2715**, or any combination of elements. For E-UTRA access the E2 nodes include the O-e/gNB **2710**. As shown in FIG. **27**, the E2 interface also connects the O-e/gNB **2710** to the Near-RT RIC **2714**. The protocols over E2 interface are based exclusively on Control Plane (CP) protocols. The E2 functions are grouped into the following categories: (a) near-RT RIC **2714** services (REPORT. INSERT. CONTROL and POLICY); and (b) near-RT RIC **2714** support functions, which include E2 Interface Management (E2 Setup. E2 Reset. Reporting of General Error Situations, etc.) and Near-RT RIC Service Update (e.g., capability exchange related to the list of E2 Node functions exposed over E2).

FIG. **27** shows the Uu interface between a UE **2701** and O-e/gNB **2710** as well as between the UE **2701** and O-RAN components. The Uu interface is a 3GPP defined interface, which includes a complete protocol stack from L1 to L3 and terminates in the NG-RAN or E-UTRAN. The O-e/gNB **2710** is an LTE eNB, a 5G gNB or ng-eNB that supports the E2 interface. The O-e/gNB **2710** may be the same or similar as AN **2308** and/or AN **2404** discussed previously. The UE **2701** may correspond to UE **2302** and/or UE **2402** discussed with respect to FIGS. **23** and **24**, and/or the like. There may be multiple UEs **2701** and/or multiple O-e/gNB **2710**, each of which may be connected to one another the via respective Uu interfaces. Although not shown in FIG. **27**, the O-e/gNB **2710** supports O-DU **2715** and O-RU **2716** functions with an Open Fronthaul interface between them.

The Open Fronthaul (OF) interface(s) is/are between O-DU **2715** and O-RU **2716** functions. The OF interface(s) includes the Control User Synchronization (CUS) Plane and Management (M) Plane. FIGS. **26** and **27** also show that the O-RU **2716** terminates the OF M-Plane interface towards the O-DU **2715** and optionally towards the SMO **2702**. The O-RU **2716** terminates the OF CUS-Plane interface towards the O-DU **2715** and the SMO **2702**.

The F1-c interface connects the O-CU-CP **2721** with the O-DU **2715**. As defined by 3GPP, the F1-c interface is between the gNB-CU-CP and gNB-DU nodes. However, for purposes of O-RAN, the F1-c interface is adopted between the O-CU-CP **2721** with the O-DU **2715** functions while reusing the principles and protocol stack defined by 3GPP and the definition of interoperability profile specifications.

The F1-u interface connects the O-CU-UP **2722** with the O-DU **2715**. As defined by 3GPP, the F1-u interface is between the gNB-CU-UP and gNB-DU nodes. However, for purposes of O-RAN, the F1-u interface is adopted between the O-CU-UP **2722** with the O-DU **2715** functions while reusing the principles and protocol stack defined by 3GPP and the definition of interoperability profile specifications.

The NG-c interface is defined by 3GPP as an interface between the gNB-CU-CP and the AMF in the 5GC. The NG-c is also referred as the N2 interface. The NG-u interface is defined by 3GPP, as an interface between the gNB-CU-UP and the UPF in the 5GC. The NG-u interface is referred as the N3 interface. In O-RAN, NG-c and NG-u protocol stacks defined by 3GPP are reused and may be adapted for O-RAN purposes.

The X2-c interface is defined in 3GPP for transmitting control plane information between eNBs or between eNB and en-gNB in EN-DC. The X2-u interface is defined in 3GPP for transmitting user plane information between eNBs or between eNB and en-gNB in EN-DC. In O-RAN, X2-c and X2-u protocol stacks defined by 3GPP are reused and may be adapted for O-RAN purposes

The Xn-c interface is defined in 3GPP for transmitting control plane information between gNBs, ng-eNBs, or between an ng-eNB and gNB. The Xn-u interface is defined in 3GPP for transmitting user plane information between gNBs, ng-eNBs, or between ng-eNB and gNB. In O-RAN, Xn-c and Xn-u protocol stacks defined by 3GPP are reused and may be adapted for O-RAN purposes

The E1 interface is defined by 3GPP as being an interface between the gNB-CU-CP and gNB-CU-UP. In O-RAN, E1 protocol stacks defined by 3GPP are reused and adapted as being an interface between the O-CU-CP **2721** and the O-CU-UP **2722** functions.

The O-RAN Non-Real Time (RT) RAN Intelligent Controller (RIC) **2712** is a logical function within the SMO framework **2602**, **2702** that enables non-real-time control and optimization of RAN elements and resources: AI/machine learning (ML) workflow(s) including model training, inferences, and updates: and policy-based guidance of applications/features in the Near-RT RIC **2714**.

The O-RAN near-RT RIC **2714** is a logical function that enables near-real-time control and optimization of RAN elements and resources via fine-grained data collection and actions over the E2 interface. The near-RT RIC **2714** may include one or more AI/ML workflows including model training, inferences, and updates.

The non-RT RIC **2712** can be an ML training host to host the training of one or more ML models. ML training can be performed offline using data collected from the RIC. O-DU **2715** and O-RU **2716**. For supervised learning, non-RT RIC **2712** is part of the SMO **2702**, and the ML training host and/or ML model host/actor can be part of the non-RT RIC **2712** and/or the near-RT RIC **2714**. For unsupervised learning, the ML training host and ML model host/actor can be part of the non-RT RIC **2712** and/or the near-RT RIC **2714**. For reinforcement learning, the ML training host and ML model host/actor may be co-located as part of the non-RT RIC **2712** and/or the near-RT RIC **2714**. In some implementations, the non-RT RIC **2712** may request or trigger ML model training in the training hosts regardless of where the model is deployed and executed. ML models may be trained and not currently deployed.

In some implementations, the non-RT RIC **2712** provides a query-able catalog for an ML designer/developer to publish/install trained ML models (e.g., executable software components). In these implementations, the non-RT RIC **2712** may provide discovery mechanism if a particular ML model can be executed in a target ML inference host (MF), and what number and type of ML models can be executed in the MF. For example, there may be three types of ML catalogs made discoverable by the non-RT RIC **2712**: a design-time catalog (e.g., residing outside the non-RT RIC **2712** and hosted by some other ML platform(s)), a training/deployment-time catalog (e.g., residing inside the non-RT RIC **2712**), and a run-time catalog (e.g., residing inside the non-RT RIC **2712**). The non-RT RIC **2712** supports necessary capabilities for ML model inference in support of ML assisted solutions running in the non-RT RIC **2712** or some other ML inference host. These capabilities enable executable software to be installed such as VMs, containers, etc. The non-RT RIC **2712** may also include and/or operate one or more ML engines, which are packaged software executable libraries that provide methods, routines, data types, etc., used to run ML models. The non-RT RIC **2712** may also implement policies to switch and activate ML model instances under different operating conditions.

The non-RT RIC **272** is be able to access feedback data (e.g., FM and PM statistics) over the O1 interface on ML model performance and perform necessary evaluations. If the ML model fails during runtime, an alarm can be generated as feedback to the non-RT RIC **2712**. How well the ML model is performing in terms of prediction accuracy or other operating statistics it produces can also be sent to the non-RT RIC **2712** over O1. The non-RT RIC **2712** can also scale ML model instances running in a target MF over the O1 interface by observing resource utilization in MF. The environment where the ML model instance is running (e.g., the MF) monitors resource utilization of the running ML model. This can be done, for example, using an ORAN-SC component called ResourceMonitor in the near-RT RIC **2714** and/or in the non-RT RIC **2712**, which continuously monitors resource utilization. If resources are low or fall below a certain threshold, the runtime environment in the near-RT RIC **2714** and/or the non-RT RIC **2712** provides a scaling mechanism to add more ML instances. The scaling mechanism may include a scaling factor such as an number, percentage, and/or other like data used to scale up/down the number of ML instances. ML model instances running in the target ML inference hosts may be automatically scaled by observing resource utilization in the MF. For example, the Kubernetes® (K8s) runtime environment typically provides an auto-scaling feature.

The A1 interface is between the non-RT RIC **2712** (within or outside the SMO **2702**) and the near-RT RIC **2714**. The A1 interface supports three types of services, including a Policy Management Service, an Enrichment Information Service, and ML Model Management Service. A1 policies have the following characteristics compared to persistent configuration: A1 policies are not critical to traffic: A1 policies have temporary validity: A1 policies may handle individual UE or dynamically defined groups of UEs: A1 policies act within and take precedence over the configuration: and A1 policies are non-persistent, e.g., do not survive a restart of the near-RT RIC.

## EXAMPLE PROCEDURES

In some embodiments, the electronic device(s), network(s), system(s), chip(s) or component(s), or portions or implementations thereof, of FIGS. **10-12**, or some other figure herein, may be configured to perform one or more processes, techniques, or methods as described herein, or portions thereof. One such process is depicted in FIG. **28**. In this example, process **2800**) includes, at **2805**, retrieving, from a memory, policy statement information for a plurality of radio access network (RAN) automation applications (rApps), wherein the policy statement information includes respective policy scope identifiers for respective rApps in the plurality of rApps. The process further includes, at **2810**, identifying a conflict associated with common or overlapping policy scope identifiers between two or more rApps from the plurality of rApps. The process further includes, at **2815**, modifying one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict. The process further includes, at **2820**, notifying the two or more rApps of the modification of the one or more A1 policies.

Another such example is illustrated in FIG. **29**. In this example, process **2900** includes, at **2905**, identifying a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps. The process further includes, at **2910**, modifying one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict. The process further includes, at **2915**, notifying the two or more rApps of the modification of the one or more A1 policies.

Another such example is illustrated in FIG. **30**. In this example, process **3000** includes, at **3005**, Identifying a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps. The process further includes, at **3010**, modifying one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict. The process further includes, at **3015**, notifying the two or more rApps of the modification of the one or more A1 policies. The process further includes, at **3020**, tracking a supported A1 policy type for each of a plurality of near-RT RICs, and either: notifying an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier, or notifying an rApp of a change in supported policy types for a near-RT RIC

For one or more embodiments, at least one of the components set forth in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, and/or methods as set forth in the example section below. For example, the baseband circuitry as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below. For another example, circuitry associated with a UE, base station, network element, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below in the example section.

EXAMPLES

Example 1 may include a method of an A1 policy function to provide one or more of the following functionalities in the Non-RT RIC framework:

It hosts RAN intent repository. A1 policy function stores injected RAN intent.

It hosts A1 policy repository. A1 policy function stores A1 policy created by policy generation rApps. A1 policy function response to rApp's query about A1 policies. A1 policy function notifies rApps about their subscribed events on A1 policies.

It performs conflict mitigation for A1 policies. If A1 policies created by policy generation rApps set contradicting policy statements to the same or overlapping policy scope identifiers, the A1 policy function resolves the conflicts by modifying one or more A1 policies. A1 policy function notifies the change of A1 policies to their generation rApp.

It tracks the enforcement status of all A1 policies created in the Non-RT RIC. A1 policy function notifies rApp when the enforcement status of subscribed A1 policy changes.

It tracks the supported A1 policy types of each connected Near-RT RIC.

Example 2 may include the method of example 1 or some other example herein, wherein A1 policy function provides one or more of the following services to rApps:

RAN intent discovery.

A1 policy management

Supported A1 policy management discovery

Example 3 may include rApp can query RAN intent available in the Non-RT RIC framework.

Example 4 may include rApp can subscribe, unsubscribe RAN intent from A1 policy function. A1 policy function notifies rApp of its subscribed events on RAN intent, e.g.,

New RAN intent available;

RAN intent is modified: and/or

RAN intent is deleted, etc.

Example 5 may include rApp can generate A1 policy based on RAN intent and send it to the A1 policy function (via POST method). A1 policy function assigns an identifier for generated A1 policy in the response. The identifier is not the same of the A1 policy ID used in A1 specification

Example 6 rApp can modify A1 policy based on RAN intent and network performance observation (O1 data). rApp send the updated A1 policy to the A1 policy function (e.g., via PUT method).

Example 7 may include rApp can delete A1 policy (e.g., via DELETE method).

Example 8 may include rApp can query A1 policy (e.g., via GET method) by providing filtering on the scope identifier (e.g., UE id, Cell id, etc.). A1 policy function returns a list of A1 policy ID, which satisfies the filtering in the query request.

Example 9 may include rApp can subscribe (e.g., via POST method), unsubscribe (e.g., via DELETE method) notifications about A1 policy from A1 policy function. A1 policy function notifies rApp of its subscribed events on A1 policies, e.g.,

New A1 policy is created;

A1 policy is modified;

A1 policy is deleted; and/or

The enforcement status of A1 policy is changed, etc.

Example 10 may include rApp can query supported A1 policy (e.g., via GET method) by providing filtering on the scope identifier (e.g., UE id, Cell id, Near-RT RIC id, etc.). A1 policy function returns a list of supported A1 policy type ID.

Example 11 may include rApp can subscribe (e.g., via POST method), unsubscribe (e.g., via DELETE method) notifications about supported A1 policy types from A1 policy function. A1 policy function notifies rApp of its subscribed events on supported A1 policy types, e.g.,

New supported A1 policy types are added

Some of supported A1 policy types are removed, etc.

Example 12 may include A1 policy related procedures are proposed for RAN intent subscription, rApp querying supported A1 policy types, A1 policy generation, rApp subscribing policy enforcement status feedback, A1 policy update, A1 deletion.

Example X1 includes an apparatus comprising:

memory to store policy statement information for a plurality of radio access network (RAN) automation applications (rApps); and

processing circuitry, coupled with the memory, to:

retrieve the policy statement information from the memory, wherein the policy statement information includes respective policy scope identifiers for respective rApps in the plurality of rApps;

identify a conflict associated with common or overlapping policy scope identifiers between two or more rApps from the plurality of rApps;

modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict; and

notify the two or more rApps of the modification of the one or more A1 policies.

Example X2 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to:

track an enforcement status of an A1 policy created in the non-RT RIC; and notify an rApp subscribed to the tracked A1 policy when the enforcement status of the tracked A1 policy changes.

Example X3 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to track a supported A1 policy type for each of a plurality of near-RT RICs.

Example X4 includes the apparatus of example X3 or some other example herein, wherein the processing circuitry is to notify an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier.

Example X5 includes the apparatus of example X3 or some other example herein, wherein the processing circuitry is to notify an rApp of a change in supported policy types for a near-RT RIC.

Example X6 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to generate an A1-related service that includes: a RAN intent discovery service, an A1 policy management service, or an A1 policy type discovery service.

Example X7 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to:

receive a supported A1 policy types query request via an R1 termination; and

in response to the supported A1 policy types query request, send a query response that includes an indication of supported A1 policy types to a policy generation rApp via the R1 termination.

Example X8 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to:

receive an A1 policy feedback message from the near-RT RIC via an A1 termination to indicate a change of A1 policy enforcement status; and

send a notification of the A1 policy enforcement status change to a policy generation rApp via an R1 termination.

Example X9 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to:

receive an A1 policy modification request from a policy generation rApp via an R1 termination;

update an A1 policy over the A1 interface based on the A1 policy modification request; and

send an A1 policy modification response to the policy generation rApp via the R1 termination.

Example X10 includes the apparatus of example X1 or some other example herein, wherein the processing circuitry is further to:

receive an A1 policy deletion request from a policy generation rApp via an R1 termination;

remove an A1 policy over the A1 interface based on the A1 policy deletion request; and

send an A1 policy deletion response to the policy generation rApp via the R1 termination.

Example X11 includes the apparatus of any of examples X1-X10 or some other example herein, wherein the processing circuitry is to implement an A1 policy function.

Example X12 includes one or more computer-readable media storing instructions that, when executed by one or more processors, cause an A1 policy function to:

identify a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps;

modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict; and

notify the two or more rApps of the modification of the one or more A1 policies.

Example X13 includes the one or more computer-readable media of example X12 or some other example herein, wherein the media further stores instructions to:

track an enforcement status of an A1 policy created in the non-RT RIC; and

notify an rApp subscribed to the tracked A1 policy when the enforcement status of the tracked A1 policy changes.

Example X14 includes the one or more computer-readable media of example X12 or some other example herein, wherein the media further stores instructions to: track a supported A1 policy type for each of a plurality of near-RT RICs

Example X15 includes the one or more computer-readable media of example X14 or some other example herein, wherein the media further stores instructions to:

notify an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier; or

notify an rApp of a change in supported policy types for a near-RT RIC.

Example X16 includes the one or more computer-readable media of example X12 or some other example herein, wherein the media further stores instructions to:

receive a supported A1 policy types query request via an R1 termination; and

in response to the supported A1 policy types query request, send a query response that includes an indication of supported A1 policy types to a policy generation rApp via the R1 termination.

Example X17 includes the one or more computer-readable media of example X12 or some other example herein, wherein the media further stores instructions to:

receive an A1 policy feedback message from the near-RT RIC via an A1 termination to indicate a change of A1 policy enforcement status; and

send a notification of the A1 policy enforcement status change to a policy generation rApp via an R1 termination.

Example X18 includes the one or more computer-readable media of example X12 or some other example herein, wherein the media further stores instructions to:

receive an A1 policy modification request from a policy generation rApp via an R1 termination;

update an A1 policy over the A1 interface based on the A1 policy modification request; and

send an A1 policy modification response to the policy generation rApp via the R1 termination.

Example X19 includes the one or more computer-readable media of example X12 or some other example herein, wherein the media further stores instructions to:

receive an A1 policy deletion request from a policy generation rApp via an R1 termination;

remove an A1 policy over the A1 interface based on the A1 policy deletion request; and

send an A1 policy deletion response to the policy generation rApp via the R1 termination.

Example X20 includes the one or more computer-readable media of any of Examples X12-X19 or some other example herein, wherein the media further stores instructions to generate an A1-related service that includes: a RAN intent discovery service, an A1 policy management service, or an A1 policy type discovery service.

Example X21 includes one or more computer-readable media storing instructions that, when executed by one or more processors, cause an A1 policy function to:

identify a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps;

modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict;

notify the two or more rApps of the modification of the one or more A1 policies; and

track a supported A1 policy type for each of a plurality of near-RT RICs, and either: notify an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier, or notify an rApp of a change in supported policy types for a near-RT RIC.

Example X22 includes the one or more computer-readable media of example X21 or some other example herein, wherein the media further stores instructions to send, to an rApp via an R1 termination, one or more of;

an indication of a change in an enforcement status of a tracked A1 policy;

an indication of supported A1 policy types; and

a notification of an A1 policy enforcement status change.

Example X23 includes the one or more computer-readable media of example X21 or some other example herein, wherein the media further stores instructions to send, to:

update an A1 policy over the A1 interface based on an A1 policy modification request; or

remove an A1 policy over the A1 interface based on an A1 policy deletion request.

Example X24 includes the one or more computer-readable media of any of examples X21-X23 or some other example herein, wherein the media further stores instructions to generate an A1-related service that includes: a RAN intent discovery service, an A1 policy management service, or an A1 policy type discovery service.

Example Z01 may include an apparatus comprising means to perform one or more elements of a method described in or related to any of examples 1-X24, or any other method or process described herein.

Example Z02 may include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of a method described in or related to any of examples 1-X24, or any other method or process described herein.

Example Z03 may include an apparatus comprising logic, modules, or circuitry to perform one or more elements of a method described in or related to any of examples 1-X24, or any other method or process described herein.

Example Z04 may include a method, technique, or process as described in or related to any of examples 1-X24, or portions or parts thereof.

Example Z05 may include an apparatus comprising: one or more processors and one or more computer-readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-X24, or portions thereof.

Example Z06 may include a signal as described in or related to any of examples 1-X24, or portions or parts thereof.

Example Z07 may include a datagram, packet, frame, segment, protocol data unit (PDU), or message as described in or related to any of examples 1-X24, or portions or parts thereof, or otherwise described in the present disclosure.

Example Z08 may include a signal encoded with data as described in or related to any of examples 1-X24, or portions or parts thereof, or otherwise described in the present disclosure.

Example Z09 may include a signal encoded with a datagram, packet, frame, segment, protocol data unit (PDU), or message as described in or related to any of examples 1-X24, or portions or parts thereof, or otherwise described in the present disclosure.

Example Z10 may include an electromagnetic signal carrying computer-readable instructions, wherein execution of the computer-readable instructions by one or more processors is to cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-X24, or portions thereof.

Example Z11 may include a computer program comprising instructions, wherein execution of the program by a processing element is to cause the processing element to carry out the method, techniques, or process as described in or related to any of examples 1-X24, or portions thereof.

Example Z12 may include a signal in a wireless network as shown and described herein.

Example Z13 may include a method of communicating in a wireless network as shown and described herein.

Example Z14 may include a system for providing wireless communication as shown and described herein.

Example Z15 may include a device for providing wireless communication as shown and described herein.

Any of the above-described examples may be combined with any other example (or combination of examples), unless explicitly stated otherwise. The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

Abbreviations

Unless used differently herein, terms, definitions, and abbreviations may be consistent with terms, definitions, and abbreviations defined in 3GPP TR 21.905 v16.0.0 (2019-06). For the purposes of the present document, the following abbreviations may apply to the examples and embodiments discussed herein.

| | |
|---|---|
| 3GPP | Third Generation Partnership Project |
| 4G | Fourth Generation |
| 5G | Fifth Generation |
| 5GC | 5G Core network |
| AC | Application Client |
| ACR | Application Context Relocation |
| ACK | Acknowledgement |
| ACID | Application Client Identification |
| AF | Application Function |
| AM | Acknowledged Mode |
| AMBR | Aggregate Maximum Bit Rate |
| AMF | Access and Mobility Management Function |

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| | |
|---|---|
| AN | Access Network |
| ANR | Automatic Neighbour Relation |
| AOA | Angle of Arrival |
| AP | Application Protocol, Antenna Port, Access Point |
| API | Application Programming Interface |
| APN | Access Point Name |
| ARP | Allocation and Retention Priority |
| ARQ | Automatic Repeat Request |
| AS | Access Stratum |
| ASP | Application Service Provider |
| ASN.1 | Abstract Syntax Notation One |
| AUSF | Authentication Server Function |
| AWGN | Additive White Gaussian Noise |
| BAP | Backhaul Adaptation Protocol |
| BCH | Broadcast Channel |
| BER | Bit Error Ratio |
| BFD | Beam Failure Detection |
| BLER | Block Error Rate |
| BPSK | Binary Phase Shift Keying |
| BRAS | Broadband Remote Access Server |
| BSS | Business Support System |
| BS | Base Station |
| BSR | Buffer Status Report |
| BW | Bandwidth |
| BWP | Bandwidth Part |
| C-RNTI | Cell Radio Network Temporary Identity |
| CA | Carrier Aggregation, Certification Authority |
| CAPEX | CAPital EXpenditure |
| CBRA | Contention Based Random Access |
| CC | Component Carrier, Country Code, Cryptographic Checksum |
| CCA | Clear Channel Assessment |
| CCE | Control Channel Element |
| CCCH | Common Control Channel |
| CE | Coverage Enhancement |
| CDM | Content Delivery Network |
| CDMA | Code-Division Multiple Access |
| CDR | Charging Data Request |
| CDR | Charging Data Response |
| CFRA | Contention Free Random Access |
| CG | Cell Group |
| CGF | Charging Gateway Function |
| CHF | Charging Function |
| CI | Cell Identity |
| CID | Cell-ID (e g., positioning method) |
| CIM | Common Information Model |
| CIR | Carrier to Interference Ratio |
| CK | Cipher Key |
| CM | Connection Management, Conditional Mandatory |
| CMAS | Commercial Mobile Alert Service |
| CMD | Command |
| CMS | Cloud Management System |
| CO | Conditional Optional |
| CoMP | Coordinated Multi-Point |
| CORESET | Control Resource Set |
| COTS | Commercial Off-The-Shelf |
| CP | Control Plane, Cyclic Prefix, Connection Point |
| CPD | Connection Point Descriptor |
| CPE | Customer Premise Equipment |
| CPICH | Common Pilot Channel |
| CQI | Channel Quality Indicator |
| CPU | CSI processing unit, Central Processing Unit |
| C/R | Command/Response field bit |
| CRAN | Cloud Radio Access Network, Cloud RAN |
| CRB | Common Resource Block |
| CRC | Cyclic Redundancy Check |
| CRI | Channel-State Information Resource Indicator, CSI-RS Resource Indicator |
| C-RNTI | Cell RNTI |
| CS | Circuit Switched |
| CSCF | call session control function |
| CSAR | Cloud Service Archive |
| CSI | Channel-State Information |
| CSI-IM | CSI Interference Measurement |
| CSI-RS | CSI Reference Signal |
| CSI-RSRP | CSI reference signal received power |
| CSI-RSRQ | CSI reference signal received quality |
| CSI-SINR | CSI signal-to-noise and interference ratio |
| CSMA | Carrier Sense Multiple Access |

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| | |
|---|---|
| CSMA/CA | CSMA with collision avoidance |
| CSS | Common Search Space, Cell-specific Search Space |
| CTF | Charging Trigger Function |
| CTS | Clear-to-Send |
| CW | Codeword |
| CWS | Contention Window Size |
| D2D | Device-to-Device |
| DC | Dual Connectivity, Direct Current |
| DCI | Downlink Control Information |
| DF | Deployment Flavour |
| DL | Downlink |
| DMTF | Distributed Management Task Force |
| DPDK | Data Plane Development Kit |
| DM-RS, DMRS | Demodulation Reference Signal |
| DN | Data network |
| DNN | Data Network Name |
| DNAI | Data Network Access Identifier |
| DRB | Data Radio Bearer |
| DRS | Discovery Reference Signal |
| DRX | Discontinuous Reception |
| DSL | Domain Specific Language. Digital Subscriber Line |
| DSLAM | DSL Access Multiplexer |
| DwPTS | Downlink Pilot Time Slot |
| E-LAN | Ethernet Local Area Network |
| E2E | End-to-End |
| EAS | Edge Application Server |
| ECCA | extended clear channel assessment, extended CCA |
| ECCE | Enhanced Control Channel Element, Enhanced CCE |
| ED | Energy Detection |
| EDGE | Enhanced Datarates for GSM Evolution (GSM Evolution) |
| EAS | Edge Application Server |
| EASID | Edge Application Server Identification |
| ECS | Edge Configuration Server |
| ECSP | Edge Computing Service Provider |
| EDN | Edge Data Network |
| EEC | Edge Enabler Client |
| EECID | Edge Enabler Client Identification |
| EES | Edge Enabler Server |
| EESID | Edge Enabler Server Identification |
| EHE | Edge Hosting Environment |
| EGMF | Exposure Governance Management Function |
| EGPRS | Enhanced GPRS |
| EIR | Equipment Identity Register |
| eLAA | enhanced Licensed Assisted Access, enhanced LAA |
| EM | Element Manager |
| eMBB | Enhanced Mobile Broadband |
| EMS | Element Management System |
| eNB | evolved NodeB, E-UTRAN Node B |
| EN-DC | E-UTRA-NR Dual Connectivity |
| EPC | Evolved Packet Core |
| EPDCCH | enhanced PDCCH, enhanced Physical Downlink Control Cannel |
| EPRE | Energy per resource element |
| EPS | Evolved Packet System |
| EREG | enhanced REG, enhanced resource element groups |
| ETSI | European Telecommunications Standards Institute |
| ETWS | Earthquake and Tsunami Warning System |
| eUICC | embedded UICC, embedded Universal Integrated Circuit Card |
| E-UTRA | Evolved UTRA |
| E-UTRAN | Evolved UTRAN |
| EV2X | Enhanced V2X |
| F1AP | F1 Application Protocol |
| F1-C | F1 Control plane interface |
| F1-U | F1 User plane interface |
| FACCH | Fast Associated Control CHannel |
| FACCH/F | Fast Associated Control Channel/Full rate |
| FACCH/H | Fast Associated Control Channel/Half rate |
| FACH | Forward Access Channel |
| FAUSCH | Fast Uplink Signalling Channel |
| FB | Functional Block |
| FBI | Feedback Information |
| FCC | Federal Communications Commission |
| FCCH | Frequency Correction CHannel |
| FDD | Frequency Division Duplex |
| FDM | Frequency Division Multiplex |
| FDMA | Frequency Division Multiple Access |
| FE | Front End |

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|---|---|
| FEC | Forward Error Correction |
| FFS | For Further Study |
| FFT | Fast Fourier Transformation |
| feLAA | further enhanced Licensed Assisted Access, further enhanced LAA |
| FN | Frame Number |
| FPGA | Field-Programmable Gate Array |
| FR | Frequency Range |
| FQDN | Fully Qualified Domain Name |
| G-RNTI | GERAN Radio Network Temporary Identity |
| GERAN | GSM EDGE RAN, GSM EDGE Radio Access Network |
| GGSN | Gateway GPRS Support Node |
| GLONASS | GLObal'naya NAvigatsionnaya Sputnikovaya Sistema (Engl.: Global Navigation Satellite System) |
| gNB | Next Generation NodeB |
| gNB-CU | gNB-centralized unit, Next Generation NodeB centralized unit |
| gNB-DU | gNB-distributed unit, Next Generation NodeB distributed unit |
| GNSS | Global Navigation Satellite System |
| GPRS | General Packet Radio Service |
| GPSI | Generic Public Subscription Identifier |
| GSM | Global System for Mobile Communications, Groupe Spécial Mobile |
| GTP | GPRS Tunneling Protocol |
| GTP-U | GPRS Tunnelling Protocol for User Plane |
| GTS | Go To Sleep Signal (related to WUS) |
| GUMMEI | Globally Unique MME Identifier |
| GUTI | Globally Unique Temporary UE Identity |
| HARQ | Hybrid ARQ, Hybrid Automatic Repeat Request |
| HANDO | Handover |
| HFN | HyperFrame Number |
| HHO | Hard Handover |
| HLR | Home Location Register |
| HN | Home Network |
| HO | Handover |
| HPLMN | Home Public Land Mobile Network |
| HSDPA | High Speed Downlink Packet Access |
| HSN | Hopping Sequence Number |
| HSPA | High Speed Packet Access |
| HSS | Home Subscriber Server |
| HSUPA | High Speed Uplink Packet Access |
| HTTP | Hyper Text Transfer Protocol |
| HTTPS | Hyper Text Transfer Protocol Secure (https is http/1.1 over SSL, i.e. port 443) |
| I-Block | Information Block |
| ICCID | Integrated Circuit Card Identification |
| IAB | Integrated Access and Backhaul |
| ICIC | Inter-Cell Interference Coordination |
| ID | Identity, identifier |
| IDFT | Inverse Discrete Fourier Transform |
| IE | Information element |
| IBE | In-Band Emission |
| IEEE | Institute of Electrical and Electronics Engineers |
| IEI | Information Element Identifier |
| IEIDL | Information Element Identifier Data Length |
| IETF | Internet Engineering Task Force |
| IF | Infrastructure |
| IIOT | Industrial Internet of Things |
| IM | Interference Measurement, Intermodulation, IP Multimedia |
| IMC | IMS Credentials |
| IMEI | International Mobile Equipment Identity |
| IMGI | International mobile group identity |
| IMPI | IP Multimedia Private Identity |
| IMPU | IP Multimedia PUblic identity |
| IMS | IP Multimedia Subsystem |
| IMSI | International Mobile Subscriber Identity |
| IoT | Internet of Things |
| IP | Internet Protocol |
| Ipsec | IP Security, Internet Protocol Security |
| IP-CAN | IP-Connectivity Access Network |
| IP-M | IP Multicast |
| IPv4 | Internet Protocol Version 4 |
| IPv6 | Internet Protocol Version 6 |
| IR | Infrared |
| IS | In Sync |
| IRP | Integration Reference Point |

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|---|---|
| ISDN | Integrated Services Digital Network |
| ISIM | IM Services Identity Module |
| ISO | International Organisation for Standardisation |
| ISP | Internet Service Provider |
| IWF | Interworking-Function |
| I-WLAN | Interworking WLAN Constraint length of the convolutional code, USIM Individual key |
| kB | Kilobyte (1000 bytes) |
| kbps | kilo-bits per second |
| Kc | Ciphering key |
| Ki | Individual subscriber authentication key |
| KPI | Key Performance Indicator |
| KQI | Key Quality Indicator |
| KSI | Key Set Identifier |
| ksps | kilo-symbols per second |
| KVM | Kernel Virtual Machine |
| L1 | Layer 1 (physical layer) |
| L1-RSRP | Layer 1 reference signal received power |
| L2 | Layer 2 (data link layer) |
| L3 | Layer 3 (network layer) |
| LAA | Licensed Assisted Access |
| LAN | Local Area Network |
| LADN | Local Area Data Network |
| LBT | Listen Before Talk |
| LCM | LifeCycle Management |
| LCR | Low Chip Rate |
| LCS | Location Services |
| LCID | Logical Channel ID |
| LI | Layer Indicator |
| LLC | Logical Link Control, Low Layer Compatibility |
| LMF | Location Management Function |
| LOS | Line of Sight |
| LPLMN | Local PLMN |
| LPP | LTE Positioning Protocol |
| LSB | Least Significant Bit |
| LTE | Long Term Evolution |
| LWA | LTE-WLAN aggregation |
| LWIP | LTE/WLAN Radio Level Integration with IPsec Tunnel |
| LTE | Long Term Evolution |
| M2M | Machine-to-Machine |
| MAC | Medium Access Control (protocol layering context) |
| MAC | Message authentication code (security/encryption context) |
| MAC-A | MAC used for authentication and key agreement (TSG T WG3 context) |
| MAC-I | MAC used for data integrity of signalling messages (TSG T WG3 context) |
| MANO | Management and Orchestration |
| MBMS | Multimedia Broadcast and Multicast Service |
| MBSFN | Multimedia Broadcast multicast service Single Frequency Network |
| MCC | Mobile Country Code |
| MCG | Master Cell Group |
| MCOT | Maximum Channel Occupancy Time |
| MCS | Modulation and coding scheme |
| MDAF | Management Data Analytics Function |
| MDAS | Management Data Analytics Service |
| MDT | Minimization of Drive Tests |
| ME | Mobile Equipment |
| MeNB | master eNB |
| MER | Message Error Ratio |
| MGL | Measurement Gap Length |
| MGRP | Measurement Gap Repetition Period |
| MIB | Master Information Block, Management Information Base |
| MIMO | Multiple Input Multiple Output |
| MLC | Mobile Location Centre |
| MM | Mobility Management |
| MME | Mobility Management Entity |
| MN | Master Node |
| MNO | Mobile Network Operator |
| MO | Measurement Object, Mobile Originated |
| MPBCH | MTC Physical Broadcast CHannel |
| MPDCCH | MTC Physical Downlink Control CHannel |
| MPDSCH | MTC Physical Downlink Shared CHannel |
| MPRACH | MTC Physical Random Access CHannel |
| MPUSCH | MTC Physical Uplink Shared Channel |
| MPLS | MultiProtocol Label Switching |

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|---|---|
| MS | Mobile Station |
| MSB | Most Significant Bit |
| MSC | Mobile Switching Centre |
| MSI | Minimum System Information, MCH Scheduling Information |
| MSID | Mobile Station Identifier |
| MSIN | Mobile Station Identification Number |
| MSISDN | Mobile Subscriber ISDN Number |
| MT | Mobile Terminated, Mobile Termination |
| MTC | Machine-Type Communications |
| mMTCmassive | MTC, massive Machine-Type Communications |
| MU-MIMO | Multi User MIMO |
| MWUS | MTC wake-up signal, MTC WUS |
| NACK | Negative Acknowledgement |
| NAI | Network Access Identifier |
| NAS | Non-Access Stratum, Non-Access Stratum layer |
| NCT | Network Connectivity Topology |
| NC-JT | Non-Coherent Joint Transmission |
| NEC | Network Capability Exposure |
| NE-DC | NR-E-UTRA Dual Connectivity |
| NEF | Network Exposure Function |
| NF | Network Function |
| NFP | Network Forwarding Path |
| NFPD | Network Forwarding Path Descriptor |
| NFV | Network Functions Virtualization |
| NFVI | NFV Infrastructure |
| NFVO | NFV Orchestrator |
| NG | Next Generation, Next Gen |
| NGEN-DC | NG-RAN E-UTRA-NR Dual Connectivity |
| NM | Network Manager |
| NMS | Network Management System |
| N-PoP | Network Point of Presence |
| NMIB, N-MIB | Narrowband MIB |
| NPBCH | Narrowband Physical Broadcast CHannel |
| NPDCCH | Narrowband Physical Downlink Control CHannel |
| NPDSCH | Narrowband Physical Downlink Shared CHannel |
| NPRACH | Narrowband Physical Random Access CHannel |
| NPUSCH | Narrowband Physical Uplink Shared CHannel |
| NPSS | Narrowband Primary Synchronization Signal |
| NSSS | Narrowband Secondary Synchronization Signal |
| NR | New Radio, Neighbour Relation |
| NRF | NF Repository Function |
| NRS | Narrowband Reference Signal |
| NS | Network Service |
| NSA | Non-Standalone operation mode |
| NSD | Network Service Descriptor |
| NSR | Network Service Record |
| NSSAI | Network Slice Selection Assistance Information |
| S-NNSAI | Single-NSSAI |
| NSSF | Network Slice Selection Function |
| NW | Network |
| NWUS | Narrowband wake-up signal, Narrowband WUS |
| NZP | Non-Zero Power |
| O&M | Operation and Maintenance |
| ODU2 | Optical channel Data Unit - type 2 |
| OFDM | Orthogonal Frequency Division Multiplexing |
| OFDMA | Orthogonal Frequency Division Multiple Access |
| OOB | Out-of-band |
| OOS | Out of Sync |
| OPEX | OPerating EXpense |
| OSI | Other System Information |
| OSS | Operations Support System |
| OTA | over-the-air |
| PAPR | Peak-to-Average Power Ratio |
| PAR | Peak to Average Ratio |
| PBCH | Physical Broadcast Channel |
| PC | Power Control, Personal Computer |
| PCC | Primary Component Carrier, Primary CC |
| P-CSCF | Proxy CSCF |
| PCell | Primary Cell |
| PCI | Physical Cell ID, Physical Cell Identity |
| PCEF | Policy and Charging Enforcement Function |
| PCF | Policy Control Function |
| PCRF | Policy Control and Charging Rules Function |
| PDCP | Packet Data Convergence Protocol, Packet Data Convergence Protocol layer |
| PDCCH | Physical Downlink Control Channel |
| PDCP | Packet Data Convergence Protocol |
| PDN | Packet Data Network, Public Data Network |

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|---|---|
| PDSCH | Physical Downlink Shared Channel |
| PDU | Protocol Data Unit |
| PEI | Permanent Equipment Identifiers |
| PFD | Packet Flow Description |
| P-GW | PDN Gateway |
| PHICH | Physical hybrid-ARQ indicator channel |
| PHY | Physical layer |
| PLMN | Public Land Mobile Network |
| PIN | Personal Identification Number |
| PM | Performance Measurement |
| PMI | Precoding Matrix Indicator |
| PNF | Physical Network Function |
| PNFD | Physical Network Function Descriptor |
| PNFR | Physical Network Function Record |
| POC | PTT over Cellular |
| PP, PTP | Point-to-Point |
| PPP | Point-to-Point Protocol |
| PRACH | Physical RACH |
| PRB | Physical resource block |
| PRG | Physical resource block group |
| ProSe | Proximity Services, Proximity-Based Service |
| PRS | Positioning Reference Signal |
| PRR | Packet Reception Radio |
| PS | Packet Services |
| PSBCH | Physical Sidelink Broadcast Channel |
| PSDCH | Physical Sidelink Downlink Channel |
| PSCCH | Physical Sidelink Control Channel |
| PSSCH | Physical Sidelink Shared Channel |
| PSCell | Primary SCell |
| PSS | Primary Synchronization Signal |
| PSTN | Public Switched Telephone Network |
| PT-RS | Phase-tracking reference signal |
| PTT | Push-to-Talk |
| PUCCH | Physical Uplink Control Channel |
| PUSCH | Physical Uplink Shared Channel |
| QAM | Quadrature Amplitude Modulation |
| QCI | QoS class of identifier |
| QCL | Quasi co-location |
| QFI | QoS Flow ID, QoS Flow Identifier |
| QoS | Quality of Service |
| QPSK | Quadrature (Quaternary) Phase Shift Keying |
| QZSS | Quasi-Zenith Satellite System |
| RA-RNTI | Random Access RNTI |
| RAB | Radio Access Bearer, Random Access Burst |
| RACH | Random Access Channel |
| RADIUS | Remote Authentication Dial In User Service |
| RAN | Radio Access Network |
| RAND | RANDom number (used for authentication) |
| RAR | Random Access Response |
| RAT | Radio Access Technology |
| RAU | Routing Area Update |
| RB | Resource block, Radio Bearer |
| RBG | Resource block group |
| REG | Resource Element Group |
| Rel | Release |
| REQ | REQuest |
| RF | Radio Frequency |
| RI | Rank Indicator |
| RIV | Resource indicator value |
| RL | Radio Link |
| RLC | Radio Link Control, Radio Link Control layer |
| RLC AM | RLC Acknowledged Mode |
| RLC UM | RLC Unacknowledged Mode |
| RLF | Radio Link Failure |
| RLM | Radio Link Monitoring |
| RLM-RS | Reference Signal for RLM |
| RM | Registration Management |
| RMC | Reference Measurement Channel |
| RMSI | Remaining MSI, Remaining Minimum System Information |
| RN | Relay Node |
| RNC | Radio Network Controller |
| RNL | Radio Network Layer |
| RNTI | Radio Network Temporary Identifier |
| ROHC | RObust Header Compression |
| RRC | Radio Resource Control, Radio Resource Control layer |
| RRM | Radio Resource Management |
| RS | Reference Signal |

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| | |
|---|---|
| RSRP | Reference Signal Received Power |
| RSRQ | Reference Signal Received Quality |
| RSSI | Received Signal Strength Indicator |
| RSU | Road Side Unit |
| RSTD | Reference Signal Time difference |
| RTP | Real Time Protocol |
| RTS | Ready-To-Send |
| RTT | Round Trip Time |
| Rx | Reception, Receiving, Receiver |
| S1AP | S1 Application Protocol |
| S1-MME | S1 for the control plane |
| S1-U | S1 for the user plane |
| S-CSCF | serving CSCF |
| S-GW | Serving Gateway |
| S-RNTI | SRNC Radio Network Temporary Identity |
| S-TMSI | SAE Temporary Mobile Station Identifier |
| SA | Standalone operation mode |
| SAE | System Architecture Evolution |
| SAP | Service Access Point |
| SAPD | Service Access Point Descriptor |
| SAPI | Service Access Point Identifier |
| SCC | Secondary Component Carrier, Secondary CC |
| SCell | Secondary Cell |
| SCEF | Service Capability Exposure Function |
| SC-FDMA | Single Carrier Frequency Division Multiple Access |
| SCG | Secondary Cell Group |
| SCM | Security Context Management |
| SCS | Subcarrier Spacing |
| SCTP | Stream Control Transmission Protocol |
| SDAP | Service Data Adaptation Protocol, Service Data Adaptation Protocol layer |
| SDL | Supplementary Downlink |
| SDNF | Structured Data Storage Network Function |
| SDP | Session Description Protocol |
| SDSF | Structured Data Storage Function |
| SDT | Small Data Transmission |
| SDU | Service Data Unit |
| SEAF | Security Anchor Function |
| SeNB | secondary eNB |
| SEPP | Security Edge Protection Proxy |
| SFI | Slot format indication |
| SFTD | Space-Frequency Time Diversity, SFN and frame timing difference |
| SFN | System Frame Number |
| SgNB | Secondary gNB |
| SGSN | Serving GPRS Support Node |
| S-GW | Serving Gateway |
| SI | System Information |
| SI-RNTI | System Information RNTI |
| SIB | System Information Block |
| SIM | Subscriber Identity Module |
| SIP | Session Initiated Protocol |
| SiP | System in Package |
| SL | Sidelink |
| SLA | Service Level Agreement |
| SM | Session Management |
| SMF | Session Management Function |
| SMS | Short Message Service |
| SMSF | SMS Function |
| SMTC | SSB-based Measurement Timing Configuration |
| SN | Secondary Node, Sequence Number |
| SoC | System on Chip |
| SON | Self-Organizing Network |
| SpCell | Special Cell |
| SP-CSI-RNTI | Semi-Persistent CSI RNTI |
| SPS | Semi-Persistent Scheduling |
| SQN | Sequence number |
| SR | Scheduling Request |
| SRB | Signalling Radio Bearer |
| SRS | Sounding Reference Signal |
| SS | Synchronization Signal |
| SSB | Synchronization Signal Block |
| SSID | Service Set Identifier |
| SS/PBCH Block SSBRI | SS/PBCH Block Resource Indicator, Synchronization Signal Block Resource Indicator |
| SSC | Session and Service Continuity |
| SS-RSRP | Synchronization Signal based Reference Signal Received Power |

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|---|---|
| SS-RSRQ | Synchronization Signal based Reference Signal Received Quality |
| SS-SINR | Synchronization Signal based Signal to Noise and Interference Ratio |
| SSS | Secondary Synchronization Signal |
| SSSG | Search Space Set Group |
| SSSIF | Search Space Set Indicator |
| SST | Slice/Service Types |
| SU-MIMO | Single User MIMO |
| SUL | Supplementary Uplink |
| TA | Timing Advance, Tracking Area |
| TAC | Tracking Area Code |
| TAG | Timing Advance Group |
| TAI | Tracking Area Identity |
| TAU | Tracking Area Update |
| TB | Transport Block |
| TBS | Transport Block Size |
| TBD | To Be Defined |
| TCI | Transmission Configuration Indicator |
| TCP | Transmission Communication Protocol |
| TDD | Time Division Duplex |
| TDM | Time Division Multiplexing |
| TDMA | Time Division Multiple Access |
| TE | Terminal Equipment |
| TEID | Tunnel End Point Identifier |
| TFT | Traffic Flow Template |
| TMSI | Temporary Mobile Subscriber Identity |
| TNL | Transport Network Layer |
| TPC | Transmit Power Control |
| TPMI | Transmitted Precoding Matrix Indicator |
| TR | Technical Report |
| TRP, TRxP | Transmission Reception Point |
| TRS | Tracking Reference Signal |
| TRx | Transceiver |
| TS | Technical Specifications, Technical Standard |
| TTI | Transmission Time Interval |
| Tx | Transmission, Transmitting, Transmitter |
| U-RNTI | UTRAN Radio Network Temporary Identity |
| UART | Universal Asynchronous Receiver and Transmitter |
| UCI | Uplink Control Information |
| UE | User Equipment |
| UDM | Unified Data Management |
| UDP | User Datagram Protocol |
| UDSF | Unstructured Data Storage Network Function |
| UICC | Universal Integrated Circuit Card |
| UL | Uplink |
| UM | Unacknowledged Mode |
| UML | Unified Modelling Language |
| UMTS | Universal Mobile Telecommunications System |
| UP | User Plane |
| UPF | User Plane Function |
| URI | Uniform Resource Identifier |
| URL | Uniform Resource Locator |
| URLLC | Ultra-Reliable and Low Latency |
| USB | Universal Serial Bus |
| USIM | Universal Subscriber Identity Module |
| USS | UE-specific search space |
| UTRA | UMTS Terrestrial Radio Access |
| UTRAN | Universal Terrestrial Radio Access Network |
| UwPTS | Uplink Pilot Time Slot |
| V2I | Vehicle-to-Infrastruction |
| V2P | Vehicle-to-Pedestrian |
| V2V | Vehicle-to-Vehicle |
| V2X | Vehicle-to-everything |
| VIM | Virtualized Infrastructure Manager |
| VL | Virtual Link, |
| VLAN | Virtual LAN, Virtual Local Area Network |
| VM | Virtual Machine |
| VNF | Virtualized Network Function |
| VNFFG | VNF Forwarding Graph |
| VNFFGD | VNF Forwarding Graph Descriptor |
| VNFM | VNF Manager |
| VoIP | Voice-over-IP, Voice-over-Internet Protocol |
| VPLMN | Visited Public Land Mobile Network |
| VPN | Virtual Private Network |
| VRB | Virtual Resource Block |
| WiMAX | Worldwide Interoperability for Microwave Access |
| WLAN | Wireless Local Area Network |
| WMAN | Wireless Metropolitan Area Network |

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| | |
|---|---|
| WPAN | Wireless Personal Area Network |
| X2-C | X2-Control plane |
| X2-U | X2-User plane |
| XML | eXtensible Markup Language |
| XRES | EXpected user RESponse |
| XOR | eXclusive OR |
| ZC | Zadoff-Chu |
| ZP | Zero Power |

Terminology

For the purposes of the present document, the following terms and definitions are applicable to the examples and embodiments discussed herein.

The term "circuitry" as used herein refers to, is part of, or includes hardware components such as an electronic circuit, a logic circuit, a processor (shared, dedicated, or group) and/or memory (shared, dedicated, or group), an Application Specific Integrated Circuit (ASIC), a field-programmable device (FPD) (e.g., a field-programmable gate array (FPGA), a programmable logic device (PLD), a complex PLD (CPLD), a high-capacity PLD (HCPLD), a structured ASIC, or a programmable SoC), digital signal processors (DSPs), etc., that are configured to provide the described functionality. In some embodiments, the circuitry may execute one or more software or firmware programs to provide at least some of the described functionality. The term "circuitry" may also refer to a combination of one or more hardware elements (or a combination of circuits used in an electrical or electronic system) with the program code used to carry out the functionality of that program code. In these embodiments, the combination of hardware elements and program code may be referred to as a particular type of circuitry.

The term "processor circuitry." as used herein refers to, is part of, or includes circuitry capable of sequentially and automatically carrying out a sequence of arithmetic or logical operations, or recording, storing, and/or transferring digital data. Processing circuitry may include one or more processing cores to execute instructions and one or more memory structures to store program and data information. The term "processor circuitry" may refer to one or more application processors, one or more baseband processors, a physical central processing unit (CPU), a single-core processor, a dual-core processor, a triple-core processor, a quad-core processor, and/or any other device capable of executing or otherwise operating computer-executable instructions, such as program code, software modules, and/or functional processes. Processing circuitry may include more hardware accelerators, which may be microprocessors, programmable processing devices, or the like. The one or more hardware accelerators may include, for example, computer vision (CV) and/or deep learning (DL) accelerators. The terms "application circuitry" and/or "baseband circuitry" may be considered synonymous to, and may be referred to as, "processor circuitry."

The term "interface circuitry" as used herein refers to, is part of, or includes circuitry that enables the exchange of information between two or more components or devices. The term "interface circuitry" may refer to one or more hardware interfaces, for example, buses, I/O interfaces, peripheral component interfaces, network interface cards, and/or the like.

The term "user equipment" or "UE" as used herein refers to a device with radio communication capabilities and may describe a remote user of network resources in a communications network. The term "user equipment" or "UE" may be considered synonymous to, and may be referred to as, client, mobile, mobile device, mobile terminal, user terminal, mobile unit, mobile station, mobile user, subscriber, user, remote station, access agent, user agent, receiver, radio equipment, reconfigurable radio equipment, reconfigurable mobile device, etc. Furthermore, the term "user equipment" or "UE" may include any type of wireless/wired device or any computing device including a wireless communications interface.

The term "network element" as used herein refers to physical or virtualized equipment and/or infrastructure used to provide wired or wireless communication network services. The term "network element" may be considered synonymous to and/or referred to as a networked computer, networking hardware, network equipment, network node, router, switch, hub, bridge, radio network controller, RAN device, RAN node, gateway, server, virtualized VNF, NFVI, and/or the like.

The term "computer system" as used herein refers to any type interconnected electronic devices, computer devices, or components thereof. Additionally, the term "computer system" and/or "system" may refer to various components of a computer that are communicatively coupled with one another. Furthermore, the term "computer system" and/or "system" may refer to multiple computer devices and/or multiple computing systems that are communicatively coupled with one another and configured to share computing and/or networking resources.

The term "appliance." "computer appliance," or the like, as used herein refers to a computer device or computer system with program code (e.g., software or firmware) that is specifically designed to provide a specific computing resource. A "virtual appliance" is a virtual machine image to be implemented by a hypervisor-equipped device that virtualizes or emulates a computer appliance or otherwise is dedicated to provide a specific computing resource.

The term "resource" as used herein refers to a physical or virtual device, a physical or virtual component within a computing environment, and/or a physical or virtual component within a particular device, such as computer devices, mechanical devices, memory space, processor/CPU time, processor/CPU usage, processor and accelerator loads, hardware time or usage, electrical power, input/output operations, ports or network sockets, channel/link allocation, throughput, memory usage, storage, network, database and applications, workload units, and/or the like. A "hardware resource" may refer to compute, storage, and/or network resources provided by physical hardware element(s). A "virtualized resource" may refer to compute, storage, and/or network resources provided by virtualization infrastructure to an application, device, system, etc. The term "network resource" or "communication resource" may refer to resources that are accessible by computer devices/systems via a communications network. The term "system resources" may refer to any kind of shared entities to provide services, and may include computing and/or network resources. System resources may be considered as a set of coherent functions, network data objects or services, accessible through a server where such system resources reside on a single host or multiple hosts and are clearly identifiable.

The term "channel" as used herein refers to any transmission medium, either tangible or intangible, which is used to communicate data or a data stream. The term "channel" may be synonymous with and/or equivalent to "communications channel," "data communications channel," "transmission channel," "data transmission channel," "access channel," "data access channel," "link," "data link," "carrier," "radiofrequency carrier," and/or any other like term denoting a pathway or medium through which data is communicated. Additionally, the term "link" as used herein refers to a connection between two devices through a RAT for the purpose of transmitting and receiving information.

The terms "instantiate," "instantiation," and the like as used herein refers to the creation of an instance. An "instance" also refers to a concrete occurrence of an object, which may occur, for example, during execution of program code.

The terms "coupled," "communicatively coupled," along with derivatives thereof are used herein. The term "coupled" may mean two or more elements are in direct physical or electrical contact with one another, may mean that two or more elements indirectly contact each other but still cooperate or interact with each other, and/or may mean that one or more other elements are coupled or connected between the elements that are said to be coupled with each other. The term "directly coupled" may mean that two or more elements are in direct contact with one another. The term "communicatively coupled" may mean that two or more elements may be in contact with one another by a means of communication including through a wire or other interconnect connection, through a wireless communication channel or link, and/or the like.

The term "information element" refers to a structural element containing one or more fields. The term "field" refers to individual contents of an information element, or a data element that contains content.

The term "SMTC" refers to an SSB-based measurement timing configuration configured by SSB-MeasurementTimingConfiguration.

The term "SSB" refers to an SS/PBCH block.

The term "a "Primary Cell" refers to the MCG cell, operating on the primary frequency, in which the UE either performs the initial connection establishment procedure or initiates the connection re-establishment procedure.

The term "Primary SCG Cell" refers to the SCG cell in which the UE performs random access when performing the Reconfiguration with Sync procedure for DC operation.

The term "Secondary Cell" refers to a cell providing additional radio resources on top of a Special Cell for a UE configured with CA.

The term "Secondary Cell Group" refers to the subset of serving cells comprising the PSCell and zero or more secondary cells for a UE configured with DC.

The term "Serving Cell" refers to the primary cell for a UE in RRC_CONNECTED not configured with CA/DC there is only one serving cell comprising of the primary cell.

The term "serving cell" or "serving cells" refers to the set of cells comprising the Special Cell(s) and all secondary cells for a UE in RRC_CONNECTED configured with CA/.

The term "Special Cell" refers to the PCell of the MCG or the PSCell of the SCG for DC operation: otherwise, the term "Special Cell" refers to the Pcell.

The term "application" may refer to a complete and deployable package, environment to achieve a certain function in an operational environment. The term "AI/ML application" or the like may be an application that contains some AI/ML models and application-level descriptions.

The term "machine learning" or "ML" refers to the use of computer systems implementing algorithms and/or statistical models to perform specific task(s) without using explicit instructions, but instead relying on patterns and inferences. ML algorithms build or estimate mathematical model(s) (referred to as "ML models" or the like) based on sample data (referred to as "training data." "model training information," or the like) in order to make predictions or decisions without being explicitly programmed to perform such tasks. Generally, an ML algorithm is a computer program that learns from experience with respect to some task and some performance measure, and an ML model may be any object or data structure created after an ML algorithm is trained with one or more training datasets. After training, an ML model may be used to make predictions on new datasets. Although the term "ML algorithm" refers to different concepts than the term "ML model," these terms as discussed herein may be used interchangeably for the purposes of the present disclosure.

The term "machine learning model." "ML model," or the like may also refer to ML methods and concepts used by an ML-assisted solution. An "ML-assisted solution" is a solution that addresses a specific use case using ML algorithms during operation. ML models include supervised learning (e.g., linear regression, k-nearest neighbor (KNN), descision tree algorithms, support machine vectors. Bayesian algorithm, ensemble algorithms, etc.) unsupervised learning (e.g., K-means clustering, principle component analysis (PCA), etc.), reinforcement learning (e.g., Q-learning, multi-armed bandit learning, deep RL, etc.), neural networks, and the like. Depending on the implementation a specific ML model could have many sub-models as components and the ML model may train all sub-models together. Separately trained ML models can also be chained together in an ML pipeline during inference. An "ML pipeline" is a set of functionalities, functions, or functional entities specific for an ML-assisted solution; an ML pipeline may include one or several data sources in a data pipeline, a model training pipeline, a model evaluation pipeline, and an actor. The "actor" is an entity that hosts an ML assisted solution using the output of the ML model inference). The term "ML training host" refers to an entity, such as a network function, that hosts the training of the model. The term "ML inference host" refers to an entity, such as a network function, that hosts model during inference mode (which includes both the model execution as well as any online learning if applicable). The ML-host informs the actor about the output of the ML algorithm, and the actor takes a decision for an action (an "action" is performed by an actor as a result of the output of an ML assisted solution). The term "model inference information" refers to information used as an input to the ML model for determining inference(s): the data used to train an ML model and the data used to determine inferences may overlap, however, "training data" and "inference data" refer to different concepts.

What is claimed is:

1. An apparatus comprising:

memory to store policy statement information for a plurality of radio access network (RAN) automation applications (rApps); and

processor circuitry, coupled with the memory, to:

retrieve the policy statement information from the memory, wherein the policy statement information includes respective policy scope identifiers for respective rApps in the plurality of rApps;

identify a conflict associated with common or overlapping policy scope identifiers between two or more rApps from the plurality of rApps;

modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict; and

notify the two or more rApps of the modification of the one or more A1 policies.

2. The apparatus of claim 1, wherein the processor circuitry is further to:

track an enforcement status of an A1 policy created in the non-RT RIC; and

notify an rApp subscribed to the tracked A1 policy when the enforcement status of the tracked A1 policy changes.

3. The apparatus of claim 1, wherein the processor circuitry is further to track a supported A1 policy type for each of a plurality of near-RT RICs.

4. The apparatus of claim 3, wherein the processor circuitry is to notify an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier.

5. The apparatus of claim 3, wherein the processor circuitry is to notify an rApp of a change in supported policy types for a near-RT RIC.

6. The apparatus of claim 1, wherein the processor circuitry is further to generate an A1-related service that includes: a RAN intent discovery service, an A1 policy management service, or an A1 policy type discovery service.

7. The apparatus of claim 1, wherein the processor circuitry is further to:

receive a supported A1 policy types query request via an R1 termination; and

in response to the supported A1 policy types query request, send a query response that includes an indication of supported A1 policy types to a policy generation rApp via the R1 termination.

8. The apparatus of claim 1, wherein the processor circuitry is further to:

receive an A1 policy feedback message from the near-RT RIC via an A1 termination to indicate a change of A1 policy enforcement status; and

send a notification of the A1 policy enforcement status change to a policy generation rApp via an R1 termination.

9. The apparatus of claim 1, wherein the processor circuitry is further to:

receive an A1 policy modification request from a policy generation rApp via an R1 termination;

update an A1 policy over the A1 interface based on the A1 policy modification request; and

send an A1 policy modification response to the policy generation rApp via the R1 termination.

10. The apparatus of claim 1, wherein the processor circuitry is further to:

receive an A1 policy deletion request from a policy generation rApp via an R1 termination;

remove an A1 policy over the A1 interface based on the A1 policy deletion request; and

send an A1 policy deletion response to the policy generation rApp via the R1 termination.

11. One or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause an A1 policy function to:

identify a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps;

modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict; and

notify the two or more rApps of the modification of the one or more A1 policies.

12. The one or more non-transitory computer-readable media of claim 11, wherein the media further stores instructions to:

track an enforcement status of an A1 policy created in the non-RT RIC; and

notify an rApp subscribed to the tracked A1 policy when the enforcement status of the tracked A1 policy changes.

13. The one or more non-transitory computer-readable media of claim 11, wherein the media further stores instructions to:

track a supported A1 policy type for each of a plurality of near-RT RICs; and

notify an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier; or

notify an rApp of a change in supported policy types for a near-RT RIC.

14. The one or more non-transitory computer-readable media of claim 11, wherein the media further stores instructions to:

receive a supported A1 policy types query request via an R1 termination; and

in response to the supported A1 policy types query request, send a query response that includes an indication of supported A1 policy types to a policy generation rApp via the R1 termination.

15. The one or more non-transitory computer-readable media of claim 11, wherein the media further stores instructions to:

receive an A1 policy feedback message from the near-RT RIC via an A1 termination to indicate a change of A1 policy enforcement status; and

send a notification of the A1 policy enforcement status change to a policy generation rApp via an R1 termination.

16. The one or more non-transitory computer-readable media of claim 11, wherein the media further stores instructions to:

receive an A1 policy modification request from a policy generation rApp via an R1 termination;

update an A1 policy over the A1 interface based on the A1 policy modification request; and

send an A1 policy modification response to the policy generation rApp via the R1 termination.

17. The one or more non-transitory computer-readable media of claim 11, wherein the media further stores instructions to generate an A1-related service that includes: a RAN intent discovery service, an A1 policy management service, or an A1 policy type discovery service.

18. One or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause an A1 policy function to:

identify a conflict associated with common or overlapping policy scope identifiers between two or more radio access network (RAN) automation applications (rApps) from a plurality of rApps based on policy statement information that includes respective policy scope identifiers for respective rApps in the plurality of rApps;

modify one or more A1 policies associated with an A1 interface connecting a non-real-time (non-RT) RAN intelligence controller (RIC) and a near-real-time (near-RT) RIC to resolve the conflict;

notify the two or more rApps of the modification of the one or more A1 policies; and track a supported A1 policy type for each of a plurality of near-RT RICs, and either: notify an rApp of a supported A1 policy type for a near-RT RIC based on a policy identifier, or notify an rApp of a change in supported policy types for a near-RT RIC.

**19**. The one or more non-transitory computer-readable media of claim **18**, wherein the media further stores instructions to send, to an rApp via an R1 termination, one or more of;

an indication of a change in an enforcement status of a tracked A1 policy;

an indication of supported A1 policy types; and

a notification of an A1 policy enforcement status change.

**20**. The one or more non-transitory computer-readable media of claim **18**, wherein the media further stores instructions to generate an A1-related service that includes: a RAN intent discovery service, an A1 policy management service, or an A1 policy type discovery service.

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